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PART I

Appendices

About the appendices

Appendices A–K are the Study 1 appendices, carried over verbatim. The remaining appendices hold the complete ledgers of every Study 2 and Amendment 8 run, the variance components of every registered model, the commit chronology, the registration hashes, the preregistration documents verbatim, and a reproduction cookbook.

A

Complete SIPaKMeD classifier run ledger

The following 120 rows are the entire accepted classifier matrix. Accuracy, macro-F1, and balanced accuracy are final-epoch test values. This appendix makes every seed visible and prevents mean-only reporting from hiding an outlying run.

Table A.1: Complete SIPaKMeD classification run ledger. All values are final-epoch results.

Architecture	Protocol	Seed	Train	Test	Acc. (percent)	Macro-F1 (percent)	Bal. acc. (percent)	min	GPU
densenet121	P1	17	3239	810	98.519	98.521	98.527	4.66	3
densenet121	P1	29	3239	810	98.025	98.023	98.027	4.62	2
densenet121	P1	43	3239	810	97.778	97.776	97.803	4.54	1
densenet121	P1	71	3239	810	98.025	98.031	98.018	4.61	3
densenet121	P1	101	3239	810	98.272	98.284	98.276	4.70	2
densenet121	P1-prime	17	3239	810	98.519	98.521	98.527	4.61	1
densenet121	P1-prime	29	3239	810	98.025	98.023	98.027	4.48	3
densenet121	P1-prime	43	3239	810	97.778	97.776	97.803	4.49	2
densenet121	P1-prime	71	3239	810	98.025	98.031	98.018	4.48	1
densenet121	P1-prime	101	3239	810	98.272	98.284	98.276	4.45	3
densenet121	P2	17	3239	810	95.926	95.972	95.946	4.47	2
densenet121	P2	29	3239	810	97.778	97.766	97.798	4.55	1
densenet121	P2	43	3239	810	96.914	96.927	96.906	4.51	3
densenet121	P2	71	3239	810	96.543	96.561	96.566	4.47	2
densenet121	P2	101	3239	810	96.420	96.446	96.395	4.59	1
densenet121	P3	17	3239	810	97.778	97.781	97.774	4.54	3
densenet121	P3	29	3239	810	97.901	97.895	97.904	4.64	2
densenet121	P3	43	3239	810	97.531	97.538	97.553	4.56	1
densenet121	P3	71	3239	810	96.790	96.824	96.801	4.20	3
densenet121	P3	101	3239	810	96.914	96.945	96.894	4.00	2
efficientnet_b0	P1	17	3239	810	97.654	97.659	97.662	2.84	1
efficientnet_b0	P1	29	3239	810	98.272	98.276	98.273	2.89	3
efficientnet_b0	P1	43	3239	810	97.160	97.183	97.177	2.92	2
efficientnet_b0	P1	71	3239	810	97.531	97.540	97.541	2.98	1
efficientnet_b0	P1	101	3239	810	98.272	98.279	98.288	2.87	3
efficientnet_b0	P1-prime	17	3239	810	97.654	97.659	97.662	2.87	2
efficientnet_b0	P1-prime	29	3239	810	98.272	98.276	98.273	3.00	1
efficientnet_b0	P1-prime	43	3239	810	97.160	97.183	97.177	2.96	3
efficientnet_b0	P1-prime	71	3239	810	97.531	97.540	97.541	2.88	2

Architecture	Protocol	Seed	Train	Test	Acc. (percent)	Macro-F1 (percent)	Bal. acc. (percent)	min	GPU
efficientnet_b0	P1-prime	101	3239	810	98.272	98.279	98.288	2.86	1
efficientnet_b0	P2	17	3239	810	97.778	97.790	97.799	3.01	3
efficientnet_b0	P2	29	3239	810	95.185	95.259	95.163	2.92	2
efficientnet_b0	P2	43	3239	810	98.642	98.641	98.643	2.92	1
efficientnet_b0	P2	71	3239	810	95.432	95.450	95.407	2.97	3
efficientnet_b0	P2	101	3239	810	96.420	96.413	96.422	2.93	2
efficientnet_b0	P3	17	3239	810	98.148	98.144	98.170	2.89	1
efficientnet_b0	P3	29	3239	810	97.901	97.920	97.910	2.98	3
efficientnet_b0	P3	43	3239	810	96.296	96.316	96.325	2.99	2
efficientnet_b0	P3	71	3239	810	97.407	97.428	97.427	2.94	1
efficientnet_b0	P3	101	3239	810	97.284	97.300	97.285	2.80	3
mobilenet_v2	P1	17	3239	810	97.284	97.287	97.278	2.72	2
mobilenet_v2	P1	29	3239	810	97.407	97.406	97.403	2.76	1
mobilenet_v2	P1	43	3239	810	96.914	96.933	96.944	2.74	3
mobilenet_v2	P1	71	3239	810	97.284	97.296	97.289	2.71	2
mobilenet_v2	P1	101	3239	810	98.148	98.149	98.167	2.72	1
mobilenet_v2	P1-prime	17	3239	810	97.284	97.287	97.278	2.69	3
mobilenet_v2	P1-prime	29	3239	810	97.407	97.406	97.403	2.74	2
mobilenet_v2	P1-prime	43	3239	810	96.914	96.933	96.944	2.72	1
mobilenet_v2	P1-prime	71	3239	810	97.284	97.296	97.289	2.71	3
mobilenet_v2	P1-prime	101	3239	810	98.148	98.149	98.167	2.73	2
mobilenet_v2	P2	17	3239	810	96.420	96.432	96.453	2.75	1
mobilenet_v2	P2	29	3239	810	93.704	93.732	93.714	2.73	3
mobilenet_v2	P2	43	3239	810	96.667	96.679	96.637	2.74	2
mobilenet_v2	P2	71	3239	810	93.457	93.486	93.458	2.78	1
mobilenet_v2	P2	101	3239	810	95.556	95.553	95.534	2.77	3
mobilenet_v2	P3	17	3239	810	97.407	97.408	97.433	2.68	2
mobilenet_v2	P3	29	3239	810	96.173	96.163	96.163	2.74	1
mobilenet_v2	P3	43	3239	810	96.667	96.669	96.696	2.74	3
mobilenet_v2	P3	71	3239	810	95.802	95.823	95.813	2.92	2
mobilenet_v2	P3	101	3239	810	94.938	94.930	94.900	2.79	1
resnet18	P1	17	3239	810	97.901	97.900	97.930	2.57	1
resnet18	P1	29	3239	810	97.037	97.028	97.051	2.54	2
resnet18	P1	43	3239	810	96.914	96.922	96.938	2.48	3

Architecture	Protocol	Seed	Train	Test	Acc. (percent)	Macro-F1 (percent)	Bal. acc. (percent)	min	GPU
resnet18	P1	71	3239	810	96.420	96.455	96.410	2.61	1
resnet18	P1	101	3239	810	97.654	97.661	97.674	2.54	2
resnet18	P1-prime	17	3239	810	97.901	97.900	97.930	2.53	3
resnet18	P1-prime	29	3239	810	97.037	97.028	97.051	2.62	1
resnet18	P1-prime	43	3239	810	96.914	96.922	96.938	2.61	2
resnet18	P1-prime	71	3239	810	96.420	96.455	96.410	2.48	3
resnet18	P1-prime	101	3239	810	97.654	97.661	97.674	2.62	1
resnet18	P2	17	3239	810	96.420	96.444	96.447	2.58	2
resnet18	P2	29	3239	810	96.049	96.051	96.058	2.47	3
resnet18	P2	43	3239	810	96.543	96.535	96.502	2.49	1
resnet18	P2	71	3239	810	96.543	96.563	96.528	2.56	2
resnet18	P2	101	3239	810	94.444	94.436	94.384	2.50	3
resnet18	P3	17	3239	810	98.395	98.395	98.400	2.55	1
resnet18	P3	29	3239	810	95.062	95.069	95.014	2.55	3
resnet18	P3	43	3239	810	96.173	96.165	96.215	2.52	2
resnet18	P3	71	3239	810	96.543	96.578	96.561	2.79	1
resnet18	P3	101	3239	810	95.556	95.576	95.526	2.79	3
resnet50	P1	17	3239	810	97.778	97.781	97.792	3.73	2
resnet50	P1	29	3239	810	97.284	97.304	97.275	3.29	1
resnet50	P1	43	3239	810	96.790	96.801	96.805	3.32	3
resnet50	P1	71	3239	810	97.901	97.903	97.904	3.48	2
resnet50	P1	101	3239	810	97.778	97.776	97.777	3.45	1
resnet50	P1-prime	17	3239	810	97.778	97.781	97.792	3.26	3
resnet50	P1-prime	29	3239	810	97.284	97.304	97.275	3.45	2
resnet50	P1-prime	43	3239	810	96.790	96.801	96.805	3.52	1
resnet50	P1-prime	71	3239	810	97.901	97.903	97.904	3.34	3
resnet50	P1-prime	101	3239	810	97.778	97.776	97.777	3.54	2
resnet50	P2	17	3239	810	97.037	97.067	97.053	3.47	1
resnet50	P2	29	3239	810	96.914	96.931	96.916	3.24	3
resnet50	P2	43	3239	810	97.284	97.298	97.290	3.45	2

Architecture	Protocol	Seed	Train	Test	Acc. (percent)	Macro-F1 (percent)	Bal. acc. (percent)	min	GPU
resnet50	P2	71	3239	810	96.420	96.422	96.433	3.47	1
resnet50	P2	101	3239	810	96.296	96.303	96.288	3.29	3
resnet50	P3	17	3239	810	96.543	96.519	96.573	3.47	2
resnet50	P3	29	3239	810	97.037	97.071	97.049	3.46	1
resnet50	P3	43	3239	810	95.679	95.701	95.703	3.25	3
resnet50	P3	71	3239	810	95.926	95.945	95.958	3.73	2
resnet50	P3	101	3239	810	96.173	96.168	96.161	3.89	1
vit_b_16	P1	17	3239	810	97.407	97.414	97.430	9.30	3
vit_b_16	P1	29	3239	810	97.037	97.050	97.061	9.25	2
vit_b_16	P1	43	3239	810	95.556	95.541	95.606	9.35	1
vit_b_16	P1	71	3239	810	98.025	98.036	98.023	9.31	3
vit_b_16	P1	101	3239	810	97.654	97.666	97.684	9.27	2
vit_b_16	P1-prime	17	3239	810	97.407	97.414	97.430	9.29	1
vit_b_16	P1-prime	29	3239	810	97.037	97.050	97.061	9.19	3
vit_b_16	P1-prime	43	3239	810	95.556	95.541	95.606	9.24	2
vit_b_16	P1-prime	71	3239	810	98.025	98.036	98.023	9.30	1
vit_b_16	P1-prime	101	3239	810	97.654	97.666	97.684	9.24	3
vit_b_16	P2	17	3239	810	95.556	95.618	95.526	9.27	2
vit_b_16	P2	29	3239	810	95.802	95.821	95.788	9.36	1
vit_b_16	P2	43	3239	810	96.667	96.693	96.688	9.32	3
vit_b_16	P2	71	3239	810	95.432	95.432	95.392	9.30	2
vit_b_16	P2	101	3239	810	94.321	94.421	94.283	9.28	1
vit_b_16	P3	17	3239	810	96.296	96.258	96.328	9.19	3
vit_b_16	P3	29	3239	810	95.185	95.218	95.214	9.23	2
vit_b_16	P3	43	3239	810	92.840	92.915	92.770	9.26	1
vit_b_16	P3	71	3239	810	97.284	97.292	97.301	9.00	3
vit_b_16	P3	101	3239	810	96.049	96.039	96.066	8.81	2

B

Complete parent-probe run ledger

These 30 rows include five preprocessing arms and the label-shuffled negative control. All use exactly 424 parent classes, 848 training crops, and 848 held-out sibling crops.

Table B.1: Complete parent-identity probe ledger.

Mode	Shuffle	Seed	Parents	Train	Test	Chance (percent)	Accuracy (percent)	min	GPU
background_only	no	17	424	848	848	0.2358	18.514	21.00	3
background_only	no	29	424	848	848	0.2358	20.047	20.55	1
background_only	no	43	424	848	848	0.2358	19.811	20.23	2
background_only	no	71	424	848	848	0.2358	17.925	89.59	3
background_only	no	101	424	848	848	0.2358	17.099	89.40	2
grayscale	no	17	424	848	848	0.2358	15.212	0.93	3
grayscale	no	29	424	848	848	0.2358	13.208	0.88	2
grayscale	no	43	424	848	848	0.2358	14.741	1.02	1
grayscale	no	71	424	848	848	0.2358	13.208	0.97	2
grayscale	no	101	424	848	848	0.2358	12.618	1.02	3
highpass	no	17	424	848	848	0.2358	16.863	15.73	1
highpass	no	29	424	848	848	0.2358	16.156	15.87	1
highpass	no	43	424	848	848	0.2358	16.863	15.93	1
highpass	no	71	424	848	848	0.2358	16.038	15.80	1
highpass	no	101	424	848	848	0.2358	15.448	16.04	1
rgb	no	17	424	848	848	0.2358	25.943	0.89	1
rgb	no	29	424	848	848	0.2358	26.769	0.76	2
rgb	no	43	424	848	848	0.2358	25.236	0.76	3
rgb	no	71	424	848	848	0.2358	25.590	0.77	1
rgb	no	101	424	848	848	0.2358	25.118	0.79	2
rgb	yes	17	424	848	848	0.2358	0.590	0.78	3
rgb	yes	29	424	848	848	0.2358	0.000	0.88	1
rgb	yes	43	424	848	848	0.2358	0.118	0.86	3
rgb	yes	71	424	848	848	0.2358	0.000	0.76	2
rgb	yes	101	424	848	848	0.2358	0.118	0.82	1
stain_normalized	no	17	424	848	848	0.2358	17.335	2.49	1
stain_normalized	no	29	424	848	848	0.2358	17.099	2.41	2
stain_normalized	no	43	424	848	848	0.2358	18.396	2.26	3
stain_normalized	no	71	424	848	848	0.2358	16.745	2.42	1
stain_normalized	no	101	424	848	848	0.2358	16.981	2.20	2

C

Complete Cx22 model run ledger

The following 175 rows are all accepted Cx22 cells: five semantic architectures and two instance architectures, each under five protocols and five seeds. "Primary" is semantic macro-Dice or matched-instance Dice; "Nuc./R70" is nucleus Dice for semantic rows and Recall@0.70 for instance rows.

Table C.1: Complete Cx22 run ledger. Primary is macro Dice for semantic models and matched-instance Dice for instance models; the ninth column is nucleus Dice or recall at Dice 0.70.

Task	Architecture	Protocol	Seed	Train	Test	Primary (percent)	IoU (percent)	Nuc./R70 (percent)	hours
instance	maskrcnn_resnet50_fpn	OFFICIAL	17	400	100	92.117	87.060	96.991	14.35
instance	maskrcnn_resnet50_fpn	OFFICIAL	29	400	100	92.372	87.354	97.068	14.92
instance	maskrcnn_resnet50_fpn	OFFICIAL	43	400	100	91.844	86.861	96.682	15.89
instance	maskrcnn_resnet50_fpn	OFFICIAL	71	400	100	92.669	87.541	97.608	0.49
instance	maskrcnn_resnet50_fpn	OFFICIAL	101	400	100	92.433	87.559	97.145	1.29
instance	maskrcnn_resnet50_fpn	P1	17	1056	264	92.683	87.370	96.540	33.85
instance	maskrcnn_resnet50_fpn	P1	29	1056	264	91.354	86.327	95.009	33.71
instance	maskrcnn_resnet50_fpn	P1	43	1056	264	92.282	87.191	96.854	35.84
instance	maskrcnn_resnet50_fpn	P1	71	1056	264	92.343	87.320	96.550	37.54
instance	maskrcnn_resnet50_fpn	P1	101	1056	264	91.819	86.735	96.471	2.55
instance	maskrcnn_resnet50_fpn	P1-prime	17	1056	264	92.683	87.370	96.540	33.94
instance	maskrcnn_resnet50_fpn	P1-prime	29	1056	264	91.354	86.327	95.009	33.95
instance	maskrcnn_resnet50_fpn	P1-prime	43	1056	264	92.282	87.191	96.854	35.19
instance	maskrcnn_resnet50_fpn	P1-prime	71	1055	264	91.733	86.958	95.483	38.29
instance	maskrcnn_resnet50_fpn	P1-prime	101	1053	264	92.213	87.257	97.248	2.56
instance	maskrcnn_resnet50_fpn	P2	17	1056	264	91.815	86.503	95.983	33.84
instance	maskrcnn_resnet50_fpn	P2	29	1056	264	91.313	86.228	95.159	34.29
instance	maskrcnn_resnet50_fpn	P2	43	1056	264	92.024	86.559	96.750	35.88
instance	maskrcnn_resnet50_fpn	P2	71	1055	265	90.993	85.998	92.239	0.26
instance	maskrcnn_resnet50_fpn	P2	101	1053	267	91.793	86.639	94.671	2.57
instance	maskrcnn_resnet50_fpn	P3	17	891	429	89.855	84.435	94.937	47.44
instance	maskrcnn_resnet50_fpn	P3	29	825	495	91.606	86.357	95.207	49.79
instance	maskrcnn_resnet50_fpn	P3	43	894	426	90.023	84.807	94.415	47.25
instance	maskrcnn_resnet50_fpn	P3	71	773	547	92.181	86.751	96.205	1.16
instance	maskrcnn_resnet50_fpn	P3	101	791	529	91.910	86.399	96.806	1.98
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	17	400	100	91.970	87.274	95.910	1.31
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	29	400	100	92.548	87.588	96.991	1.33
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	43	400	100	91.041	86.290	94.367	1.32
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	71	400	100	92.442	87.459	97.068	1.69
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	101	400	100	92.886	87.943	97.531	1.40
instance	maskrcnn_resnet50_fpn_v2	P1	17	1056	264	91.235	86.322	91.816	2.63
instance	maskrcnn_resnet50_fpn_v2	P1	29	1056	264	91.543	86.726	92.276	2.64
instance	maskrcnn_resnet50_fpn_v2	P1	43	1056	264	92.536	87.688	94.870	2.65
instance	maskrcnn_resnet50_fpn_v2	P1	71	1056	264	92.342	87.566	94.856	2.63
instance	maskrcnn_resnet50_fpn_v2	P1	101	1056	264	92.318	87.546	95.131	2.99
instance	maskrcnn_resnet50_fpn_v2	P1-prime	17	1056	264	91.235	86.322	91.816	2.63
instance	maskrcnn_resnet50_fpn_v2	P1-prime	29	1056	264	91.543	86.726	92.276	2.63
instance	maskrcnn_resnet50_fpn_v2	P1-prime	43	1056	264	92.536	87.688	94.870	2.64
instance	maskrcnn_resnet50_fpn_v2	P1-prime	71	1055	264	91.938	87.241	93.915	2.62
instance	maskrcnn_resnet50_fpn_v2	P1-prime	101	1053	264	92.502	87.584	95.272	2.85
instance	maskrcnn_resnet50_fpn_v2	P2	17	1056	264	92.036	87.121	95.303	2.64
instance	maskrcnn_resnet50_fpn_v2	P2	29	1056	264	91.737	86.754	94.870	2.65
instance	maskrcnn_resnet50_fpn_v2	P2	43	1056	264	92.056	86.980	95.297	2.65

Task	Architecture	Protocol	Seed	Train	Test	Primary (percent)	IoU (percent)	Nuc./R70 (percent)	hours
instance	maskrcnn_resnet50_fpn_v2	P2	71	1055	265	90.232	85.363	87.492	2.64
instance	maskrcnn_resnet50_fpn_v2	P2	101	1053	267	90.762	85.672	92.038	2.70
instance	maskrcnn_resnet50_fpn_v2	P3	17	891	429	91.242	85.867	94.762	2.26
instance	maskrcnn_resnet50_fpn_v2	P3	29	825	495	89.960	84.956	91.273	2.10
instance	maskrcnn_resnet50_fpn_v2	P3	43	894	426	90.317	85.094	91.622	2.29
instance	maskrcnn_resnet50_fpn_v2	P3	71	773	547	90.474	85.440	90.769	1.98
instance	maskrcnn_resnet50_fpn_v2	P3	101	791	529	91.026	85.770	94.358	2.02
semantic	deeplabv3plus_resnet50	OFFICIAL	17	400	100	82.697	72.771	71.084	1.02
semantic	deeplabv3plus_resnet50	OFFICIAL	29	400	100	83.602	73.897	72.700	1.13
semantic	deeplabv3plus_resnet50	OFFICIAL	43	400	100	84.126	74.580	73.775	1.08
semantic	deeplabv3plus_resnet50	OFFICIAL	71	400	100	83.972	74.385	73.418	1.12
semantic	deeplabv3plus_resnet50	OFFICIAL	101	400	100	80.382	70.180	66.995	1.11
semantic	deeplabv3plus_resnet50	P1	17	1056	264	88.570	80.909	81.412	2.09
semantic	deeplabv3plus_resnet50	P1	29	1056	264	88.464	80.728	81.373	1.91
semantic	deeplabv3plus_resnet50	P1	43	1056	264	89.196	81.695	82.572	2.02
semantic	deeplabv3plus_resnet50	P1	71	1056	264	89.542	82.165	83.162	2.14
semantic	deeplabv3plus_resnet50	P1	101	1056	264	88.738	81.195	81.662	2.32
semantic	deeplabv3plus_resnet50	P1-prime	17	1056	264	88.570	80.909	81.412	2.13
semantic	deeplabv3plus_resnet50	P1-prime	29	1056	264	88.464	80.728	81.373	1.93
semantic	deeplabv3plus_resnet50	P1-prime	43	1056	264	89.196	81.695	82.572	2.03
semantic	deeplabv3plus_resnet50	P1-prime	71	1055	264	89.443	82.040	82.958	2.14
semantic	deeplabv3plus_resnet50	P1-prime	101	1053	264	88.514	80.848	81.162	2.28
semantic	deeplabv3plus_resnet50	P2	17	1056	264	88.304	80.549	81.070	1.90
semantic	deeplabv3plus_resnet50	P2	29	1056	264	88.930	81.407	82.016	2.03
semantic	deeplabv3plus_resnet50	P2	43	1056	264	88.731	80.967	81.799	1.99
semantic	deeplabv3plus_resnet50	P2	71	1055	265	88.988	81.513	82.202	2.22
semantic	deeplabv3plus_resnet50	P2	101	1053	267	88.268	80.470	80.952	2.28
semantic	deeplabv3plus_resnet50	P3	17	891	429	88.332	80.528	81.111	2.13
semantic	deeplabv3plus_resnet50	P3	29	825	495	87.707	79.682	80.088	2.57
semantic	deeplabv3plus_resnet50	P3	43	894	426	88.621	80.940	81.705	2.70
semantic	deeplabv3plus_resnet50	P3	71	773	547	88.041	80.050	80.594	3.46
semantic	deeplabv3plus_resnet50	P3	101	791	529	87.790	79.677	80.185	3.43
semantic	fpn_resnet34	OFFICIAL	17	400	100	84.378	74.779	74.746	1.08
semantic	fpn_resnet34	OFFICIAL	29	400	100	84.801	75.377	75.422	1.12
semantic	fpn_resnet34	OFFICIAL	43	400	100	84.389	74.760	74.723	1.10
semantic	fpn_resnet34	OFFICIAL	71	400	100	84.630	75.136	75.152	1.07
semantic	fpn_resnet34	OFFICIAL	101	400	100	83.934	74.231	73.850	1.09
semantic	fpn_resnet34	P1	17	1056	264	88.606	80.936	81.657	2.30
semantic	fpn_resnet34	P1	29	1056	264	88.376	80.651	81.204	2.17
semantic	fpn_resnet34	P1	43	1056	264	89.242	81.731	82.802	2.29
semantic	fpn_resnet34	P1	71	1056	264	89.415	81.991	83.139	2.29
semantic	fpn_resnet34	P1	101	1056	264	88.854	81.286	81.903	2.31
semantic	fpn_resnet34	P1-prime	17	1056	264	88.606	80.936	81.657	2.23
semantic	fpn_resnet34	P1-prime	29	1056	264	88.376	80.651	81.204	2.19
semantic	fpn_resnet34	P1-prime	43	1056	264	89.242	81.731	82.802	2.30
semantic	fpn_resnet34	P1-prime	71	1055	264	89.760	82.473	83.695	2.30
semantic	fpn_resnet34	P1-prime	101	1053	264	88.880	81.350	81.891	2.26

Task	Architecture	Protocol	Seed	Train	Test	Primary (percent)	IoU (percent)	Nuc./R70 (percent)	hours
semantic	fpn_resnet34	P2	17	1056	264	88.539	80.901	81.452	2.18
semantic	fpn_resnet34	P2	29	1056	264	89.006	81.470	82.385	2.20
semantic	fpn_resnet34	P2	43	1056	264	89.498	82.080	83.321	2.22
semantic	fpn_resnet34	P2	71	1055	265	89.102	81.618	82.649	2.24
semantic	fpn_resnet34	P2	101	1053	267	88.296	80.471	81.211	2.25
semantic	fpn_resnet34	P3	17	891	429	88.029	80.133	80.698	2.84
semantic	fpn_resnet34	P3	29	825	495	88.080	80.127	80.967	3.16
semantic	fpn_resnet34	P3	43	894	426	88.744	81.048	82.174	2.75
semantic	fpn_resnet34	P3	71	773	547	88.510	80.711	81.583	3.46
semantic	fpn_resnet34	P3	101	791	529	88.287	80.367	81.266	3.45
semantic	manet_resnet34	OFFICIAL	17	400	100	83.189	73.517	71.864	1.06
semantic	manet_resnet34	OFFICIAL	29	400	100	86.409	77.571	77.856	1.08
semantic	manet_resnet34	OFFICIAL	43	400	100	86.521	77.608	78.266	1.07
semantic	manet_resnet34	OFFICIAL	71	400	100	86.256	77.306	77.825	1.15
semantic	manet_resnet34	OFFICIAL	101	400	100	85.118	75.878	75.519	1.19
semantic	manet_resnet34	P1	17	1056	264	89.317	81.998	82.837	2.13
semantic	manet_resnet34	P1	29	1056	264	89.510	82.294	83.125	2.19
semantic	manet_resnet34	P1	43	1056	264	90.535	83.620	85.003	2.09
semantic	manet_resnet34	P1	71	1056	264	90.674	83.752	85.088	2.06
semantic	manet_resnet34	P1	101	1056	264	90.128	83.049	84.158	2.14
semantic	manet_resnet34	P1-prime	17	1056	264	89.317	81.998	82.837	2.17
semantic	manet_resnet34	P1-prime	29	1056	264	89.510	82.294	83.125	2.06
semantic	manet_resnet34	P1-prime	43	1056	264	90.535	83.620	85.003	2.05
semantic	manet_resnet34	P1-prime	71	1055	264	90.764	83.957	85.467	2.05
semantic	manet_resnet34	P1-prime	101	1053	264	90.221	83.230	84.212	2.07
semantic	manet_resnet34	P2	17	1056	264	89.337	81.962	83.086	2.14
semantic	manet_resnet34	P2	29	1056	264	90.185	83.160	84.409	2.08
semantic	manet_resnet34	P2	43	1056	264	90.170	83.032	84.460	2.01
semantic	manet_resnet34	P2	71	1055	265	90.458	83.608	84.919	2.09
semantic	manet_resnet34	P2	101	1053	267	89.687	82.410	83.599	2.00
semantic	manet_resnet34	P3	17	891	429	89.762	82.615	83.748	2.75
semantic	manet_resnet34	P3	29	825	495	89.392	82.029	83.305	2.62
semantic	manet_resnet34	P3	43	894	426	89.895	82.761	83.970	2.16
semantic	manet_resnet34	P3	71	773	547	89.469	82.176	83.199	3.03
semantic	manet_resnet34	P3	101	791	529	89.625	82.349	83.609	1.96
semantic	unet_resnet34	OFFICIAL	17	400	100	85.332	76.145	75.734	1.03
semantic	unet_resnet34	OFFICIAL	29	400	100	85.420	76.382	75.839	1.11
semantic	unet_resnet34	OFFICIAL	43	400	100	85.102	75.810	75.351	1.03
semantic	unet_resnet34	OFFICIAL	71	400	100	86.580	77.740	78.197	1.13
semantic	unet_resnet34	OFFICIAL	101	400	100	85.749	76.664	76.744	1.06
semantic	unet_resnet34	P1	17	1056	264	90.196	83.235	84.389	2.09
semantic	unet_resnet34	P1	29	1056	264	89.654	82.414	83.452	2.16
semantic	unet_resnet34	P1	43	1056	264	90.616	83.797	85.081	2.01
semantic	unet_resnet34	P1	71	1056	264	90.880	84.082	85.657	2.09
semantic	unet_resnet34	P1	101	1056	264	90.093	82.967	84.100	2.04
semantic	unet_resnet34	P1-prime	17	1056	264	90.196	83.235	84.389	2.12
semantic	unet_resnet34	P1-prime	29	1056	264	89.654	82.414	83.452	2.20

Task	Architecture	Protocol	Seed	Train	Test	Primary (percent)	IoU (percent)	Nuc./R70 (percent)	hours
semantic	unet_resnet34	P1-prime	43	1056	264	90.616	83.797	85.081	1.98
semantic	unet_resnet34	P1-prime	71	1055	264	91.020	84.317	85.849	2.06
semantic	unet_resnet34	P1-prime	101	1053	264	90.092	83.064	84.113	1.95
semantic	unet_resnet34	P2	17	1056	264	89.429	82.217	83.000	2.10
semantic	unet_resnet34	P2	29	1056	264	90.249	83.272	84.602	2.17
semantic	unet_resnet34	P2	43	1056	264	90.674	83.773	85.275	2.03
semantic	unet_resnet34	P2	71	1055	265	89.519	82.235	83.280	2.16
semantic	unet_resnet34	P2	101	1053	267	90.082	82.951	84.446	2.01
semantic	unet_resnet34	P3	17	891	429	89.911	82.788	83.948	2.23
semantic	unet_resnet34	P3	29	825	495	89.629	82.392	83.516	2.85
semantic	unet_resnet34	P3	43	894	426	89.892	82.748	83.984	2.82
semantic	unet_resnet34	P3	71	773	547	89.840	82.694	83.704	3.12
semantic	unet_resnet34	P3	101	791	529	89.576	82.274	83.557	2.96
semantic	unetplusplus_resnet34	OFFICIAL	17	400	100	85.933	77.036	76.615	1.06
semantic	unetplusplus_resnet34	OFFICIAL	29	400	100	86.864	78.288	78.526	1.15
semantic	unetplusplus_resnet34	OFFICIAL	43	400	100	86.771	78.140	78.396	1.11
semantic	unetplusplus_resnet34	OFFICIAL	71	400	100	88.276	79.991	81.686	1.08
semantic	unetplusplus_resnet34	OFFICIAL	101	400	100	88.645	80.557	82.123	1.08
semantic	unetplusplus_resnet34	P1	17	1056	264	90.270	83.391	84.440	1.97
semantic	unetplusplus_resnet34	P1	29	1056	264	89.890	82.832	83.858	1.88
semantic	unetplusplus_resnet34	P1	43	1056	264	90.909	84.191	85.525	2.05
semantic	unetplusplus_resnet34	P1	71	1056	264	91.190	84.587	86.038	1.89
semantic	unetplusplus_resnet34	P1	101	1056	264	90.564	83.715	84.868	2.26
semantic	unetplusplus_resnet34	P1-prime	17	1056	264	90.270	83.391	84.440	1.97
semantic	unetplusplus_resnet34	P1-prime	29	1056	264	89.890	82.832	83.858	1.85
semantic	unetplusplus_resnet34	P1-prime	43	1056	264	90.909	84.191	85.525	2.03
semantic	unetplusplus_resnet34	P1-prime	71	1055	264	91.266	84.711	86.229	1.85
semantic	unetplusplus_resnet34	P1-prime	101	1053	264	90.576	83.692	84.939	2.27
semantic	unetplusplus_resnet34	P2	17	1056	264	90.077	83.064	84.058	1.87
semantic	unetplusplus_resnet34	P2	29	1056	264	90.324	83.431	84.639	1.91
semantic	unetplusplus_resnet34	P2	43	1056	264	90.205	83.068	84.221	2.03
semantic	unetplusplus_resnet34	P2	71	1055	265	90.977	84.343	85.735	1.81
semantic	unetplusplus_resnet34	P2	101	1053	267	90.300	83.261	84.697	2.02
semantic	unetplusplus_resnet34	P3	17	891	429	90.011	82.987	84.055	2.10
semantic	unetplusplus_resnet34	P3	29	825	495	89.719	82.531	83.578	2.58
semantic	unetplusplus_resnet34	P3	43	894	426	90.126	83.176	84.281	2.19
semantic	unetplusplus_resnet34	P3	71	773	547	90.283	83.365	84.578	2.56
semantic	unetplusplus_resnet34	P3	101	791	529	90.099	83.076	84.302	2.83

D

Owner data files and integrity hashes

This table is generated from `data_sources.json`. A reproduction refuses a file whose byte length or SHA-256 differs, even if its archive name is unchanged.

Table D.1: Declared owner-source files, exact byte sizes, checksums, and use notes.

Dataset	File	Bytes	SHA-256	Terms note
sipakmed	im_Superficial-Intermediate.7z	762,939,737	3b73e406df14d60b1bd9c8ac3b01b594e70bb51ba32fb080141fc6a7f489a599	Public for experimental use; cite Plissiti et al., ICIP 2018
sipakmed	im_Parabasal.7z	548,667,699	13e9c29f016a5b83731e038cfe518671a7815df28836dee706acba6ecd2369a8	Public for experimental use; cite Plissiti et al., ICIP 2018
sipakmed	im_Koilocytotic.7z	1,290,478,065	653578e878e3cc63a8510cd7ec26d918637d9881e913e29a5bb205d0d9cfb70d	Public for experimental use; cite Plissiti et al., ICIP 2018
sipakmed	im_Metaplastic.7z	1,396,480,123	97f1badc5f606f127d2af0d18a008ebd41a2b60f44785cb332a6eaa33004d2af	Public for experimental use; cite Plissiti et al., ICIP 2018
sipakmed	im_Dyskeratotic.7z	1,182,980,549	a8768bc03c9e814063ff946b7ef7548067e16b760168267d2b12e95a66955cce	Public for experimental use; cite Plissiti et al., ICIP 2018
cx22	Cx22-Multi-Test.zip	1,780,461	6d444e6e4bf8734073bc90e6e2bb29b0e36c66f7b783cbad9f969208621624bb	Non-commercial research and education only
cx22	Cx22-Multi-Train.zip	6,361,959	171be9caf6d0709f3fce2b81a07e1974229c5faae355f197efe553ec1d6d1992	Non-commercial research and education only
cx22	Cx22-Pair.zip	3,705,743	130b6a92db9811151a60e27dc70f4180965ecf0f1dbc84146c87d00c41d7ba9f	Non-commercial research and education only
ccedd	CCEDD.zip	1,507,832,267	d85e6a380a15923f0c2b38e13dd31467f75d11b49e8d21740f16ee69082cdcb1	Non-commercial research and education only; redistribution prohibited
cric	classifications.csv	1,224,400	e1f50183998ddb7392d2a0d4c96dce44c00dbe3833df03f266cc36e047445de5	CC BY 4.0; cite Rezende et al. and the Figshare record
herlev	Cervical Cell Image Database.zip	44,688,829	5b5b740abb599710b9d97fe72954540df9ca7255b2b40a52c8b25ac90d0c50da	CC BY 4.0; repackaging of the original Herlev cervical-cell images

E

Released split hashes

Each CSV contains crop ID, parent ID, partition, protocol, and seed. Hashes identify the exact semantic assignment rather than merely naming a random seed.

Table E.1: Released SIPaKMeD split manifests and hashes.

Protocol	Seed	File	SHA-256
P1	17	sipakmed_p1_seed17.csv	28abd1605d72d888f561cecbc4d987b0dafff671653c37d0460bccac8d3c0cac
P1-prime	17	sipakmed_p1_prime_seed17.csv	611fcb7e6ed98e66a86f9cbdbcb27644850e15d93a4143249dc611d00364d784
P2	17	sipakmed_p2_seed17.csv	488077976eb9397b2a772597e4453e2a31e809d67cb1a21595ece5b143b35262
P3	17	sipakmed_p3_seed17.csv	797a4c0d73fd0722aebec95a14cf798ac3cf94575ac7e93812a85b738df0ea14
P1	29	sipakmed_p1_seed29.csv	adef646d2edabf02bf92a00a7a8e68eefee876159ee2d43d1ab4c25fc80e1e75
P1-prime	29	sipakmed_p1_prime_seed29.csv	72f36aa2713e5dbd4a7d16f471ec466f88dacefbf597d5831bf3b409cc9cb256
P2	29	sipakmed_p2_seed29.csv	4088e6c4b03fca9e16ecb7db57b221e1769e09a5bf6f9c2f36b52519332aed21
P3	29	sipakmed_p3_seed29.csv	3722ec8802c1bbfdf4fe7d9aad2ecd99be4ca66da83add1e331f6d816ed936cc
P1	43	sipakmed_p1_seed43.csv	d5087a1636cceebd8943accfbeb2129046772df01276a693b202f02340846beb
P1-prime	43	sipakmed_p1_prime_seed43.csv	a7417cb3eeccbc4629bd212afffc2c3f42045e9d864f10b232b2c1f15e3a4b1b
P2	43	sipakmed_p2_seed43.csv	bd1a079eadacd51adbeb0dc169d0bcca5cf3b011d3d6b71bb98e4fa71988a62f
P3	43	sipakmed_p3_seed43.csv	877f62465214f160aaa2776bd4ba92c2054c78d744261ae9ada9a4ce66343de8
P1	71	sipakmed_p1_seed71.csv	5148ddbc11d90a927f6cf7d9bc5213d0b1431b9d80cb58fa0d870de76c889198
P1-prime	71	sipakmed_p1_prime_seed71.csv	21ce3cab719badee5dad0bb192c2e55e695fe44d35d58c0a9c6921e4aa7c5b3a
P2	71	sipakmed_p2_seed71.csv	52095a0a39a27f439d64646ed50b07f15bfc729dee195a42f0dcb4c4739d0e25
P3	71	sipakmed_p3_seed71.csv	bbf975f39f790159dcc903bb3d53eb87968e6d612066bfe0cd5c1e8fb6cc860e
P1	101	sipakmed_p1_seed101.csv	1dadec3ecdbce04080b796d735ca7ecfe84cc4b74342d8480326b7c95dd9ff44
P1-prime	101	sipakmed_p1_prime_seed101.csv	d9e096b00642b40eb979f3d75e827f6929e15d0c62a3a9a3738eb4490526185f
P2	101	sipakmed_p2_seed101.csv	178418b059a5c924b305ca5a05472b2076f3423626538e27416ad39f6e1f891f
P3	101	sipakmed_p3_seed101.csv	269ac8e1badbad7c11a5a45cd72b2c2d17ac0fa2c09cb0fa007b45ce51bbef90

F

Git decision chronology

The chronology below is descriptive provenance for major decisions and corrections. The authoritative ordering remains the Git object database. In particular, the first entry is the original preregistration commit.

Table F.1: Complete repository decision chronology through the frozen result set.

Commit	Date	Recorded change
67e8ded	2026-08-22	preregister confirmatory analysis
8ebc6d5	2026-08-22	build leakage verification pipeline
12c9de7	2026-08-22	add parent identity probe and complete source checksums
4ba6a96	2026-08-22	make image provenance recovery independent
1dbd6ac	2026-08-22	separate provenance recovery stages
c551f2d	2026-08-22	add matched classification harness
14a1569	2026-08-22	add near duplicate audit and P3 grouping
79b798a	2026-08-22	amend SIPaKMeD class order before analysis
70b5ffa	2026-08-22	add bounded GPU probe launcher
4aa3e60	2026-08-22	fail closed on image provenance confidence
c18d393	2026-08-22	validate Cx22 against declared source images
f7ca331	2026-08-22	add bounded matched evaluation launcher
318120f	2026-08-22	summarize matched runs in one correction family
8f6f76e	2026-08-22	complete preregistered probe ablations
680fa53	2026-08-22	record reproducible model run metadata
0c5166e	2026-08-22	checkpoint exhaustive provenance matching
ea053e9	2026-08-22	verify structural protocol ordering
65b0230	2026-08-22	export fixed protocol split assets
3261dc0	2026-08-22	render fail-closed verification contract
79e5b31	2026-08-22	require named provenance evidence in report
cded044	2026-08-22	measure literal cross-split pixel overlap
55708d4	2026-08-22	keep training commands free of analysis-only imports
6369698	2026-08-22	generate self-contained thesis figure
da548a4	2026-08-22	bound clean reproduction matcher to three workers
83d56af	2026-08-22	verify published class crop counts
03e7442	2026-08-22	document fail-closed release workflow
be46a53	2026-08-22	separate canonical and audit provenance maps
c617374	2026-08-22	bound provenance image cache to class working set
4c6282e	2026-08-22	use multiscale provenance matching only as fallback
67892b7	2026-08-22	align provenance validation with registered criteria
268e56a	2026-08-22	build explicit section 11 acceptance ledger
438fa3f	2026-08-22	measure duplicate detector precision and recall
62a51ef	2026-08-22	record virgin-host reproduction evidence
dd0a56e	2026-08-22	generate five-seed parent probe figure
b90805a	2026-08-22	reproduce P3 splits and pixel overlap end to end
82e541d	2026-08-22	add matched protocol results figure
adf5539	2026-08-22	document full matched classifier matrix
bf8a4ec	2026-08-22	enforce paper table reporting contract
ab40681	2026-08-22	separate Figure 2 control labels
a72fb5f	2026-08-22	record split and image library versions
4da9f62	2026-08-22	pin CUDA 12.8 training environment
f936802	2026-08-22	verify classification matrix integrity
157b882	2026-08-22	report secondary ordering violations

Commit	Date	Recorded change
10fb2a3	2026-08-22	pin reproducible split runtime
970c74e	2026-08-22	gate acceptance on matrix integrity
8ebb2b9	2026-08-22	simulate exact P1 allocation rule
525df1d	2026-08-22	report architecture ranking stability
1c2da7f	2026-08-22	record preregistered provenance and five-seed results
2c5d5c3	2026-08-22	record post-result acceptance evidence
d347b07	2026-08-24	preregister Cx22 segmentation rescore
a58f72f	2026-08-24	add resumable Cx22 segmentation extension
635e660	2026-08-24	decode Cx22 instance labels with bounded memory
2fdb136	2026-08-24	refresh Cx22 controller heartbeat during epochs
d66dba4	2026-08-24	add memory-mapped Cx22 training cache
a18ddef	2026-08-24	preregister mixed-precision Cx22 execution
f10c9c5	2026-08-24	enable preregistered mixed precision for Cx22
48c9b65	2026-08-28	stabilize Cx22 MAnet training in FP32
373276a	2026-08-28	enforce deterministic CUDA workspace for Cx22
a9594b3	2026-09-06	remove redundant Cx22 instance runtime overhead
01c4c31	2026-09-06	stage Cx22 checkpoints off network storage

G

Exact command cookbook

G.1 Local checks

```
uv sync --extra dev
uv run pytest
uv run ruff check .
uv run cytofair --help
```

G.2 Recover and verify structural evidence

```
uv run cytofair recover-sipakmed data/extracted/sipakmed --workers 3
uv run cytofair recover-cx22 data/extracted/cx22 \
  --image-root data/extracted/ccedd/png
uv run cytofair recover-cric data/raw/cric/classifications.csv
uv run cytofair recover-herlev data/extracted/herlev

uv run cytofair audit-duplicates \
  provenance/sipakmed_parent_map.parquet \
  --hashes results/duplicates/sipakmed_hashes.parquet \
  --pairs results/duplicates/sipakmed_near_duplicate_pairs.csv \
  --output provenance/sipakmed_parent_map_p3.parquet

uv run cytofair make-splits \
  provenance/sipakmed_parent_map_p3.parquet \
  --dataset SIPaKMeD --output splits/generated

uv run cytofair pixel-overlap \
  provenance/sipakmed_branch_b.parquet --dataset SIPaKMeD
```

G.3 Launch the probe and classifier matrices

```
python scripts/launch_probe_matrix.py \
  provenance/sipakmed_parent_map_p3.parquet results/probe \
  --max-gpus 3 \
  --modes rgb grayscale stain_normalized background_only highpass

```

```
python scripts/launch_classification_matrix.py \
  --dataset-map SIPaKMeD=provenance/sipakmed_parent_map_p3.parquet \
  --output results/classification --max-gpus 3 \
  --architectures resnet18 resnet50 densenet121 \
    efficientnet_b0 mobilenet_v2 vit_b_16 \
  --protocols P1 P1-prime P2 P3
```

G.4 Materialize and launch Cx22

```
cytofair materialize-cx22 provenance/cx22_parent_map.parquet \
  --labels /path/to/extracted/cx22 \
  --assets /path/to/cx22_assets \
  --output provenance/cx22_model_map.parquet

cytofair audit-duplicates provenance/cx22_model_map.parquet \
  --hashes results/cx22_duplicates/hashes.parquet \
  --pairs results/cx22_duplicates/pairs.csv \
  --output provenance/cx22_model_map_p3.parquet

cytofair cache-cx22 provenance/cx22_model_map_p3.parquet \
  --cache /path/to/cx22_assets/cache256 \
  --output provenance/cx22_cached_model_map_p3.parquet

nohup python scripts/launch_cx22_segmentation_matrix.py \
  --map provenance/cx22_cached_model_map_p3.parquet \
  --output results/cx22_segmentation --max-gpus 3 \
  > results/cx22_segmentation/controller.log 2>&1 < /dev/null &

python scripts/check_cx22_segmentation.py results/cx22_segmentation
```

H

Artifact index and evidence status

Table H.1: Principal repository artifacts.

Artifact	Status	Purpose
`PREREGISTRATION.md`	Frozen plan	Endpoints, seeds, protocols, exclusions, corrections, amendments.
`data_sources.json`	Versioned	Owner URLs, filenames, sizes, hashes, and terms notes.
`provenance/*.parquet`	Complete derived maps	Observation-to-parent identities, match diagnostics, boxes, hashes, and P3 components.
`splits/*.csv`	Versioned interface	Exact per-seed assignments joined by crop ID.
`results/classification/*.json`	120 accepted	Classifier metrics, environment, split hash, and history.
`results/probe/*.json`	30 accepted	Mechanism probe metrics, chance, environment, and history.
`results/cx22_segmentation/*.json`	175 accepted	Semantic/instance metrics, manifests, runtime, and history.
`results/attempts/`	Excluded, retained	Twelve incompatible split-library runs and three smoke tests.
`reports/generated/acceptance_contract.*`	Current	Machine-readable PASS/FAIL/PENDING gates.
`scripts/reproduce_table1.sh`	Implemented	GPU-free raw-download-to-structural-table workflow.
`scripts/verify_clean_clone.sh`	Implemented, untested	Non-local clone wrapper and optional virgin-host attestation.
`reports/lab_paper_self_audit.md`	Pending	Nine-record title/protocol crosswalk requiring source-paper checks.
`reports/figure1_reader_test.md`	Pending	Independent reader protocol and blank response record.

I

Complete derived-result ledgers

These tables expose every structural seed, literal-overlap seed/protocol, registered classifier contrast, registered Cx22 contrast, and rank-stability result. The model run ledgers in Appendices A–C are the underlying run-level evidence; this appendix is the complete derived statistical layer.

I.1 Structural and literal-overlap seeds

Table I.1: Complete structural seed ledger.

Dataset	Seed	P1 parent straddle (percent)	P1 test parent in train (percent)	P2 straddle (percent)	P3 straddle (percent)
SIPaKMeD	17	45.549	92.222	0.000	0.000
SIPaKMeD	29	45.445	92.840	0.000	0.000
SIPaKMeD	43	44.203	92.469	0.000	0.000
SIPaKMeD	71	47.101	93.210	0.000	0.000
SIPaKMeD	101	45.859	92.346	0.000	0.000
Cx22	17	34.241	85.985	0.000	0.000
Cx22	29	32.685	80.682	0.000	0.000
Cx22	43	35.214	89.773	0.000	0.000
Cx22	71	34.825	86.742	0.000	0.000
Cx22	101	33.074	82.955	0.000	0.000
CRIC	17	94.750	99.827	0.000	0.000
CRIC	29	95.000	100.000	0.000	0.000
CRIC	43	94.250	100.000	0.000	0.000
CRIC	71	92.500	100.000	0.000	0.000
CRIC	101	95.250	100.000	0.000	0.000
Herlev	17	0.000	0.000	0.000	0.000
Herlev	29	0.000	0.000	0.000	0.000
Herlev	43	0.000	0.000	0.000	0.000
Herlev	71	0.000	0.000	0.000	0.000
Herlev	101	0.000	0.000	0.000	0.000

Table I.2: Complete SIPaKMeD literal source-pixel overlap ledger.

Protocol	Seed	Cross-split pairs	Overlapping pairs	Pair overlap (percent)	Test crops overlapping train	Test-crop overlap (percent)
P1	17	4294	673	15.673	411	50.741
P2	17	0	0	0.000	0	0.000
P3	17	0	0	0.000	0	0.000
P1	29	4153	695	16.735	415	51.235
P2	29	0	0	0.000	0	0.000
P3	29	0	0	0.000	0	0.000
P1	43	4222	749	17.740	443	54.691
P2	43	0	0	0.000	0	0.000
P3	43	0	0	0.000	0	0.000
P1	71	4240	705	16.627	418	51.605
P2	71	0	0	0.000	0	0.000
P3	71	0	0	0.000	0	0.000
P1	101	4199	724	17.242	444	54.815
P2	101	0	0	0.000	0	0.000
P3	101	0	0	0.000	0	0.000

I.2 Classifier contrasts

Table I.3: Complete SIPaKMeD classifier contrast ledger. Secondary tests have no confirmatory BH q-value.

Family	Architecture	Left	Right	n	Mean diff. (pp)	SD (pp)	Test	p	BH q	Effect	Value	Effect 95 percent CI	Shapiro p
primary	densenet121	P1	P2	5	1.407	0.901	paired_t	0.0251	0.1109	cohens_dz	1.561	[0.985, 4.747]	0.9922
primary	efficientnet_b0	P1	P2	5	1.086	1.849	paired_t	0.2591	0.2591	cohens_dz	0.588	[-0.316, 4.668]	0.6215
primary	mobilenet_v2	P1	P2	5	2.247	1.632	paired_t	0.0370	0.1109	cohens_dz	1.377	[0.809, 6.686]	0.3183
primary	resnet18	P1	P2	5	1.185	1.285	paired_t	0.1081	0.1511	cohens_dz	0.922	[0.442, 2.365]	0.6256
primary	resnet50	P1	P2	5	0.716	0.830	paired_t	0.1260	0.1511	cohens_dz	0.863	[0.085, 4.025]	0.4504
primary	vit_b_16	P1	P2	5	1.580	1.698	paired_t	0.1059	0.1511	cohens_dz	0.931	[0.057, 4.950]	0.6216
primary	densenet121	P1-prime	P2	5	1.407	0.901	paired_t	0.0251	0.1109	cohens_dz	1.561	[0.932, 4.747]	0.9922
primary	efficientnet_b0	P1-prime	P2	5	1.086	1.849	paired_t	0.2591	0.2591	cohens_dz	0.588	[-0.316, 4.610]	0.6215
primary	mobilenet_v2	P1-prime	P2	5	2.247	1.632	paired_t	0.0370	0.1109	cohens_dz	1.377	[0.809, 6.686]	0.3183
primary	resnet18	P1-prime	P2	5	1.185	1.285	paired_t	0.1081	0.1511	cohens_dz	0.922	[0.426, 2.442]	0.6256
primary	resnet50	P1-prime	P2	5	0.716	0.830	paired_t	0.1260	0.1511	cohens_dz	0.863	[0.085, 4.025]	0.4504
primary	vit_b_16	P1-prime	P2	5	1.580	1.698	paired_t	0.1059	0.1511	cohens_dz	0.931	[0.120, 4.950]	0.6216
secondary	densenet121	P2	P3	5	-0.667	0.691	paired_t	0.0971	–	cohens_dz	-0.965	[-3.266, -0.725]	0.0863
secondary	efficientnet_b0	P2	P3	5	-0.716	1.943	paired_t	0.4562	–	cohens_dz	-0.369	[-2.708, 0.467]	0.6101
secondary	mobilenet_v2	P2	P3	5	-1.037	1.376	paired_t	0.1673	–	cohens_dz	-0.753	[-3.329, 0.075]	0.4329
secondary	resnet18	P2	P3	5	-0.346	1.189	paired_t	0.5510	–	cohens_dz	-0.291	[-1.516, 1.255]	0.7941
secondary	resnet50	P2	P3	5	0.519	0.661	paired_t	0.1545	–	cohens_dz	0.784	[0.361, 2.534]	0.2922
secondary	vit_b_16	P2	P3	5	0.025	2.345	paired_t	0.9823	–	cohens_dz	0.011	[-3.338, 0.817]	0.2169

I.3 Cx22 contrasts

Table I.4: Complete Cx22 contrast ledger. Semantic and instance primary families were corrected separately by task.

Family	Architecture	Left	Right	n	Mean diff. (pp)	SD (pp)	Test	p	BH q	Effect	Value	Effect 95 percent CI	Shapiro p
primary	deeplabv3plus_resnet50	P1	P2	5	0.258	0.418	wilcoxon	0.3125	0.8458	rank_biserial	0.600	[-0.200, 1.000]	0.0338
primary	fpn_resnet34	P1	P2	5	0.010	0.468	paired_t	0.9625	0.9625	cohens_dz	0.022	[-1.157, 1.799]	0.9413
primary	manet_resnet34	P1	P2	5	0.065	0.450	paired_t	0.7612	0.8458	cohens_dz	0.146	[-0.653, 3.986]	0.2308
primary	unet_resnet34	P1	P2	5	0.297	0.767	paired_t	0.4353	0.8458	cohens_dz	0.387	[-0.840, 1.537]	0.7640
primary	unetplusplus_resnet34	P1	P2	5	0.188	0.406	paired_t	0.3582	0.8458	cohens_dz	0.464	[-0.447, 7.482]	0.4491
primary	deeplabv3plus_resnet50	P1-prime	P2	5	0.193	0.382	wilcoxon	0.6250	0.8458	rank_biserial	0.333	[-0.600, 1.000]	0.0411
primary	fpn_resnet34	P1-prime	P2	5	0.085	0.549	paired_t	0.7474	0.8458	cohens_dz	0.154	[-1.157, 2.035]	0.5968
primary	manet_resnet34	P1-prime	P2	5	0.102	0.479	paired_t	0.6583	0.8458	cohens_dz	0.213	[-0.694, 4.502]	0.2960
primary	unet_resnet34	P1-prime	P2	5	0.325	0.817	paired_t	0.4240	0.8458	cohens_dz	0.398	[-0.842, 1.506]	0.7316
primary	unetplusplus_resnet34	P1-prime	P2	5	0.206	0.409	paired_t	0.3233	0.8458	cohens_dz	0.503	[-0.375, 6.594]	0.4512
primary	maskrcnn_resnet50_fpn	P1	P2	5	0.509	0.581	paired_t	0.1220	0.2440	cohens_dz	0.875	[0.516, 2.175]	0.2370
primary	maskrcnn_resnet50_fpn_v2	P1	P2	5	0.630	1.204	paired_t	0.3068	0.3116	cohens_dz	0.523	[-0.568, 2.360]	0.7974
primary	maskrcnn_resnet50_fpn	P1-prime	P2	5	0.466	0.340	paired_t	0.0377	0.1508	cohens_dz	1.368	[0.769, 4.031]	0.7993
primary	maskrcnn_resnet50_fpn_v2	P1-prime	P2	5	0.586	1.133	paired_t	0.3116	0.3116	cohens_dz	0.518	[-0.568, 2.651]	0.4050
secondary	deeplabv3plus_resnet50	P2	P3	5	0.546	0.534	paired_t	0.0843	–	cohens_dz	1.022	[0.461, 3.165]	0.6257
secondary	fpn_resnet34	P2	P3	5	0.558	0.346	paired_t	0.0226	–	cohens_dz	1.613	[0.738, 7.765]	0.6155
secondary	manet_resnet34	P2	P3	5	0.339	0.569	paired_t	0.2537	–	cohens_dz	0.596	[-0.284, 2.625]	0.8303
secondary	unet_resnet34	P2	P3	5	0.221	0.579	paired_t	0.4421	–	cohens_dz	0.381	[-0.533, 6.386]	0.1888
secondary	unetplusplus_resnet34	P2	P3	5	0.329	0.299	paired_t	0.0696	–	cohens_dz	1.101	[0.722, 2.785]	0.1372
secondary	maskrcnn_resnet50_fpn	P2	P3	5	0.473	1.435	paired_t	0.5021	–	cohens_dz	0.330	[-1.325, 1.664]	0.2311
secondary	maskrcnn_resnet50_fpn_v2	P2	P3	5	0.761	1.006	paired_t	0.1661	–	cohens_dz	0.756	[-0.093, 3.627]	0.1440

I.4 Rank stability

Table I.5: Complete secondary P1-versus-P2 rank-stability ledger.

Task	Arch.	Seeds	Kendall tau	95 percent CI	p	Reordered pairs	Tie changes
classification	6	5	0.467	[0.071, 0.733]	0.2722	4/15	0
semantic segmentation	5	5	0.800	[0.600, 1.000]	0.0833	1/10	0
instance segmentation	2	5	1.000	[-1.000, 1.000]	1.0000	0/1	0

J

Excluded attempt ledger

No row below was excluded for its score or direction. The twelve twenty-epoch rows belong to the incompatible scikit-learn 1.7.2 split matrix; all were excluded when the entire matrix restarted. The three one-epoch rows are execution smoke tests that were never eligible for confirmatory summaries.

Table J.1: Complete excluded-attempt ledger. Metrics are shown only to demonstrate that exclusion was rule-based, not score-based.

Reason	Architecture/mode	Protocol	Seed	Epochs	Train	Test	Recorded metric (percent)	Eligible
split-library failure	resnet18	P1-prime	101	20	3223	810	98.395	No
split-library failure	resnet18	P1-prime	17	20	3214	810	96.914	No
split-library failure	resnet18	P1-prime	29	20	3237	810	97.901	No
split-library failure	resnet18	P1-prime	43	20	3215	810	97.284	No
split-library failure	resnet18	P1-prime	71	20	3236	810	98.148	No
split-library failure	resnet18	P1	101	20	3239	810	97.654	No
split-library failure	resnet18	P1	17	20	3239	810	97.901	No
split-library failure	resnet18	P1	29	20	3239	810	97.037	No
split-library failure	resnet18	P1	43	20	3239	810	96.914	No
split-library failure	resnet18	P1	71	20	3239	810	96.420	No
split-library failure	resnet18	P2	17	20	3214	835	96.647	No
split-library failure	resnet18	P2	29	20	3237	812	95.936	No
smoke test	resnet18/rgb	–	17	1	848	848	0.590	No
smoke test	resnet18	P1	17	1	3239	810	95.556	No
smoke test	vit_b_16	P1	17	1	3239	810	95.679	No

K

Reading guide for future maintainers

Start with the acceptance contract, not the paper prose. If a gate fails, follow its evidence path into the structured output. Regenerate a summary rather than editing a number. If a library upgrade changes a split, do not overwrite the published CSV: create a new version and preserve the previous hash. If new data supply patient IDs, introduce a stronger protocol rather than relabeling source-disjoint as patient-disjoint. If a model is rerun, retain its per-sample predictions and checkpoint so confusion matrices and qualitative panels become reproducible first-class artifacts.

The simplest invariant is also the most important: an observation may be the tensor consumed by a network, but it is not automatically the independent unit represented by the test set.

L

Study 2 run ledgers

Every registered Study 2 run, with its metrics in percent, wall time in minutes, and the GPU it ran on.

Table L.1: Complete Phase 1 run ledger (720 runs; metrics in percent; minutes of wall time; GPU index).

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-18	137	P1	97.41	97.42	97.43	1.36	0.9	3
ResNet-18	137	P2	96.05	96.08	96.05	2.29	0.9	4
ResNet-18	137	P3	95.80	95.84	95.80	2.53	0.9	5
ResNet-18	173	P1	97.41	97.39	97.45	1.66	0.9	3
ResNet-18	173	P2	96.05	96.08	96.07	2.06	0.9	0
ResNet-18	173	P3	96.42	96.41	96.45	1.86	0.9	3
ResNet-18	211	P1	96.91	96.91	96.92	1.61	0.9	4
ResNet-18	211	P2	95.80	95.83	95.79	2.33	0.9	0
ResNet-18	211	P3	94.69	94.73	94.69	3.21	0.9	4
ResNet-18	257	P1	96.67	96.67	96.70	2.53	0.9	3
ResNet-18	257	P2	96.54	96.56	96.54	2.11	0.9	5
ResNet-18	257	P3	97.04	97.05	97.05	1.65	0.9	3
ResNet-18	307	P1	98.64	98.64	98.65	0.65	0.9	0
ResNet-18	307	P2	96.67	96.67	96.64	1.50	0.9	5
ResNet-18	307	P3	97.78	97.80	97.78	1.28	0.9	0
ResNet-18	349	P1	96.54	96.55	96.55	1.95	0.9	5
ResNet-18	349	P2	96.42	96.46	96.42	1.74	0.9	0
ResNet-18	349	P3	98.27	98.29	98.29	0.97	0.9	5
ResNet-18	401	P1	97.28	97.28	97.32	1.47	0.9	4
ResNet-18	401	P2	98.15	98.17	98.16	1.12	0.9	3
ResNet-18	401	P3	97.53	97.55	97.52	1.08	0.9	4
ResNet-18	457	P1	98.02	98.03	98.05	0.98	0.9	3
ResNet-18	457	P2	95.68	95.65	95.69	2.81	0.9	0
ResNet-18	457	P3	97.65	97.66	97.67	1.36	0.9	3
ResNet-18	503	P1	96.67	96.70	96.69	2.06	0.9	4
ResNet-18	503	P2	96.54	96.54	96.58	2.56	0.9	0
ResNet-18	503	P3	96.54	96.54	96.55	2.35	0.9	4
ResNet-18	557	P1	97.65	97.67	97.67	1.29	0.9	0
ResNet-18	557	P2	97.53	97.53	97.52	1.55	0.9	5
ResNet-18	557	P3	97.41	97.41	97.42	1.51	0.9	0
ResNet-18	601	P1	97.78	97.78	97.80	1.13	0.9	3
ResNet-18	601	P2	96.17	96.19	96.19	1.83	0.9	0
ResNet-18	601	P3	96.91	96.95	96.92	1.58	0.9	3
ResNet-18	653	P1	96.79	96.81	96.81	1.68	0.9	0
ResNet-18	653	P2	97.28	97.30	97.31	1.86	0.9	3
ResNet-18	653	P3	96.42	96.36	96.38	1.96	0.9	0
ResNet-18	701	P1	97.53	97.53	97.54	1.99	0.9	3
ResNet-18	701	P2	95.68	95.71	95.70	2.77	0.9	5
ResNet-18	701	P3	93.33	93.38	93.35	4.43	0.9	3
ResNet-18	751	P1	97.53	97.53	97.56	1.92	0.9	5
ResNet-18	751	P2	96.05	96.05	96.01	2.75	0.9	3
ResNet-18	751	P3	96.05	96.06	96.04	1.94	0.9	4
ResNet-18	809	P1	98.15	98.14	98.18	1.87	0.9	4

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-18	809	P2	96.17	96.17	96.17	2.79	0.9	0
ResNet-18	809	P3	96.30	96.28	96.34	2.72	0.9	3
ResNet-18	853	P1	98.02	98.02	98.04	1.10	0.9	5
ResNet-18	853	P2	97.16	97.16	97.17	1.73	0.9	4
ResNet-18	853	P3	96.17	96.16	96.17	2.50	0.9	5
ResNet-18	907	P1	97.78	97.79	97.79	1.23	0.9	0
ResNet-18	907	P2	96.67	96.65	96.66	1.95	0.9	3
ResNet-18	907	P3	98.02	98.04	98.01	1.14	0.9	0
ResNet-18	953	P1	97.16	97.17	97.17	1.25	0.9	3
ResNet-18	953	P2	96.67	96.70	96.65	1.72	0.9	0
ResNet-18	953	P3	98.15	98.16	98.14	1.29	0.9	3
ResNet-18	1009	P1	98.02	98.04	98.03	1.36	0.9	4
ResNet-18	1009	P2	96.91	96.91	96.92	2.03	0.9	0
ResNet-18	1009	P3	97.65	97.65	97.62	1.00	0.9	4
ResNet-18	1051	P1	97.41	97.42	97.41	1.38	0.9	5
ResNet-18	1051	P2	97.53	97.53	97.54	1.06	0.9	0
ResNet-18	1051	P3	97.28	97.28	97.30	1.42	0.9	5
ResNet-50	137	P1	97.65	97.66	97.68	1.05	2.6	0
ResNet-50	137	P2	95.19	95.24	95.18	2.87	2.6	3
ResNet-50	137	P3	97.28	97.30	97.31	1.62	2.6	4
ResNet-50	173	P1	97.78	97.80	97.79	1.03	2.6	5
ResNet-50	173	P2	95.93	95.97	95.96	2.05	2.6	4
ResNet-50	173	P3	97.28	97.32	97.29	1.46	2.6	0
ResNet-50	211	P1	97.41	97.42	97.42	1.47	2.6	0
ResNet-50	211	P2	96.30	96.32	96.27	2.20	2.6	3
ResNet-50	211	P3	95.31	95.32	95.29	3.36	2.6	5
ResNet-50	257	P1	97.41	97.42	97.44	1.59	2.6	4
ResNet-50	257	P2	96.67	96.68	96.66	2.07	2.6	0
ResNet-50	257	P3	97.65	97.65	97.67	1.18	2.6	5
ResNet-50	307	P1	98.27	98.28	98.28	1.29	2.6	3
ResNet-50	307	P2	97.28	97.28	97.29	1.75	2.6	5
ResNet-50	307	P3	98.64	98.66	98.64	1.10	2.6	4
ResNet-50	349	P1	96.67	96.69	96.67	2.11	2.6	4
ResNet-50	349	P2	96.42	96.45	96.42	2.23	2.6	0
ResNet-50	349	P3	95.43	95.47	95.43	2.67	2.6	3
ResNet-50	401	P1	98.02	98.03	98.03	1.38	2.6	0
ResNet-50	401	P2	97.41	97.42	97.43	1.32	2.6	3
ResNet-50	401	P3	93.33	93.36	93.20	3.95	2.6	5
ResNet-50	457	P1	98.15	98.14	98.18	1.18	2.6	5
ResNet-50	457	P2	94.32	94.31	94.35	3.88	2.6	4
ResNet-50	457	P3	96.30	96.30	96.33	2.63	2.6	0
ResNet-50	503	P1	97.53	97.54	97.56	1.35	2.6	3
ResNet-50	503	P2	96.79	96.79	96.78	2.09	2.6	0
ResNet-50	503	P3	96.17	96.16	96.17	2.54	2.6	5
ResNet-50	557	P1	97.53	97.53	97.55	1.29	2.6	4
ResNet-50	557	P2	98.40	98.40	98.40	1.16	2.6	3
ResNet-50	557	P3	96.30	96.35	96.28	2.26	2.6	5
ResNet-50	601	P1	97.16	97.18	97.19	1.88	2.6	5
ResNet-50	601	P2	97.41	97.42	97.42	1.68	2.6	0
ResNet-50	601	P3	97.28	97.29	97.31	1.50	2.6	4
ResNet-50	653	P1	97.28	97.31	97.29	2.18	2.6	4
ResNet-50	653	P2	96.67	96.69	96.68	2.23	2.6	5
ResNet-50	653	P3	95.68	95.66	95.61	3.04	2.6	3
ResNet-50	701	P1	97.53	97.54	97.53	1.47	2.6	5
ResNet-50	701	P2	96.79	96.81	96.81	2.25	2.7	3
ResNet-50	701	P3	95.31	95.32	95.31	3.15	2.7	0
ResNet-50	751	P1	97.53	97.56	97.55	1.31	2.6	0
ResNet-50	751	P2	95.56	95.55	95.55	3.75	2.7	5
ResNet-50	751	P3	95.93	95.95	95.92	2.45	2.7	3
ResNet-50	809	P1	97.78	97.77	97.80	1.37	2.7	4
ResNet-50	809	P2	97.04	97.05	97.02	1.90	2.7	5
ResNet-50	809	P3	96.67	96.64	96.70	2.43	2.7	0
ResNet-50	853	P1	97.41	97.41	97.44	2.18	2.7	3
ResNet-50	853	P2	96.17	96.17	96.18	2.40	2.7	0
ResNet-50	853	P3	95.93	95.96	95.91	2.35	2.7	4
ResNet-50	907	P1	98.64	98.65	98.65	0.84	2.6	5
ResNet-50	907	P2	97.41	97.41	97.43	1.60	2.7	3
ResNet-50	907	P3	97.53	97.56	97.53	1.40	2.7	4
ResNet-50	953	P1	97.53	97.55	97.53	1.28	2.6	0
ResNet-50	953	P2	95.80	95.87	95.79	2.85	2.6	4
ResNet-50	953	P3	98.52	98.52	98.51	0.83	2.6	5
ResNet-50	1009	P1	97.16	97.18	97.16	1.72	2.6	3
ResNet-50	1009	P2	97.41	97.42	97.41	1.58	2.6	5

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	1009	P3	97.16	97.16	97.13	2.12	2.6	0
ResNet-50	1051	P1	98.27	98.29	98.28	0.81	2.7	0
ResNet-50	1051	P2	97.41	97.40	97.43	0.99	2.6	3
ResNet-50	1051	P3	96.30	96.28	96.28	2.12	2.6	4
DenseNet-121	137	P1	97.90	97.91	97.92	1.44	3.3	5
DenseNet-121	137	P2	95.56	95.54	95.57	2.34	3.3	0
DenseNet-121	137	P3	96.67	96.69	96.66	2.19	3.3	3
DenseNet-121	173	P1	97.78	97.79	97.79	1.43	3.3	3
DenseNet-121	173	P2	98.52	98.52	98.51	1.03	3.3	5
DenseNet-121	173	P3	98.52	98.53	98.52	0.83	3.3	4
DenseNet-121	211	P1	98.40	98.40	98.41	1.04	3.3	4
DenseNet-121	211	P2	97.16	97.18	97.15	1.83	3.3	0
DenseNet-121	211	P3	96.67	96.69	96.66	2.43	3.3	3
DenseNet-121	257	P1	97.16	97.17	97.19	1.74	3.3	3
DenseNet-121	257	P2	96.54	96.54	96.52	1.39	3.3	4
DenseNet-121	257	P3	97.78	97.78	97.78	0.90	3.3	0
DenseNet-121	307	P1	98.64	98.65	98.65	0.70	3.3	0
DenseNet-121	307	P2	97.53	97.53	97.51	1.61	3.3	3
DenseNet-121	307	P3	98.27	98.29	98.28	0.88	3.3	5
DenseNet-121	349	P1	97.53	97.53	97.53	1.17	3.3	5
DenseNet-121	349	P2	97.78	97.79	97.79	1.24	3.3	4
DenseNet-121	349	P3	98.15	98.15	98.14	1.18	3.3	0
DenseNet-121	401	P1	97.16	97.18	97.19	1.83	3.3	4
DenseNet-121	401	P2	97.28	97.29	97.30	1.36	3.3	0
DenseNet-121	401	P3	96.91	96.90	96.86	1.59	3.3	3
DenseNet-121	457	P1	97.41	97.39	97.45	1.48	3.3	3
DenseNet-121	457	P2	95.43	95.43	95.43	3.19	3.3	5
DenseNet-121	457	P3	96.30	96.30	96.32	1.75	3.3	4
DenseNet-121	503	P1	97.90	97.91	97.93	1.22	3.3	4
DenseNet-121	503	P2	97.65	97.65	97.67	1.54	3.3	3
DenseNet-121	503	P3	96.05	96.05	96.05	2.20	3.3	0
DenseNet-121	557	P1	97.41	97.43	97.42	0.92	3.3	0
DenseNet-121	557	P2	97.90	97.90	97.92	1.18	3.3	4
DenseNet-121	557	P3	97.90	97.91	97.89	0.88	3.3	3
DenseNet-121	601	P1	97.04	97.05	97.06	1.45	3.3	3
DenseNet-121	601	P2	97.16	97.17	97.18	1.39	3.3	5
DenseNet-121	601	P3	96.91	96.94	96.92	1.53	3.3	0
DenseNet-121	653	P1	96.67	96.70	96.67	1.99	3.3	0
DenseNet-121	653	P2	97.16	97.18	97.15	1.27	3.3	4
DenseNet-121	653	P3	97.41	97.42	97.39	1.92	3.3	5
DenseNet-121	701	P1	97.90	97.90	97.91	1.39	3.4	4
DenseNet-121	701	P2	96.79	96.80	96.83	2.02	3.4	5
DenseNet-121	701	P3	95.43	95.43	95.44	2.90	3.4	3
DenseNet-121	751	P1	98.27	98.29	98.30	0.91	3.4	4
DenseNet-121	751	P2	96.79	96.78	96.81	1.87	3.4	0
DenseNet-121	751	P3	96.79	96.80	96.78	1.56	3.4	5
DenseNet-121	809	P1	98.40	98.40	98.41	0.96	3.4	3
DenseNet-121	809	P2	97.41	97.39	97.39	1.62	3.4	4
DenseNet-121	809	P3	97.16	97.15	97.20	2.00	3.4	5
DenseNet-121	853	P1	97.41	97.41	97.43	1.44	3.4	5
DenseNet-121	853	P2	96.17	96.19	96.17	2.47	3.3	3
DenseNet-121	853	P3	96.30	96.34	96.30	2.43	3.4	0
DenseNet-121	907	P1	97.53	97.54	97.55	1.59	3.3	0
DenseNet-121	907	P2	96.91	96.91	96.91	1.25	3.3	5
DenseNet-121	907	P3	97.90	97.93	97.90	1.97	3.3	3
DenseNet-121	953	P1	98.02	98.03	98.04	1.33	3.3	3
DenseNet-121	953	P2	97.28	97.31	97.26	1.54	3.3	0
DenseNet-121	953	P3	97.53	97.54	97.51	1.13	3.3	4
DenseNet-121	1009	P1	98.02	98.03	98.03	1.47	3.3	4
DenseNet-121	1009	P2	96.79	96.80	96.80	1.75	3.3	3
DenseNet-121	1009	P3	97.65	97.64	97.63	0.63	3.3	5
DenseNet-121	1051	P1	97.53	97.55	97.55	1.06	3.3	5
DenseNet-121	1051	P2	97.04	97.03	97.06	1.77	3.3	0
DenseNet-121	1051	P3	98.02	98.03	98.02	0.57	3.3	3
EfficientNet-B0	137	P1	98.15	98.15	98.16	0.95	1.8	4
EfficientNet-B0	137	P2	97.16	97.16	97.16	1.94	1.8	5
EfficientNet-B0	137	P3	97.28	97.30	97.30	1.47	1.8	4
EfficientNet-B0	173	P1	96.79	96.79	96.82	1.83	1.8	0
EfficientNet-B0	173	P2	97.65	97.66	97.66	1.15	1.8	3
EfficientNet-B0	173	P3	97.04	97.06	97.05	1.67	1.8	0
EfficientNet-B0	211	P1	97.90	97.92	97.91	1.12	1.8	5
EfficientNet-B0	211	P2	97.16	97.19	97.15	1.87	1.8	4
EfficientNet-B0	211	P3	93.33	93.42	93.32	4.89	1.8	5

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
EfficientNet-B0	257	P1	97.41	97.41	97.42	1.76	1.8	5
EfficientNet-B0	257	P2	96.91	96.93	96.91	2.38	1.8	3
EfficientNet-B0	257	P3	97.28	97.28	97.27	2.10	1.8	5
EfficientNet-B0	307	P1	98.02	98.02	98.05	1.43	1.8	4
EfficientNet-B0	307	P2	96.42	96.41	96.41	1.84	1.8	0
EfficientNet-B0	307	P3	98.52	98.54	98.52	0.94	1.8	4
EfficientNet-B0	349	P1	97.65	97.67	97.67	1.63	1.8	3
EfficientNet-B0	349	P2	98.02	98.03	98.03	0.87	1.8	5
EfficientNet-B0	349	P3	97.65	97.66	97.65	0.98	1.8	3
EfficientNet-B0	401	P1	98.02	98.04	98.04	1.22	1.8	5
EfficientNet-B0	401	P2	96.54	96.57	96.52	2.20	1.8	4
EfficientNet-B0	401	P3	97.78	97.77	97.75	1.55	1.8	5
EfficientNet-B0	457	P1	97.90	97.89	97.94	1.28	1.8	0
EfficientNet-B0	457	P2	94.07	94.00	94.12	3.74	1.8	3
EfficientNet-B0	457	P3	96.30	96.28	96.34	2.66	1.8	0
EfficientNet-B0	503	P1	95.80	95.79	95.84	3.09	1.8	5
EfficientNet-B0	503	P2	97.04	97.05	97.06	1.72	1.8	4
EfficientNet-B0	503	P3	95.68	95.68	95.69	2.87	1.8	5
EfficientNet-B0	557	P1	97.78	97.78	97.78	1.24	1.8	5
EfficientNet-B0	557	P2	97.90	97.92	97.91	1.26	1.8	0
EfficientNet-B0	557	P3	97.65	97.66	97.67	1.03	1.8	5
EfficientNet-B0	601	P1	97.90	97.91	97.92	1.42	1.8	4
EfficientNet-B0	601	P2	96.79	96.81	96.80	1.58	1.8	3
EfficientNet-B0	601	P3	97.90	97.91	97.93	1.31	1.8	4
EfficientNet-B0	653	P1	98.64	98.65	98.65	0.91	1.8	3
EfficientNet-B0	653	P2	97.04	97.04	97.04	1.93	1.8	0
EfficientNet-B0	653	P3	95.80	95.81	95.79	2.28	1.8	3
EfficientNet-B0	701	P1	97.41	97.41	97.41	1.97	1.9	0
EfficientNet-B0	701	P2	96.79	96.81	96.81	1.93	1.9	4
EfficientNet-B0	701	P3	94.20	94.14	94.23	4.03	1.9	5
EfficientNet-B0	751	P1	98.27	98.28	98.29	1.16	1.9	3
EfficientNet-B0	751	P2	96.91	96.90	96.91	2.37	1.9	4
EfficientNet-B0	751	P3	97.16	97.18	97.17	1.94	1.9	3
EfficientNet-B0	809	P1	98.02	98.03	98.04	1.45	1.9	0
EfficientNet-B0	809	P2	96.54	96.55	96.54	2.17	1.8	3
EfficientNet-B0	809	P3	95.80	95.77	95.84	3.39	1.9	0
EfficientNet-B0	853	P1	97.65	97.65	97.67	1.88	1.8	4
EfficientNet-B0	853	P2	96.42	96.41	96.39	2.46	1.8	5
EfficientNet-B0	853	P3	96.54	96.55	96.53	2.31	1.8	4
EfficientNet-B0	907	P1	98.52	98.52	98.51	0.69	1.8	4
EfficientNet-B0	907	P2	97.53	97.53	97.54	1.65	1.8	0
EfficientNet-B0	907	P3	97.90	97.90	97.89	1.40	1.8	4
EfficientNet-B0	953	P1	98.02	98.04	98.03	1.15	1.8	5
EfficientNet-B0	953	P2	98.15	98.15	98.15	1.17	1.8	3
EfficientNet-B0	953	P3	98.02	98.02	98.02	1.05	1.8	5
EfficientNet-B0	1009	P1	97.78	97.79	97.79	1.38	1.8	0
EfficientNet-B0	1009	P2	97.04	97.04	97.06	2.13	1.8	4
EfficientNet-B0	1009	P3	96.91	96.92	96.89	1.58	1.8	0
EfficientNet-B0	1051	P1	97.65	97.66	97.67	1.47	1.8	4
EfficientNet-B0	1051	P2	97.53	97.53	97.56	1.13	1.8	5
EfficientNet-B0	1051	P3	98.27	98.28	98.27	0.78	1.8	4
MobileNet-V2	137	P1	96.42	96.41	96.45	2.48	1.5	0
MobileNet-V2	137	P2	96.91	96.91	96.92	1.66	1.5	5
MobileNet-V2	137	P3	97.53	97.54	97.54	1.72	1.5	3
MobileNet-V2	173	P1	97.41	97.42	97.41	1.21	1.5	3
MobileNet-V2	173	P2	96.67	96.69	96.67	1.23	1.5	5
MobileNet-V2	173	P3	97.04	97.06	97.05	1.75	1.5	4
MobileNet-V2	211	P1	97.28	97.28	97.31	1.28	1.5	4
MobileNet-V2	211	P2	93.95	94.03	93.92	3.68	1.5	0
MobileNet-V2	211	P3	94.07	94.10	94.05	3.69	1.5	3
MobileNet-V2	257	P1	96.91	96.91	96.94	1.77	1.5	3
MobileNet-V2	257	P2	96.54	96.55	96.53	1.96	1.5	4
MobileNet-V2	257	P3	96.79	96.81	96.78	1.51	1.5	0
MobileNet-V2	307	P1	97.90	97.89	97.92	0.63	1.5	0
MobileNet-V2	307	P2	94.94	94.93	94.92	2.44	1.5	3
MobileNet-V2	307	P3	97.41	97.43	97.41	1.30	1.5	5
MobileNet-V2	349	P1	96.67	96.68	96.67	1.55	1.5	5
MobileNet-V2	349	P2	95.93	95.95	95.95	2.51	1.5	0
MobileNet-V2	349	P3	97.53	97.54	97.53	1.11	1.5	4
MobileNet-V2	401	P1	97.41	97.42	97.41	1.52	1.5	0
MobileNet-V2	401	P2	96.67	96.68	96.67	1.62	1.5	4
MobileNet-V2	401	P3	96.79	96.82	96.77	2.22	1.5	3
MobileNet-V2	457	P1	96.91	96.90	96.93	1.63	1.5	3

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
MobileNet-V2	457	P2	93.83	93.82	93.85	4.12	1.5	5
MobileNet-V2	457	P3	95.93	95.92	95.97	2.52	1.5	4
MobileNet-V2	503	P1	96.54	96.54	96.57	2.28	1.5	4
MobileNet-V2	503	P2	96.67	96.67	96.68	2.39	1.5	3
MobileNet-V2	503	P3	94.20	94.22	94.19	3.67	1.5	0
MobileNet-V2	557	P1	96.67	96.68	96.69	1.63	1.5	0
MobileNet-V2	557	P2	97.78	97.79	97.78	1.23	1.5	4
MobileNet-V2	557	P3	96.42	96.45	96.40	1.47	1.5	3
MobileNet-V2	601	P1	97.41	97.42	97.42	1.07	1.5	3
MobileNet-V2	601	P2	96.30	96.31	96.31	1.96	1.5	5
MobileNet-V2	601	P3	96.91	96.92	96.93	1.61	1.5	0
MobileNet-V2	653	P1	97.53	97.54	97.53	1.33	1.5	0
MobileNet-V2	653	P2	95.43	95.47	95.42	2.16	1.5	4
MobileNet-V2	653	P3	96.54	96.53	96.53	2.42	1.5	5
MobileNet-V2	701	P1	96.67	96.66	96.69	2.22	1.6	3
MobileNet-V2	701	P2	96.17	96.19	96.19	2.08	1.5	0
MobileNet-V2	701	P3	93.70	93.68	93.69	3.99	1.5	4
MobileNet-V2	751	P1	96.54	96.56	96.57	1.92	1.6	0
MobileNet-V2	751	P2	96.79	96.79	96.81	2.51	1.5	5
MobileNet-V2	751	P3	95.19	95.21	95.15	2.44	1.5	4
MobileNet-V2	809	P1	96.91	96.92	96.92	1.46	1.5	4
MobileNet-V2	809	P2	96.42	96.42	96.43	1.45	1.6	3
MobileNet-V2	809	P3	96.05	96.02	96.10	2.36	1.5	5
MobileNet-V2	853	P1	97.65	97.65	97.66	1.37	1.5	3
MobileNet-V2	853	P2	95.93	95.91	95.92	2.05	1.5	5
MobileNet-V2	853	P3	95.06	95.08	95.03	2.90	1.5	0
MobileNet-V2	907	P1	97.90	97.91	97.90	1.14	1.5	0
MobileNet-V2	907	P2	97.04	97.05	97.04	1.57	1.5	5
MobileNet-V2	907	P3	96.67	96.70	96.64	1.76	1.5	3
MobileNet-V2	953	P1	97.41	97.43	97.42	1.51	1.5	3
MobileNet-V2	953	P2	97.16	97.19	97.16	1.05	1.5	0
MobileNet-V2	953	P3	97.41	97.42	97.37	1.04	1.5	4
MobileNet-V2	1009	P1	97.41	97.42	97.41	1.77	1.5	4
MobileNet-V2	1009	P2	95.80	95.80	95.82	1.89	1.5	3
MobileNet-V2	1009	P3	96.91	96.91	96.87	1.59	1.5	5
MobileNet-V2	1051	P1	97.28	97.31	97.30	1.68	1.5	0
MobileNet-V2	1051	P2	96.91	96.92	96.91	0.98	1.5	5
MobileNet-V2	1051	P3	97.53	97.53	97.53	0.48	1.5	3
ViT-B/16	137	P1	97.16	97.17	97.18	1.82	9.2	4
ViT-B/16	137	P2	96.67	96.69	96.66	2.06	9.1	0
ViT-B/16	137	P3	97.53	97.56	97.53	2.08	9.2	5
ViT-B/16	173	P1	97.78	97.77	97.80	1.30	9.1	0
ViT-B/16	173	P2	96.79	96.81	96.80	2.95	9.2	3
ViT-B/16	173	P3	94.69	94.80	94.70	2.84	9.2	5
ViT-B/16	211	P1	97.16	97.18	97.17	1.87	9.2	5
ViT-B/16	211	P2	94.32	94.40	94.27	4.43	9.2	4
ViT-B/16	211	P3	95.06	95.06	95.09	3.43	9.1	0
ViT-B/16	257	P1	95.56	95.54	95.54	3.36	9.2	5
ViT-B/16	257	P2	96.54	96.53	96.53	2.09	9.2	3
ViT-B/16	257	P3	97.16	97.16	97.19	2.39	9.2	4
ViT-B/16	307	P1	99.14	99.14	99.14	0.59	9.2	4
ViT-B/16	307	P2	95.80	95.80	95.78	3.32	9.1	0
ViT-B/16	307	P3	97.16	97.17	97.19	1.57	9.2	3
ViT-B/16	349	P1	94.81	94.87	94.74	2.85	9.1	3
ViT-B/16	349	P2	96.54	96.56	96.58	2.44	9.2	5
ViT-B/16	349	P3	97.28	97.29	97.30	1.60	9.1	0
ViT-B/16	401	P1	97.78	97.79	97.79	1.67	9.2	5
ViT-B/16	401	P2	94.07	93.89	94.18	4.11	9.1	0
ViT-B/16	401	P3	96.91	96.93	96.87	1.84	9.2	4
ViT-B/16	457	P1	98.64	98.65	98.65	0.97	9.1	0
ViT-B/16	457	P2	92.84	92.82	92.78	5.49	9.2	3
ViT-B/16	457	P3	96.30	96.31	96.30	2.08	9.2	5
ViT-B/16	503	P1	96.05	96.08	96.10	2.46	9.2	5
ViT-B/16	503	P2	96.17	96.17	96.21	2.67	9.2	4
ViT-B/16	503	P3	94.69	94.68	94.71	3.92	9.2	3
ViT-B/16	557	P1	97.04	97.05	97.06	2.17	9.2	5
ViT-B/16	557	P2	94.07	94.07	94.13	3.26	9.1	0
ViT-B/16	557	P3	95.68	95.69	95.72	3.13	9.2	4
ViT-B/16	601	P1	97.04	97.06	97.06	2.04	9.2	4
ViT-B/16	601	P2	97.41	97.40	97.43	2.25	9.2	3
ViT-B/16	601	P3	98.15	98.16	98.16	1.31	9.2	5
ViT-B/16	653	P1	97.53	97.54	97.54	1.86	9.2	3
ViT-B/16	653	P2	97.41	97.39	97.38	1.77	9.1	0

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ViT-B/16	653	P3	95.43	95.45	95.43	2.88	9.2	4
ViT-B/16	701	P1	97.28	97.27	97.29	1.93	9.2	5
ViT-B/16	701	P2	95.56	95.59	95.58	3.31	9.2	3
ViT-B/16	701	P3	93.09	92.99	93.15	4.48	9.1	0
ViT-B/16	751	P1	96.42	96.48	96.42	2.65	9.2	3
ViT-B/16	751	P2	95.43	95.47	95.44	2.90	9.2	0
ViT-B/16	751	P3	95.43	95.42	95.44	2.79	9.2	5
ViT-B/16	809	P1	97.28	97.30	97.33	2.11	9.2	0
ViT-B/16	809	P2	95.31	95.35	95.31	3.32	9.3	4
ViT-B/16	809	P3	96.67	96.66	96.71	2.39	9.2	3
ViT-B/16	853	P1	96.91	96.91	96.91	2.27	9.2	4
ViT-B/16	853	P2	96.42	96.39	96.42	2.64	9.3	3
ViT-B/16	853	P3	93.95	94.03	93.92	4.64	9.3	5
ViT-B/16	907	P1	98.77	98.78	98.78	0.87	9.2	4
ViT-B/16	907	P2	95.80	95.78	95.82	3.27	9.1	0
ViT-B/16	907	P3	97.04	97.03	97.07	1.89	9.2	5
ViT-B/16	953	P1	96.67	96.68	96.70	2.87	9.2	5
ViT-B/16	953	P2	96.91	96.95	96.92	2.34	9.2	3
ViT-B/16	953	P3	95.80	95.83	95.78	2.81	9.1	0
ViT-B/16	1009	P1	97.65	97.67	97.67	1.47	9.1	0
ViT-B/16	1009	P2	94.81	94.79	94.89	3.77	9.2	4
ViT-B/16	1009	P3	95.19	95.16	95.19	3.31	9.2	3
ViT-B/16	1051	P1	96.91	96.93	96.94	2.33	9.2	4
ViT-B/16	1051	P2	95.80	95.81	95.78	2.10	9.1	0
ViT-B/16	1051	P3	95.93	95.89	95.90	2.65	9.2	3
ConvNeXt-T	137	P1	98.15	98.16	98.16	1.55	5.2	3
ConvNeXt-T	137	P2	96.79	96.79	96.81	2.44	5.2	3
ConvNeXt-T	137	P3	96.67	96.67	96.68	2.41	5.2	4
ConvNeXt-T	173	P1	97.90	97.90	97.89	1.43	5.2	4
ConvNeXt-T	173	P2	96.17	96.19	96.17	2.90	5.2	4
ConvNeXt-T	173	P3	96.79	96.81	96.81	2.43	5.2	0
ConvNeXt-T	211	P1	97.90	97.90	97.91	1.64	5.2	3
ConvNeXt-T	211	P2	96.54	96.56	96.54	2.68	5.2	3
ConvNeXt-T	211	P3	95.31	95.31	95.32	4.00	5.2	5
ConvNeXt-T	257	P1	97.90	97.90	97.91	1.83	5.2	0
ConvNeXt-T	257	P2	97.16	97.16	97.16	2.07	5.2	0
ConvNeXt-T	257	P3	97.78	97.78	97.78	1.10	5.2	5
ConvNeXt-T	307	P1	98.27	98.28	98.28	1.22	5.2	5
ConvNeXt-T	307	P2	96.67	96.68	96.66	2.58	5.2	5
ConvNeXt-T	307	P3	98.27	98.27	98.29	1.32	5.2	4
ConvNeXt-T	349	P1	97.78	97.77	97.80	1.83	5.2	4
ConvNeXt-T	349	P2	93.83	93.89	93.83	5.03	5.2	4
ConvNeXt-T	349	P3	97.28	97.31	97.26	2.19	5.2	3
ConvNeXt-T	401	P1	97.53	97.55	97.56	1.51	5.2	3
ConvNeXt-T	401	P2	98.40	98.40	98.41	1.20	5.2	3
ConvNeXt-T	401	P3	97.53	97.55	97.51	1.88	5.2	5
ConvNeXt-T	457	P1	98.64	98.65	98.66	1.16	5.2	4
ConvNeXt-T	457	P2	95.56	95.55	95.57	3.12	5.2	4
ConvNeXt-T	457	P3	96.79	96.79	96.82	2.00	5.1	0
ConvNeXt-T	503	P1	97.28	97.30	97.31	1.90	5.2	0
ConvNeXt-T	503	P2	97.28	97.28	97.29	1.58	5.1	0
ConvNeXt-T	503	P3	94.69	94.70	94.74	4.10	5.2	5
ConvNeXt-T	557	P1	96.67	96.70	96.67	2.26	5.2	3
ConvNeXt-T	557	P2	98.15	98.16	98.14	1.33	5.2	3
ConvNeXt-T	557	P3	97.90	97.92	97.90	1.52	5.2	5
ConvNeXt-T	601	P1	97.65	97.67	97.68	2.02	5.2	0
ConvNeXt-T	601	P2	96.42	96.43	96.44	2.15	5.2	0
ConvNeXt-T	601	P3	98.02	98.03	98.04	1.23	5.2	4
ConvNeXt-T	653	P1	97.78	97.78	97.80	1.74	5.2	5
ConvNeXt-T	653	P2	97.28	97.26	97.31	1.68	5.2	5
ConvNeXt-T	653	P3	94.94	94.88	94.96	3.55	5.2	3
ConvNeXt-T	701	P1	97.04	97.03	97.03	2.29	5.4	4
ConvNeXt-T	701	P2	95.80	95.80	95.85	3.44	5.4	4
ConvNeXt-T	701	P3	95.19	95.14	95.25	3.72	5.3	5
ConvNeXt-T	751	P1	96.91	96.96	96.93	1.91	5.3	4
ConvNeXt-T	751	P2	96.67	96.67	96.64	2.67	5.4	4
ConvNeXt-T	751	P3	96.05	96.06	96.05	2.25	5.3	3
ConvNeXt-T	809	P1	96.17	96.25	96.16	2.72	5.4	5
ConvNeXt-T	809	P2	96.67	96.65	96.64	2.52	5.4	5
ConvNeXt-T	809	P3	97.04	97.03	97.03	2.25	5.3	0
ConvNeXt-T	853	P1	97.16	97.17	97.15	1.82	5.2	0
ConvNeXt-T	853	P2	96.79	96.79	96.75	2.51	5.2	0
ConvNeXt-T	853	P3	96.17	96.14	96.16	2.93	5.2	4

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	907	P1	98.27	98.28	98.29	1.17	5.2	3
ConvNeXt-T	907	P2	97.28	97.28	97.30	2.01	5.2	3
ConvNeXt-T	907	P3	97.28	97.29	97.29	2.04	5.2	4
ConvNeXt-T	953	P1	96.91	96.90	96.94	2.07	5.3	4
ConvNeXt-T	953	P2	96.79	96.83	96.80	2.37	5.2	4
ConvNeXt-T	953	P3	96.17	96.21	96.14	1.89	5.2	5
ConvNeXt-T	1009	P1	96.79	96.81	96.82	2.52	5.2	5
ConvNeXt-T	1009	P2	96.42	96.43	96.42	2.26	5.2	5
ConvNeXt-T	1009	P3	97.04	97.06	96.99	2.04	5.2	0
ConvNeXt-T	1051	P1	97.90	97.91	97.92	1.40	5.2	5
ConvNeXt-T	1051	P2	97.16	97.19	97.15	1.79	5.2	5
ConvNeXt-T	1051	P3	97.53	97.53	97.52	1.71	5.2	4
Swin-T	137	P1	96.05	96.01	96.11	3.18	4.6	0
Swin-T	137	P2	94.57	94.59	94.61	4.07	4.6	5
Swin-T	137	P3	97.28	97.31	97.28	2.13	4.7	3
Swin-T	173	P1	97.90	97.91	97.92	1.16	4.6	3
Swin-T	173	P2	95.31	95.37	95.32	3.21	4.7	5
Swin-T	173	P3	96.30	96.34	96.32	2.60	4.7	4
Swin-T	211	P1	97.78	97.77	97.79	1.72	4.7	4
Swin-T	211	P2	96.54	96.58	96.53	2.72	4.6	0
Swin-T	211	P3	94.20	94.24	94.14	4.49	4.6	3
Swin-T	257	P1	97.41	97.41	97.44	2.05	4.6	3
Swin-T	257	P2	97.78	97.78	97.79	1.59	4.7	4
Swin-T	257	P3	96.91	96.92	96.88	1.78	4.6	0
Swin-T	307	P1	97.41	97.40	97.43	1.77	4.6	0
Swin-T	307	P2	96.42	96.43	96.42	2.47	4.6	3
Swin-T	307	P3	97.90	97.90	97.92	1.05	4.7	5
Swin-T	349	P1	97.53	97.53	97.56	1.93	4.6	0
Swin-T	349	P2	97.04	97.05	97.05	2.27	4.6	5
Swin-T	349	P3	97.65	97.69	97.69	1.80	4.6	4
Swin-T	401	P1	98.15	98.16	98.17	1.65	4.6	0
Swin-T	401	P2	97.16	97.17	97.17	2.06	4.6	4
Swin-T	401	P3	97.78	97.77	97.76	1.71	4.7	3
Swin-T	457	P1	97.65	97.64	97.70	2.13	4.6	3
Swin-T	457	P2	94.81	94.77	94.85	4.23	4.6	5
Swin-T	457	P3	96.54	96.55	96.55	2.43	4.7	4
Swin-T	503	P1	96.05	96.03	96.10	2.84	4.7	4
Swin-T	503	P2	97.04	97.06	97.05	2.19	4.6	3
Swin-T	503	P3	95.06	95.08	95.07	3.60	4.6	0
Swin-T	557	P1	98.15	98.16	98.16	1.69	4.6	0
Swin-T	557	P2	97.16	97.16	97.20	1.70	4.6	4
Swin-T	557	P3	98.02	98.03	98.04	1.76	4.6	3
Swin-T	601	P1	97.90	97.90	97.92	1.24	4.7	3
Swin-T	601	P2	97.41	97.42	97.40	1.79	4.6	5
Swin-T	601	P3	98.77	98.78	98.78	0.81	4.6	0
Swin-T	653	P1	98.02	98.03	98.04	1.35	4.6	0
Swin-T	653	P2	96.17	96.23	96.18	2.37	4.7	4
Swin-T	653	P3	96.05	96.06	96.05	2.06	4.6	5
Swin-T	701	P1	97.28	97.29	97.29	2.32	4.7	3
Swin-T	701	P2	96.17	96.19	96.20	2.84	4.7	0
Swin-T	701	P3	93.21	93.08	93.22	5.46	4.7	4
Swin-T	751	P1	97.78	97.79	97.80	1.76	4.7	0
Swin-T	751	P2	96.17	96.20	96.14	2.84	4.7	5
Swin-T	751	P3	95.06	95.13	95.03	4.27	4.8	4
Swin-T	809	P1	95.06	95.13	95.04	3.73	4.7	4
Swin-T	809	P2	95.80	95.77	95.82	3.36	4.7	3
Swin-T	809	P3	96.42	96.41	96.46	2.67	4.7	5
Swin-T	853	P1	98.02	98.03	98.03	1.29	4.7	3
Swin-T	853	P2	97.04	97.01	97.01	2.29	4.7	5
Swin-T	853	P3	95.56	95.55	95.60	3.12	4.7	0
Swin-T	907	P1	98.40	98.40	98.42	1.04	4.6	0
Swin-T	907	P2	96.54	96.60	96.55	2.52	4.6	5
Swin-T	907	P3	98.40	98.40	98.39	1.49	4.7	3
Swin-T	953	P1	97.28	97.29	97.31	1.93	4.6	3
Swin-T	953	P2	96.79	96.80	96.81	2.44	4.6	0
Swin-T	953	P3	96.30	96.33	96.29	2.96	4.7	4
Swin-T	1009	P1	97.90	97.92	97.89	1.06	4.7	4
Swin-T	1009	P2	97.04	97.02	97.07	2.41	4.6	3
Swin-T	1009	P3	97.28	97.26	97.25	1.78	4.7	5
Swin-T	1051	P1	98.15	98.16	98.16	1.62	4.6	0
Swin-T	1051	P2	97.78	97.77	97.80	1.69	4.7	3
Swin-T	1051	P3	95.19	95.24	95.20	3.15	4.7	5
RegNetY-1.6GF	137	P1	97.90	97.91	97.91	1.50	2.5	0

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
RegNetY-1.6GF	137	P2	95.68	95.71	95.68	3.53	2.5	5
RegNetY-1.6GF	137	P3	96.05	96.06	96.06	2.35	2.5	4
RegNetY-1.6GF	173	P1	97.78	97.79	97.80	1.18	2.5	0
RegNetY-1.6GF	173	P2	97.78	97.80	97.80	1.49	2.5	3
RegNetY-1.6GF	173	P3	97.53	97.55	97.53	1.15	2.5	5
RegNetY-1.6GF	211	P1	97.65	97.67	97.67	1.33	2.5	5
RegNetY-1.6GF	211	P2	95.68	95.75	95.66	3.02	2.5	4
RegNetY-1.6GF	211	P3	95.68	95.69	95.68	2.82	2.5	0
RegNetY-1.6GF	257	P1	96.54	96.54	96.57	2.24	2.5	5
RegNetY-1.6GF	257	P2	95.80	95.82	95.78	2.47	2.5	3
RegNetY-1.6GF	257	P3	96.54	96.55	96.54	2.06	2.5	4
RegNetY-1.6GF	307	P1	97.78	97.77	97.81	1.61	2.5	0
RegNetY-1.6GF	307	P2	95.80	95.78	95.78	2.70	2.5	4
RegNetY-1.6GF	307	P3	97.78	97.82	97.78	1.34	2.5	3
RegNetY-1.6GF	349	P1	98.02	98.03	98.03	1.35	2.5	3
RegNetY-1.6GF	349	P2	97.28	97.30	97.28	1.07	2.5	0
RegNetY-1.6GF	349	P3	98.02	98.04	98.03	0.93	2.5	5
RegNetY-1.6GF	401	P1	96.79	96.81	96.80	1.75	2.5	0
RegNetY-1.6GF	401	P2	97.65	97.67	97.66	1.83	2.5	5
RegNetY-1.6GF	401	P3	97.04	97.05	96.99	1.88	2.5	4
RegNetY-1.6GF	457	P1	97.90	97.89	97.94	0.93	2.5	0
RegNetY-1.6GF	457	P2	95.06	95.01	95.10	3.76	2.5	3
RegNetY-1.6GF	457	P3	96.67	96.67	96.67	1.90	2.5	5
RegNetY-1.6GF	503	P1	97.28	97.28	97.31	1.85	2.5	5
RegNetY-1.6GF	503	P2	96.91	96.93	96.91	1.53	2.5	4
RegNetY-1.6GF	503	P3	95.43	95.44	95.42	3.12	2.5	3
RegNetY-1.6GF	557	P1	97.16	97.17	97.17	1.77	2.5	0
RegNetY-1.6GF	557	P2	97.41	97.40	97.41	1.22	2.5	5
RegNetY-1.6GF	557	P3	96.54	96.57	96.54	1.78	2.5	4
RegNetY-1.6GF	601	P1	96.05	96.08	96.05	2.62	2.5	3
RegNetY-1.6GF	601	P2	97.41	97.43	97.40	1.33	2.5	4
RegNetY-1.6GF	601	P3	97.28	97.29	97.31	1.79	2.5	5
RegNetY-1.6GF	653	P1	97.28	97.30	97.27	2.13	2.5	3
RegNetY-1.6GF	653	P2	96.91	96.94	96.92	2.07	2.5	0
RegNetY-1.6GF	653	P3	95.56	95.56	95.52	2.45	2.5	4
RegNetY-1.6GF	701	P1	97.53	97.54	97.54	1.33	2.5	5
RegNetY-1.6GF	701	P2	95.19	95.20	95.23	3.04	2.6	3
RegNetY-1.6GF	701	P3	94.81	94.80	94.85	3.17	2.6	0
RegNetY-1.6GF	751	P1	97.78	97.80	97.80	1.38	2.5	0
RegNetY-1.6GF	751	P2	96.05	96.04	96.04	2.72	2.6	3
RegNetY-1.6GF	751	P3	95.93	95.92	95.94	2.36	2.5	5
RegNetY-1.6GF	809	P1	97.90	97.92	97.91	1.40	2.6	4
RegNetY-1.6GF	809	P2	96.54	96.54	96.55	2.13	2.6	0
RegNetY-1.6GF	809	P3	96.42	96.42	96.46	2.81	2.6	3
RegNetY-1.6GF	853	P1	98.15	98.15	98.16	0.57	2.5	3
RegNetY-1.6GF	853	P2	96.42	96.41	96.39	2.59	2.5	5
RegNetY-1.6GF	853	P3	96.42	96.42	96.40	1.91	2.5	4
RegNetY-1.6GF	907	P1	98.02	98.03	98.04	1.18	2.5	0
RegNetY-1.6GF	907	P2	96.42	96.45	96.43	2.55	2.5	4
RegNetY-1.6GF	907	P3	97.90	97.91	97.88	1.54	2.5	5
RegNetY-1.6GF	953	P1	97.78	97.78	97.79	1.34	2.6	3
RegNetY-1.6GF	953	P2	97.41	97.44	97.39	1.37	2.6	5
RegNetY-1.6GF	953	P3	97.28	97.29	97.24	1.72	2.5	0
RegNetY-1.6GF	1009	P1	97.53	97.55	97.53	1.24	2.5	0
RegNetY-1.6GF	1009	P2	96.54	96.56	96.54	2.24	2.5	4
RegNetY-1.6GF	1009	P3	97.53	97.54	97.51	1.15	2.5	3
RegNetY-1.6GF	1051	P1	97.41	97.42	97.42	1.80	2.5	4
RegNetY-1.6GF	1051	P2	97.16	97.18	97.15	1.39	2.5	0
RegNetY-1.6GF	1051	P3	97.78	97.77	97.77	1.44	2.5	3
EfficientNetV2-S	137	P1	96.91	96.92	96.94	2.59	3.5	3
EfficientNetV2-S	137	P2	95.93	95.91	95.94	2.54	3.5	0
EfficientNetV2-S	137	P3	95.93	95.95	95.92	2.71	3.5	5
EfficientNetV2-S	173	P1	97.28	97.29	97.30	1.74	3.5	4
EfficientNetV2-S	173	P2	96.05	96.04	96.09	2.34	3.5	0
EfficientNetV2-S	173	P3	96.42	96.43	96.45	2.60	3.5	3
EfficientNetV2-S	211	P1	97.90	97.91	97.90	1.00	3.5	3
EfficientNetV2-S	211	P2	96.42	96.47	96.42	2.50	3.5	5
EfficientNetV2-S	211	P3	94.32	94.34	94.28	4.22	3.5	4
EfficientNetV2-S	257	P1	96.67	96.66	96.70	2.13	3.5	0
EfficientNetV2-S	257	P2	97.04	97.03	97.03	1.88	3.5	5
EfficientNetV2-S	257	P3	97.04	97.04	97.05	1.85	3.5	3
EfficientNetV2-S	307	P1	98.15	98.16	98.17	1.38	3.5	5
EfficientNetV2-S	307	P2	95.56	95.57	95.54	2.95	3.5	0

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
EfficientNetV2-S	307	P3	97.65	97.67	97.66	1.91	3.5	4
EfficientNetV2-S	349	P1	97.04	97.07	97.04	2.37	3.5	4
EfficientNetV2-S	349	P2	96.30	96.32	96.27	2.74	3.5	3
EfficientNetV2-S	349	P3	97.65	97.68	97.65	1.51	3.5	0
EfficientNetV2-S	401	P1	97.16	97.17	97.17	2.13	3.5	3
EfficientNetV2-S	401	P2	97.41	97.44	97.41	1.66	3.5	0
EfficientNetV2-S	401	P3	97.16	97.16	97.14	1.52	3.5	5
EfficientNetV2-S	457	P1	98.15	98.15	98.16	1.58	3.5	4
EfficientNetV2-S	457	P2	94.20	94.19	94.21	3.90	3.5	0
EfficientNetV2-S	457	P3	96.05	96.05	96.08	2.50	3.5	3
EfficientNetV2-S	503	P1	96.54	96.55	96.58	2.61	3.5	0
EfficientNetV2-S	503	P2	96.05	96.08	96.04	3.36	3.5	5
EfficientNetV2-S	503	P3	96.79	96.78	96.78	2.22	3.5	4
EfficientNetV2-S	557	P1	97.04	97.07	97.06	2.13	3.5	3
EfficientNetV2-S	557	P2	98.40	98.41	98.39	0.99	3.5	0
EfficientNetV2-S	557	P3	97.90	97.91	97.91	1.61	3.5	5
EfficientNetV2-S	601	P1	96.79	96.82	96.82	2.44	3.5	0
EfficientNetV2-S	601	P2	97.16	97.16	97.19	2.40	3.5	3
EfficientNetV2-S	601	P3	97.16	97.17	97.16	1.86	3.5	4
EfficientNetV2-S	653	P1	97.41	97.42	97.41	1.92	3.5	5
EfficientNetV2-S	653	P2	95.56	95.57	95.54	3.36	3.5	3
EfficientNetV2-S	653	P3	96.67	96.65	96.65	2.04	3.5	0
EfficientNetV2-S	701	P1	97.28	97.30	97.29	1.83	3.6	5
EfficientNetV2-S	701	P2	94.94	94.98	94.96	4.05	3.6	3
EfficientNetV2-S	701	P3	94.44	94.43	94.44	4.17	3.6	0
EfficientNetV2-S	751	P1	97.53	97.56	97.54	1.38	3.5	4
EfficientNetV2-S	751	P2	96.05	96.06	96.05	3.48	3.6	0
EfficientNetV2-S	751	P3	97.04	97.07	97.03	2.06	3.6	5
EfficientNetV2-S	809	P1	97.78	97.77	97.82	1.45	3.6	5
EfficientNetV2-S	809	P2	96.05	96.05	96.07	2.84	3.6	4
EfficientNetV2-S	809	P3	94.69	94.62	94.75	4.03	3.6	0
EfficientNetV2-S	853	P1	96.79	96.77	96.82	2.29	3.5	0
EfficientNetV2-S	853	P2	96.05	96.03	96.03	3.09	3.5	3
EfficientNetV2-S	853	P3	96.42	96.43	96.43	2.35	3.5	5
EfficientNetV2-S	907	P1	98.02	98.04	98.02	1.21	3.5	3
EfficientNetV2-S	907	P2	96.54	96.55	96.55	2.80	3.5	0
EfficientNetV2-S	907	P3	96.54	96.59	96.55	2.28	3.5	4
EfficientNetV2-S	953	P1	97.53	97.53	97.56	2.13	3.5	4
EfficientNetV2-S	953	P2	97.65	97.67	97.63	1.48	3.6	0
EfficientNetV2-S	953	P3	98.15	98.16	98.15	1.11	3.5	3
EfficientNetV2-S	1009	P1	96.91	96.95	96.91	2.16	3.5	5
EfficientNetV2-S	1009	P2	95.93	95.93	95.94	2.92	3.5	0
EfficientNetV2-S	1009	P3	96.30	96.30	96.27	2.42	3.5	3
EfficientNetV2-S	1051	P1	98.27	98.28	98.27	1.18	3.5	5
EfficientNetV2-S	1051	P2	97.04	97.04	97.00	1.66	3.5	4
EfficientNetV2-S	1051	P3	97.41	97.41	97.42	1.51	3.5	0
MaxViT-T	137	P1	98.02	98.02	98.04	1.40	7.8	4
MaxViT-T	137	P2	95.06	95.04	95.09	3.02	7.8	3
MaxViT-T	137	P3	95.31	95.30	95.35	3.22	7.8	0
MaxViT-T	173	P1	97.04	97.03	97.08	2.22	7.8	5
MaxViT-T	173	P2	94.57	94.61	94.62	4.09	7.8	4
MaxViT-T	173	P3	97.04	97.05	97.04	1.83	7.8	0
MaxViT-T	211	P1	98.15	98.16	98.16	1.52	7.8	0
MaxViT-T	211	P2	95.68	95.68	95.69	3.31	7.8	3
MaxViT-T	211	P3	95.80	95.84	95.79	3.59	7.8	5
MaxViT-T	257	P1	96.91	96.91	96.93	2.33	7.8	4
MaxViT-T	257	P2	97.53	97.53	97.54	2.32	7.7	0
MaxViT-T	257	P3	97.78	97.78	97.81	1.12	7.8	5
MaxViT-T	307	P1	97.78	97.76	97.81	1.65	7.8	3
MaxViT-T	307	P2	97.28	97.29	97.30	2.25	7.8	5
MaxViT-T	307	P3	97.78	97.78	97.80	1.74	7.8	0
MaxViT-T	349	P1	96.67	96.68	96.69	2.59	7.8	5
MaxViT-T	349	P2	94.57	94.61	94.59	4.08	7.8	4
MaxViT-T	349	P3	97.90	97.94	97.89	1.41	7.8	3
MaxViT-T	401	P1	97.65	97.67	97.68	1.87	7.8	4
MaxViT-T	401	P2	97.78	97.79	97.79	1.44	7.8	3
MaxViT-T	401	P3	95.80	95.76	95.84	2.90	7.8	0
MaxViT-T	457	P1	96.91	96.87	96.96	2.26	7.8	5
MaxViT-T	457	P2	94.57	94.56	94.58	4.07	7.8	4
MaxViT-T	457	P3	96.67	96.66	96.70	2.67	7.8	0
MaxViT-T	503	P1	92.72	92.69	92.80	6.48	7.8	3
MaxViT-T	503	P2	97.28	97.28	97.30	1.96	7.7	0
MaxViT-T	503	P3	95.80	95.79	95.84	3.11	7.8	5

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
MaxViT-T	557	P1	97.16	97.17	97.19	1.92	7.8	4
MaxViT-T	557	P2	97.90	97.90	97.93	1.38	7.7	3
MaxViT-T	557	P3	97.78	97.79	97.78	1.10	7.8	0
MaxViT-T	601	P1	96.91	96.94	96.93	2.42	7.8	5
MaxViT-T	601	P2	97.41	97.40	97.42	1.64	7.8	0
MaxViT-T	601	P3	96.67	96.67	96.70	2.22	7.8	3
MaxViT-T	653	P1	98.02	98.03	98.05	1.72	8.3	4
MaxViT-T	653	P2	97.04	97.04	97.06	1.92	8.7	5
MaxViT-T	653	P3	97.53	97.52	97.53	1.58	8.1	3
MaxViT-T	701	P1	97.90	97.90	97.90	1.28	8.0	4
MaxViT-T	701	P2	95.31	95.32	95.34	4.04	7.9	5
MaxViT-T	701	P3	95.06	95.02	95.10	3.72	8.0	3
MaxViT-T	751	P1	97.28	97.30	97.31	1.83	7.9	3
MaxViT-T	751	P2	96.54	96.55	96.51	2.90	8.0	4
MaxViT-T	751	P3	96.54	96.52	96.57	2.35	7.9	0
MaxViT-T	809	P1	97.04	97.02	97.06	2.40	8.0	3
MaxViT-T	809	P2	95.19	95.23	95.16	4.25	8.0	5
MaxViT-T	809	P3	96.30	96.29	96.34	3.00	8.0	4
MaxViT-T	853	P1	97.65	97.66	97.67	1.55	7.9	4
MaxViT-T	853	P2	95.43	95.42	95.36	3.46	7.8	0
MaxViT-T	853	P3	95.06	95.08	95.00	3.88	7.8	3
MaxViT-T	907	P1	98.27	98.27	98.29	1.62	7.8	5
MaxViT-T	907	P2	95.93	95.91	95.91	3.36	7.8	3
MaxViT-T	907	P3	97.65	97.66	97.66	1.52	7.8	0
MaxViT-T	953	P1	96.05	96.08	96.07	3.18	7.9	5
MaxViT-T	953	P2	97.65	97.67	97.66	1.80	7.8	4
MaxViT-T	953	P3	97.78	97.79	97.77	1.64	7.8	0
MaxViT-T	1009	P1	97.78	97.79	97.80	1.96	7.8	4
MaxViT-T	1009	P2	95.80	95.76	95.86	3.65	7.9	5
MaxViT-T	1009	P3	96.79	96.78	96.75	2.84	7.8	0
MaxViT-T	1051	P1	97.41	97.41	97.42	1.81	7.9	3
MaxViT-T	1051	P2	97.53	97.53	97.55	1.19	7.8	5
MaxViT-T	1051	P3	96.30	96.30	96.32	2.63	7.8	4
DINOv2 ViT-S/14	137	P1	90.86	90.96	90.94	3.71	4.1	5
DINOv2 ViT-S/14	137	P2	91.73	91.82	91.73	5.24	4.1	5
DINOv2 ViT-S/14	137	P3	92.84	92.86	92.88	2.65	4.1	4
DINOv2 ViT-S/14	173	P1	90.99	91.07	91.05	3.02	4.1	3
DINOv2 ViT-S/14	173	P2	92.47	92.53	92.53	3.25	4.1	3
DINOv2 ViT-S/14	173	P3	92.84	92.82	92.94	3.57	4.1	5
DINOv2 ViT-S/14	211	P1	93.58	93.60	93.63	3.08	4.1	4
DINOv2 ViT-S/14	211	P2	92.72	92.76	92.81	4.67	4.1	4
DINOv2 ViT-S/14	211	P3	90.49	90.52	90.48	4.98	4.1	0
DINOv2 ViT-S/14	257	P1	93.33	93.24	93.40	3.89	4.1	3
DINOv2 ViT-S/14	257	P2	92.84	92.82	92.95	4.11	4.1	3
DINOv2 ViT-S/14	257	P3	93.46	93.45	93.48	2.71	4.1	4
DINOv2 ViT-S/14	307	P1	93.46	93.42	93.54	3.44	4.1	4
DINOv2 ViT-S/14	307	P2	92.72	92.73	92.67	1.98	4.1	4
DINOv2 ViT-S/14	307	P3	91.48	91.48	91.48	3.68	4.1	3
DINOv2 ViT-S/14	349	P1	95.43	95.48	95.46	1.27	4.1	0
DINOv2 ViT-S/14	349	P2	92.84	92.91	92.85	2.93	4.1	0
DINOv2 ViT-S/14	349	P3	92.96	93.02	93.03	3.91	4.1	5
DINOv2 ViT-S/14	401	P1	93.46	93.50	93.55	3.28	4.1	5
DINOv2 ViT-S/14	401	P2	94.57	94.54	94.63	3.61	4.1	5
DINOv2 ViT-S/14	401	P3	90.62	90.83	90.65	4.24	4.1	4
DINOv2 ViT-S/14	457	P1	96.17	96.18	96.21	1.79	4.1	3
DINOv2 ViT-S/14	457	P2	91.85	91.92	91.86	4.79	4.1	3
DINOv2 ViT-S/14	457	P3	92.84	92.84	92.87	4.81	4.1	5
DINOv2 ViT-S/14	503	P1	93.33	93.34	93.38	3.72	4.1	4
DINOv2 ViT-S/14	503	P2	95.06	95.07	95.04	2.49	4.1	4
DINOv2 ViT-S/14	503	P3	87.78	87.89	87.88	6.13	4.1	3
DINOv2 ViT-S/14	557	P1	95.68	95.70	95.71	1.78	4.1	5
DINOv2 ViT-S/14	557	P2	94.32	94.35	94.36	2.11	4.1	5
DINOv2 ViT-S/14	557	P3	90.12	89.89	90.12	4.24	4.1	4
DINOv2 ViT-S/14	601	P1	94.81	94.85	94.84	2.80	4.1	4
DINOv2 ViT-S/14	601	P2	89.75	90.10	89.76	4.11	4.1	4
DINOv2 ViT-S/14	601	P3	94.07	94.10	94.11	2.52	4.1	5
DINOv2 ViT-S/14	653	P1	93.21	93.28	93.22	3.19	4.3	0
DINOv2 ViT-S/14	653	P2	93.33	93.39	93.35	2.63	4.1	0
DINOv2 ViT-S/14	653	P3	90.99	90.96	90.98	3.67	4.1	4
DINOv2 ViT-S/14	701	P1	94.07	94.09	94.14	2.47	4.2	0
DINOv2 ViT-S/14	701	P2	89.14	89.33	89.22	5.85	4.2	4
DINOv2 ViT-S/14	701	P3	87.90	87.92	87.92	7.21	4.2	0
DINOv2 ViT-S/14	751	P1	94.94	94.94	94.98	2.01	4.2	5

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
DINOv2 ViT-S/14	751	P2	91.98	92.02	92.06	3.51	4.2	3
DINOv2 ViT-S/14	751	P3	87.16	87.43	87.21	7.66	4.2	5
DINOv2 ViT-S/14	809	P1	93.33	93.46	93.35	1.01	4.2	0
DINOv2 ViT-S/14	809	P2	92.96	93.02	93.01	3.23	4.2	3
DINOv2 ViT-S/14	809	P3	89.88	89.96	89.96	2.90	4.2	0
DINOv2 ViT-S/14	853	P1	93.58	93.59	93.64	3.65	4.1	5
DINOv2 ViT-S/14	853	P2	91.98	91.67	92.07	4.62	4.1	5
DINOv2 ViT-S/14	853	P3	89.51	89.47	89.65	5.47	4.1	4
DINOv2 ViT-S/14	907	P1	94.57	94.64	94.56	2.42	4.1	4
DINOv2 ViT-S/14	907	P2	90.37	90.54	90.46	5.40	4.1	4
DINOv2 ViT-S/14	907	P3	90.99	91.08	91.01	3.76	4.1	5
DINOv2 ViT-S/14	953	P1	92.84	92.87	92.91	3.51	4.1	3
DINOv2 ViT-S/14	953	P2	92.47	92.45	92.51	4.31	4.1	3
DINOv2 ViT-S/14	953	P3	92.35	92.36	92.40	3.91	4.1	5
DINOv2 ViT-S/14	1009	P1	93.09	93.12	93.17	2.99	4.1	3
DINOv2 ViT-S/14	1009	P2	95.43	95.44	95.49	2.32	4.1	3
DINOv2 ViT-S/14	1009	P3	90.99	91.00	91.08	5.31	4.1	4
DINOv2 ViT-S/14	1051	P1	94.20	94.24	94.21	3.02	4.1	0
DINOv2 ViT-S/14	1051	P2	92.72	92.74	92.78	4.40	4.1	0
DINOv2 ViT-S/14	1051	P3	89.75	89.90	89.75	3.21	4.1	3

Table L.2: Complete CRIC run ledger (180 runs).

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-18	137	P1	90.29	79.91	80.99	5.17	2.4	5
ResNet-18	137	P1-prime	90.42	79.96	78.81	4.90	2.4	4
ResNet-18	137	P2	77.21	56.93	56.06	15.34	2.4	5
ResNet-18	173	P1	90.33	79.33	77.66	5.38	2.4	0
ResNet-18	173	P1-prime	89.42	77.91	78.92	6.32	2.4	0
ResNet-18	173	P2	79.19	54.72	52.76	13.90	2.4	0
ResNet-18	211	P1	89.47	79.21	77.35	5.56	2.4	5
ResNet-18	211	P1-prime	89.03	78.04	76.41	6.41	2.4	5
ResNet-18	211	P2	79.30	57.93	56.26	13.97	2.4	5
ResNet-18	257	P1	90.68	81.52	80.34	4.79	2.4	4
ResNet-18	257	P1-prime	90.51	80.46	80.79	5.23	2.4	4
ResNet-18	257	P2	76.61	56.24	54.11	16.74	2.4	4
ResNet-18	307	P1	90.16	79.43	76.56	5.47	2.4	5
ResNet-18	307	P1-prime	90.81	81.98	80.34	4.49	2.4	5
ResNet-18	307	P2	77.57	58.26	59.42	14.68	2.4	5
ResNet-18	349	P1	89.47	77.77	76.57	5.20	2.4	3
ResNet-18	349	P1-prime	90.38	78.70	77.45	4.87	2.4	3
ResNet-18	349	P2	76.45	56.43	58.27	16.90	2.4	3
ResNet-18	401	P1	89.47	78.15	74.99	6.14	2.4	0
ResNet-18	401	P1-prime	89.21	78.43	75.35	5.71	2.4	0
ResNet-18	401	P2	77.31	56.70	53.65	14.96	2.4	0
ResNet-18	457	P1	89.68	76.69	74.80	5.72	2.4	4
ResNet-18	457	P1-prime	90.51	78.75	75.63	5.12	2.4	4
ResNet-18	457	P2	76.80	58.06	60.26	15.04	2.4	4
ResNet-18	503	P1	89.81	79.25	79.18	5.98	2.4	5
ResNet-18	503	P1-prime	89.08	77.64	76.56	6.20	2.4	5
ResNet-18	503	P2	79.08	60.12	58.06	13.66	2.4	5
ResNet-18	557	P1	89.77	79.63	77.54	5.56	2.4	3
ResNet-18	557	P1-prime	89.81	77.59	73.91	5.51	2.4	3
ResNet-18	557	P2	76.66	57.06	59.07	14.67	2.4	3
ResNet-50	137	P1	89.86	80.65	81.45	6.36	7.3	4
ResNet-50	137	P1-prime	90.33	80.73	81.13	5.40	7.3	0
ResNet-50	137	P2	77.94	58.33	56.70	16.41	7.3	5
ResNet-50	173	P1	91.33	83.04	83.86	5.13	7.3	0
ResNet-50	173	P1-prime	92.02	83.50	82.32	4.72	7.3	5
ResNet-50	173	P2	81.05	64.25	63.43	13.25	7.3	0
ResNet-50	211	P1	90.03	81.34	79.72	5.84	7.3	5
ResNet-50	211	P1-prime	90.29	79.45	79.32	5.44	7.3	4
ResNet-50	211	P2	80.38	62.61	61.35	13.78	7.3	5
ResNet-50	257	P1	90.46	80.21	78.47	6.06	7.4	4
ResNet-50	257	P1-prime	91.33	80.74	80.77	5.01	7.3	3
ResNet-50	257	P2	74.84	52.48	55.44	19.13	7.3	4
ResNet-50	307	P1	91.16	82.30	79.82	6.02	7.3	5
ResNet-50	307	P1-prime	91.68	83.75	81.52	4.96	7.3	3
ResNet-50	307	P2	78.87	61.31	62.73	14.49	7.3	5

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	349	P1	90.42	78.92	78.72	5.90	7.3	3
ResNet-50	349	P1-prime	90.29	79.95	79.11	6.25	7.3	0
ResNet-50	349	P2	78.30	60.27	60.23	15.55	7.3	3
ResNet-50	401	P1	90.29	79.95	79.56	5.80	7.3	0
ResNet-50	401	P1-prime	90.59	82.00	80.65	5.37	7.3	4
ResNet-50	401	P2	79.52	61.94	58.62	13.71	7.3	0
ResNet-50	457	P1	90.59	78.99	76.56	5.66	7.3	4
ResNet-50	457	P1-prime	90.90	78.63	75.56	5.78	7.3	5
ResNet-50	457	P2	78.35	60.92	60.86	14.85	7.3	4
ResNet-50	503	P1	90.98	81.56	80.72	5.65	7.3	5
ResNet-50	503	P1-prime	90.46	81.15	80.52	5.37	7.3	3
ResNet-50	503	P2	77.78	59.65	59.78	15.47	7.3	5
ResNet-50	557	P1	90.25	77.83	75.33	6.06	7.3	3
ResNet-50	557	P1-prime	90.59	80.18	77.63	5.76	7.3	4
ResNet-50	557	P2	80.55	63.07	64.28	13.03	7.3	3
DenseNet-121	137	P1	90.77	81.40	79.36	5.65	9.2	3
DenseNet-121	137	P1-prime	90.12	78.92	78.57	5.74	9.2	4
DenseNet-121	137	P2	77.29	56.35	53.39	15.86	9.2	0
DenseNet-121	173	P1	91.55	82.97	83.22	4.36	9.2	5
DenseNet-121	173	P1-prime	90.94	81.93	81.19	5.28	9.2	0
DenseNet-121	173	P2	80.18	61.63	59.67	13.70	9.2	3
DenseNet-121	211	P1	90.12	81.50	80.12	5.08	9.2	4
DenseNet-121	211	P1-prime	90.59	79.02	75.99	5.53	9.2	0
DenseNet-121	211	P2	81.07	63.31	60.96	12.27	9.2	5
DenseNet-121	257	P1	91.07	79.90	78.63	5.07	9.2	3
DenseNet-121	257	P1-prime	90.20	80.04	77.99	5.83	9.2	4
DenseNet-121	257	P2	78.13	59.00	58.60	14.92	9.2	0
DenseNet-121	307	P1	90.20	77.44	74.42	5.57	9.2	3
DenseNet-121	307	P1-prime	91.76	81.93	79.65	4.44	9.2	5
DenseNet-121	307	P2	76.01	55.79	60.52	16.43	9.2	4
DenseNet-121	349	P1	90.20	78.60	78.54	5.42	9.1	0
DenseNet-121	349	P1-prime	91.07	80.37	78.54	4.82	9.2	3
DenseNet-121	349	P2	78.77	60.09	63.41	14.41	9.1	5
DenseNet-121	401	P1	89.12	78.75	76.82	6.43	9.2	4
DenseNet-121	401	P1-prime	90.51	79.45	76.96	5.44	9.1	0
DenseNet-121	401	P2	78.22	60.38	59.59	14.62	9.2	3
DenseNet-121	457	P1	91.20	79.85	78.11	4.71	9.1	5
DenseNet-121	457	P1-prime	90.42	78.11	75.80	5.87	9.1	0
DenseNet-121	457	P2	77.14	59.19	58.14	16.27	9.2	4
DenseNet-121	503	P1	90.16	79.54	76.27	5.40	9.2	3
DenseNet-121	503	P1-prime	90.72	82.20	80.36	5.50	9.1	5
DenseNet-121	503	P2	78.30	57.31	55.94	15.11	9.1	0
DenseNet-121	557	P1	89.64	76.65	74.17	6.54	9.1	4
DenseNet-121	557	P1-prime	90.72	80.22	78.14	5.12	9.1	3
DenseNet-121	557	P2	77.23	57.79	58.03	15.87	9.2	5
EfficientNet-B0	137	P1	90.29	80.62	79.53	5.65	5.0	5
EfficientNet-B0	137	P1-prime	89.99	79.20	78.79	6.21	5.0	3
EfficientNet-B0	137	P2	77.16	57.77	57.65	15.12	5.0	5
EfficientNet-B0	173	P1	91.89	82.77	81.66	4.80	5.0	4
EfficientNet-B0	173	P1-prime	91.72	82.68	82.66	4.72	5.0	5
EfficientNet-B0	173	P2	81.48	63.09	61.81	12.47	5.0	4
EfficientNet-B0	211	P1	90.68	79.19	77.49	5.24	5.0	3
EfficientNet-B0	211	P1-prime	91.29	81.66	81.54	4.73	5.0	4
EfficientNet-B0	211	P2	81.77	62.82	63.40	11.79	5.0	3
EfficientNet-B0	257	P1	91.55	81.41	81.44	4.73	5.0	5
EfficientNet-B0	257	P1-prime	91.72	82.07	82.64	4.60	5.0	5
EfficientNet-B0	257	P2	76.05	57.29	59.00	16.76	5.0	3
EfficientNet-B0	307	P1	90.55	81.10	79.85	5.19	5.0	0
EfficientNet-B0	307	P1-prime	91.20	81.63	82.06	5.40	5.0	3
EfficientNet-B0	307	P2	77.83	57.52	60.43	15.45	5.0	0
EfficientNet-B0	349	P1	90.46	80.08	78.06	6.16	5.0	4
EfficientNet-B0	349	P1-prime	90.29	80.27	79.07	5.81	5.0	0
EfficientNet-B0	349	P2	78.65	59.36	61.94	14.72	5.0	4
EfficientNet-B0	401	P1	90.85	80.96	80.29	5.21	5.0	5
EfficientNet-B0	401	P1-prime	90.55	80.38	81.88	5.52	5.0	4
EfficientNet-B0	401	P2	78.61	60.63	59.86	14.30	5.0	5
EfficientNet-B0	457	P1	90.51	79.73	79.60	5.56	5.0	3
EfficientNet-B0	457	P1-prime	91.24	80.90	80.88	4.81	5.0	5
EfficientNet-B0	457	P2	76.06	57.36	60.23	15.85	5.0	3
EfficientNet-B0	503	P1	90.64	81.14	81.41	5.64	5.0	4
EfficientNet-B0	503	P1-prime	90.46	80.65	80.62	5.39	5.0	3
EfficientNet-B0	503	P2	77.96	57.44	57.99	15.16	5.0	4
EfficientNet-B0	557	P1	91.33	80.42	78.46	5.08	5.0	0

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
EfficientNet-B0	557	P1-prime	91.03	80.06	78.17	4.89	5.0	4
EfficientNet-B0	557	P2	75.45	54.07	56.27	17.44	5.0	0
ViT-B/16	137	P1	90.55	79.71	78.55	5.74	25.9	5
ViT-B/16	137	P1-prime	90.20	80.75	80.50	6.32	25.8	3
ViT-B/16	137	P2	78.29	58.37	54.71	16.29	25.9	4
ViT-B/16	173	P1	89.99	78.02	79.06	6.45	26.0	4
ViT-B/16	173	P1-prime	89.60	78.42	75.28	6.52	25.7	0
ViT-B/16	173	P2	79.27	59.92	60.53	15.11	25.9	3
ViT-B/16	211	P1	90.51	80.80	79.57	6.04	25.9	3
ViT-B/16	211	P1-prime	89.94	77.45	75.29	6.05	25.7	0
ViT-B/16	211	P2	80.21	60.21	58.31	14.40	25.9	5
ViT-B/16	257	P1	89.03	77.25	75.47	7.99	25.9	3
ViT-B/16	257	P1-prime	90.12	77.30	75.69	6.20	26.0	4
ViT-B/16	257	P2	76.18	60.00	60.82	16.76	25.7	0
ViT-B/16	307	P1	90.20	80.76	79.34	5.77	25.7	0
ViT-B/16	307	P1-prime	88.99	77.43	76.11	6.22	25.9	5
ViT-B/16	307	P2	77.22	59.71	57.55	17.49	25.9	4
ViT-B/16	349	P1	89.55	73.29	70.14	7.18	26.0	4
ViT-B/16	349	P1-prime	88.77	74.45	70.14	8.36	25.8	3
ViT-B/16	349	P2	77.83	57.24	55.76	17.61	25.8	5
ViT-B/16	401	P1	88.77	77.35	71.95	8.24	25.9	5
ViT-B/16	401	P1-prime	90.03	80.54	79.28	6.51	25.7	0
ViT-B/16	401	P2	78.65	59.91	56.37	16.21	25.8	3
ViT-B/16	457	P1	90.51	79.10	77.73	5.66	25.8	3
ViT-B/16	457	P1-prime	89.81	78.89	75.15	6.03	25.7	0
ViT-B/16	457	P2	78.70	61.55	58.62	15.28	26.0	4
ViT-B/16	503	P1	89.47	75.79	75.20	6.49	26.0	4
ViT-B/16	503	P1-prime	88.08	79.19	78.85	6.87	25.8	5
ViT-B/16	503	P2	77.35	57.28	54.37	17.84	25.7	0
ViT-B/16	557	P1	89.73	78.02	76.66	5.90	25.7	0
ViT-B/16	557	P1-prime	89.86	78.34	78.26	6.38	25.8	3
ViT-B/16	557	P2	78.35	61.15	59.20	14.42	25.8	5
ConvNeXt-T	137	P1	91.55	83.50	79.91	5.85	14.3	3
ConvNeXt-T	137	P1-prime	90.77	80.41	77.64	6.69	14.4	4
ConvNeXt-T	137	P2	78.63	60.52	56.01	17.18	14.2	0
ConvNeXt-T	173	P1	91.94	83.44	83.16	5.38	14.3	0
ConvNeXt-T	173	P1-prime	90.98	82.30	79.95	5.54	14.3	3
ConvNeXt-T	173	P2	81.52	61.62	60.22	14.36	14.3	5
ConvNeXt-T	211	P1	90.38	81.37	80.77	5.95	14.3	0
ConvNeXt-T	211	P1-prime	91.29	83.19	81.53	5.62	14.3	5
ConvNeXt-T	211	P2	81.81	62.91	59.70	13.85	14.4	4
ConvNeXt-T	257	P1	91.55	82.04	81.67	5.60	14.4	4
ConvNeXt-T	257	P1-prime	91.63	81.60	80.39	5.65	14.2	0
ConvNeXt-T	257	P2	80.77	64.19	65.05	14.61	14.3	5
ConvNeXt-T	307	P1	91.68	83.12	82.39	5.21	14.3	5
ConvNeXt-T	307	P1-prime	91.68	81.84	79.38	5.67	14.3	4
ConvNeXt-T	307	P2	77.78	61.31	61.71	16.69	14.3	3
ConvNeXt-T	349	P1	91.33	80.34	76.30	6.26	14.3	3
ConvNeXt-T	349	P1-prime	90.98	77.69	75.88	5.88	14.2	5
ConvNeXt-T	349	P2	79.08	60.47	60.42	16.43	14.2	0
ConvNeXt-T	401	P1	90.59	80.19	75.89	6.90	14.2	0
ConvNeXt-T	401	P1-prime	90.94	81.39	81.60	5.84	14.3	3
ConvNeXt-T	401	P2	77.79	58.68	56.94	16.55	14.4	4
ConvNeXt-T	457	P1	90.64	78.90	76.97	6.27	14.3	0
ConvNeXt-T	457	P1-prime	90.55	77.45	75.59	6.99	14.4	4
ConvNeXt-T	457	P2	80.39	64.21	65.34	14.58	14.3	5
ConvNeXt-T	503	P1	90.98	82.89	82.26	6.52	14.3	5
ConvNeXt-T	503	P1-prime	90.68	82.10	79.29	6.61	14.2	0
ConvNeXt-T	503	P2	80.51	64.10	61.27	14.46	14.3	3
ConvNeXt-T	557	P1	92.28	82.23	81.03	5.16	14.3	3
ConvNeXt-T	557	P1-prime	91.46	79.34	77.21	5.69	14.3	5
ConvNeXt-T	557	P2	76.53	53.49	52.30	19.50	14.3	4

Table L.3: Complete 2C/2B run ledger (160 runs; protocol is the factorial cell).

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	137	clean_high	97.13	97.00	97.04	2.09	1.1	4
ResNet-50	137	clean_low	96.36	96.13	96.18	2.09	1.1	0
ResNet-50	137	leak_high	97.13	96.93	96.91	1.40	1.1	5

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	137	leak_low	97.89	97.77	97.77	1.18	1.1	5
ResNet-50	173	clean_high	95.55	95.56	95.51	1.42	1.1	4
ResNet-50	173	clean_low	94.97	94.95	95.03	2.09	1.1	0
ResNet-50	173	leak_high	96.71	96.64	96.74	1.29	1.1	3
ResNet-50	173	leak_low	96.71	96.73	96.70	1.46	1.1	5
ResNet-50	211	clean_high	94.41	94.31	94.35	3.02	1.1	4
ResNet-50	211	clean_low	94.41	94.25	94.31	2.29	1.1	0
ResNet-50	211	leak_high	96.15	96.06	96.05	2.16	1.1	3
ResNet-50	211	leak_low	97.50	97.44	97.44	2.04	1.1	5
ResNet-50	257	clean_high	94.07	93.82	93.79	3.46	1.1	4
ResNet-50	257	clean_low	95.03	94.84	94.89	2.01	1.1	3
ResNet-50	257	leak_high	96.56	96.35	96.21	1.61	1.1	3
ResNet-50	257	leak_low	96.75	96.62	96.76	2.22	1.1	5
ResNet-50	307	clean_high	94.92	94.78	94.64	1.98	1.1	4
ResNet-50	307	clean_low	93.95	93.71	93.66	3.47	1.1	3
ResNet-50	307	leak_high	95.51	95.31	95.24	1.80	1.1	0
ResNet-50	307	leak_low	97.85	97.72	97.70	1.27	1.1	5
ResNet-50	349	clean_high	92.29	92.11	92.19	3.53	1.1	4
ResNet-50	349	clean_low	94.74	94.59	94.64	2.75	1.1	3
ResNet-50	349	leak_high	94.92	94.81	94.88	1.91	1.1	0
ResNet-50	349	leak_low	97.18	97.05	97.09	1.64	1.1	5
ResNet-50	401	clean_high	92.07	91.62	91.82	4.05	1.1	4
ResNet-50	401	clean_low	93.23	93.02	93.01	4.14	1.1	0
ResNet-50	401	leak_high	95.16	94.97	94.93	2.54	1.1	0
ResNet-50	401	leak_low	94.78	94.56	94.54	3.25	1.1	5
ResNet-50	457	clean_high	96.34	96.18	96.14	2.44	1.1	4
ResNet-50	457	clean_low	95.80	95.61	95.50	3.17	1.1	0
ResNet-50	457	leak_high	96.89	96.73	96.75	1.29	1.1	3
ResNet-50	457	leak_low	96.16	96.03	95.93	1.85	1.1	5
ResNet-50	503	clean_high	95.28	95.18	95.09	2.20	1.1	4
ResNet-50	503	clean_low	95.85	95.80	95.82	1.97	1.1	3
ResNet-50	503	leak_high	96.42	96.36	96.35	2.08	1.1	3
ResNet-50	503	leak_low	97.55	97.48	97.54	1.91	1.1	5
ResNet-50	557	clean_high	93.64	93.51	93.40	3.58	1.1	4
ResNet-50	557	clean_low	94.55	94.45	94.41	2.81	1.1	3
ResNet-50	557	leak_high	95.64	95.50	95.55	2.45	1.1	0
ResNet-50	557	leak_low	95.64	95.56	95.46	2.43	1.1	5
DenseNet-121	137	clean_high	96.93	96.76	96.84	1.27	1.3	4
DenseNet-121	137	clean_low	94.25	93.97	93.92	2.00	1.3	0
DenseNet-121	137	leak_high	97.51	97.29	97.19	0.98	1.3	3
DenseNet-121	137	leak_low	96.36	96.06	96.03	1.35	1.3	5
DenseNet-121	173	clean_high	94.58	94.54	94.62	2.99	1.3	4
DenseNet-121	173	clean_low	96.32	96.26	96.36	2.13	1.3	0
DenseNet-121	173	leak_high	96.52	96.44	96.36	1.65	1.3	3
DenseNet-121	173	leak_low	97.68	97.69	97.69	1.08	1.3	5
DenseNet-121	211	clean_high	95.95	95.89	95.92	1.19	1.3	4
DenseNet-121	211	clean_low	95.18	95.03	95.06	2.73	1.3	0
DenseNet-121	211	leak_high	97.69	97.62	97.61	1.65	1.4	3
DenseNet-121	211	leak_low	97.69	97.62	97.63	1.82	1.3	5
DenseNet-121	257	clean_high	94.84	94.62	94.75	1.74	1.3	4
DenseNet-121	257	clean_low	96.75	96.63	96.80	1.67	1.4	3
DenseNet-121	257	leak_high	95.98	95.70	95.76	1.42	1.3	0
DenseNet-121	257	leak_low	97.51	97.38	97.38	0.95	1.3	5
DenseNet-121	307	clean_high	94.53	94.19	94.12	3.35	1.3	4
DenseNet-121	307	clean_low	94.14	93.89	93.88	2.61	1.3	3
DenseNet-121	307	leak_high	98.05	97.99	98.03	1.11	1.3	0
DenseNet-121	307	leak_low	97.85	97.76	97.73	1.36	1.3	5
DenseNet-121	349	clean_high	92.67	92.40	92.46	3.21	1.3	4
DenseNet-121	349	clean_low	90.41	90.56	90.38	4.62	1.3	3
DenseNet-121	349	leak_high	96.80	96.66	96.69	2.06	1.3	0
DenseNet-121	349	leak_low	96.80	96.69	96.71	2.57	1.3	5
DenseNet-121	401	clean_high	93.04	92.69	92.79	3.22	1.3	4
DenseNet-121	401	clean_low	94.58	94.40	94.42	3.65	1.3	0
DenseNet-121	401	leak_high	95.94	95.77	95.79	2.01	1.3	3
DenseNet-121	401	leak_low	95.94	95.77	95.74	2.37	1.3	5
DenseNet-121	457	clean_high	95.61	95.36	95.26	2.48	1.3	4
DenseNet-121	457	clean_low	94.33	94.15	94.03	3.07	1.3	0
DenseNet-121	457	leak_high	97.62	97.49	97.51	1.29	1.3	3
DenseNet-121	457	leak_low	97.07	96.91	96.80	1.15	1.3	5
DenseNet-121	503	clean_high	95.66	95.55	95.45	2.13	1.3	4
DenseNet-121	503	clean_low	95.85	95.78	95.85	2.13	1.3	3
DenseNet-121	503	leak_high	96.04	95.94	95.98	1.72	1.3	0
DenseNet-121	503	leak_low	97.17	97.14	97.18	1.40	1.3	5

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
DenseNet-121	557	clean_high	94.00	93.84	93.68	2.54	1.3	4
DenseNet-121	557	clean_low	94.91	94.81	94.73	3.56	1.4	3
DenseNet-121	557	leak_high	95.45	95.34	95.28	2.35	1.3	0
DenseNet-121	557	leak_low	97.09	97.00	96.97	2.05	1.3	5
ViT-B/16	137	clean_high	95.79	95.58	95.56	2.85	3.6	4
ViT-B/16	137	clean_low	96.36	96.15	96.23	2.00	3.6	0
ViT-B/16	137	leak_high	98.08	97.96	97.96	1.33	3.6	3
ViT-B/16	137	leak_low	94.83	94.52	94.35	3.53	3.6	5
ViT-B/16	173	clean_high	88.97	88.31	89.87	7.64	3.6	4
ViT-B/16	173	clean_low	90.52	90.26	90.81	6.02	3.5	0
ViT-B/16	173	leak_high	95.94	95.97	96.03	2.33	3.6	3
ViT-B/16	173	leak_low	95.16	95.20	95.13	2.87	3.6	5
ViT-B/16	211	clean_high	96.15	96.09	96.13	2.76	3.6	4
ViT-B/16	211	clean_low	95.38	95.35	95.37	2.24	3.5	0
ViT-B/16	211	leak_high	97.30	97.19	97.19	1.57	3.6	3
ViT-B/16	211	leak_low	98.46	98.41	98.39	1.27	3.6	5
ViT-B/16	257	clean_high	93.88	93.54	93.82	4.53	3.6	4
ViT-B/16	257	clean_low	93.69	93.40	94.04	4.79	3.6	3
ViT-B/16	257	leak_high	94.84	94.62	94.61	3.77	3.6	0
ViT-B/16	257	leak_low	97.32	97.21	97.29	1.43	3.6	5
ViT-B/16	307	clean_high	95.90	95.73	95.76	3.03	3.6	4
ViT-B/16	307	clean_low	93.36	93.28	93.59	4.21	3.6	3
ViT-B/16	307	leak_high	95.90	95.83	95.85	2.28	3.6	0
ViT-B/16	307	leak_low	95.51	95.30	95.33	3.71	3.5	5
ViT-B/16	349	clean_high	92.29	92.04	92.11	5.83	3.6	4
ViT-B/16	349	clean_low	94.92	94.73	94.76	3.40	3.6	3
ViT-B/16	349	leak_high	95.11	94.96	95.08	3.71	3.6	0
ViT-B/16	349	leak_low	96.80	96.68	96.69	2.30	3.6	5
ViT-B/16	401	clean_high	93.62	93.37	93.43	4.64	3.6	4
ViT-B/16	401	clean_low	92.84	92.63	92.71	4.85	3.5	0
ViT-B/16	401	leak_high	95.16	94.99	95.03	3.55	3.6	3
ViT-B/16	401	leak_low	96.13	95.95	96.00	2.92	3.6	5
ViT-B/16	457	clean_high	95.61	95.44	95.30	3.26	3.6	4
ViT-B/16	457	clean_low	95.80	95.63	95.64	3.13	3.6	0
ViT-B/16	457	leak_high	94.88	94.62	94.43	3.69	3.6	3
ViT-B/16	457	leak_low	95.98	95.82	95.77	2.07	3.6	5
ViT-B/16	503	clean_high	93.21	92.98	93.05	3.40	3.6	4
ViT-B/16	503	clean_low	92.26	91.88	92.19	4.57	3.6	3
ViT-B/16	503	leak_high	96.98	96.90	96.89	1.42	3.5	0
ViT-B/16	503	leak_low	95.47	95.44	95.46	3.08	3.6	5
ViT-B/16	557	clean_high	95.09	94.93	94.97	2.98	3.6	4
ViT-B/16	557	clean_low	96.00	95.84	95.94	2.81	3.6	3
ViT-B/16	557	leak_high	95.64	95.49	95.60	3.09	3.6	0
ViT-B/16	557	leak_low	94.55	94.45	94.47	4.34	3.6	5
ConvNeXt-T	137	clean_high	96.93	96.79	96.96	2.59	2.1	4
ConvNeXt-T	137	clean_low	94.64	94.39	94.54	3.67	2.1	0
ConvNeXt-T	137	leak_high	97.13	96.85	96.91	2.27	2.1	3
ConvNeXt-T	137	leak_low	95.98	95.84	95.94	2.26	2.1	5
ConvNeXt-T	173	clean_high	95.36	95.33	95.26	2.70	2.1	4
ConvNeXt-T	173	clean_low	94.20	94.24	94.02	3.54	2.1	0
ConvNeXt-T	173	leak_high	95.36	95.41	95.34	2.67	2.1	3
ConvNeXt-T	173	leak_low	95.16	95.20	95.41	3.65	2.1	5
ConvNeXt-T	211	clean_high	96.34	96.27	96.32	3.02	2.1	4
ConvNeXt-T	211	clean_low	97.88	97.80	97.79	1.68	2.1	0
ConvNeXt-T	211	leak_high	99.04	99.02	99.01	1.35	2.1	3
ConvNeXt-T	211	leak_low	95.95	95.84	95.90	2.32	2.1	5
ConvNeXt-T	257	clean_high	94.07	93.87	94.34	4.28	2.2	4
ConvNeXt-T	257	clean_low	94.26	94.16	94.18	3.09	2.1	3
ConvNeXt-T	257	leak_high	95.98	95.81	96.22	2.88	2.1	0
ConvNeXt-T	257	leak_low	97.13	97.06	96.91	2.09	2.1	5
ConvNeXt-T	307	clean_high	97.07	96.93	96.89	2.07	2.1	4
ConvNeXt-T	307	clean_low	96.68	96.60	96.69	2.07	2.1	3
ConvNeXt-T	307	leak_high	98.63	98.57	98.57	1.02	2.0	0
ConvNeXt-T	307	leak_low	98.24	98.14	98.09	1.53	2.0	5
ConvNeXt-T	349	clean_high	93.98	93.74	93.86	4.28	2.1	4
ConvNeXt-T	349	clean_low	96.05	95.97	96.07	3.56	2.1	3
ConvNeXt-T	349	leak_high	95.30	95.14	95.21	2.45	2.1	0
ConvNeXt-T	349	leak_low	95.11	94.97	94.93	2.78	2.1	5
ConvNeXt-T	401	clean_high	93.42	93.20	93.20	4.60	2.1	4
ConvNeXt-T	401	clean_low	93.62	93.45	93.48	5.80	2.1	0
ConvNeXt-T	401	leak_high	95.74	95.52	95.61	3.01	2.1	3
ConvNeXt-T	401	leak_low	93.62	93.39	93.28	4.99	2.1	5
ConvNeXt-T	457	clean_high	95.80	95.61	95.62	2.98	2.1	4

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	457	clean_low	91.77	91.61	91.28	5.48	2.1	0
ConvNeXt-T	457	leak_high	95.98	95.79	95.68	2.14	2.1	3
ConvNeXt-T	457	leak_low	96.53	96.34	96.42	2.14	2.1	5
ConvNeXt-T	503	clean_high	95.09	94.96	94.96	2.85	2.1	4
ConvNeXt-T	503	clean_low	95.66	95.54	95.64	2.74	2.1	3
ConvNeXt-T	503	leak_high	95.66	95.52	95.53	2.98	2.1	0
ConvNeXt-T	503	leak_low	96.23	96.13	96.25	2.83	2.2	5
ConvNeXt-T	557	clean_high	94.00	93.82	93.66	4.62	2.1	4
ConvNeXt-T	557	clean_low	94.73	94.59	94.53	4.14	2.1	3
ConvNeXt-T	557	leak_high	94.00	93.62	93.45	4.78	2.1	0
ConvNeXt-T	557	leak_low	96.00	95.88	95.90	3.00	2.1	5

Table L.4: Complete 2A gradient-reversal run ledger (280 runs).

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	137	P1	0	97.65	97.66	97.68	1.05	2.7	0
ResNet-50	137	P2	0	95.19	95.24	95.18	2.87	2.6	4
ResNet-50	173	P1	0	97.78	97.80	97.79	1.03	2.6	0
ResNet-50	173	P2	0	95.93	95.97	95.96	2.05	2.6	4
ResNet-50	211	P1	0	97.41	97.42	97.42	1.47	2.6	5
ResNet-50	211	P2	0	96.30	96.32	96.27	2.20	2.6	4
ResNet-50	257	P1	0	97.41	97.42	97.44	1.59	2.6	0
ResNet-50	257	P2	0	96.67	96.68	96.66	2.07	2.6	3
ResNet-50	307	P1	0	98.27	98.28	98.28	1.29	2.6	5
ResNet-50	307	P2	0	97.28	97.28	97.29	1.75	2.6	4
ResNet-50	349	P1	0	96.67	96.69	96.67	2.11	2.6	0
ResNet-50	349	P2	0	96.42	96.45	96.42	2.23	2.6	3
ResNet-50	401	P1	0	98.02	98.03	98.03	1.38	2.6	5
ResNet-50	401	P2	0	97.41	97.42	97.43	1.32	2.6	4
ResNet-50	457	P1	0	98.15	98.14	98.18	1.18	2.6	0
ResNet-50	457	P2	0	94.32	94.31	94.35	3.88	2.6	3
ResNet-50	503	P1	0	97.53	97.54	97.56	1.35	2.6	5
ResNet-50	503	P2	0	96.79	96.79	96.78	2.09	2.6	4
ResNet-50	557	P1	0	97.53	97.53	97.55	1.29	2.6	0
ResNet-50	557	P2	0	98.40	98.40	98.40	1.16	2.6	3
ResNet-50	137	P1	0.01	98.02	98.03	98.05	1.04	2.6	5
ResNet-50	137	P2	0.01	96.54	96.54	96.57	1.88	2.6	3
ResNet-50	173	P1	0.01	96.67	96.67	96.70	2.00	2.6	0
ResNet-50	173	P2	0.01	96.42	96.46	96.44	1.72	2.6	5
ResNet-50	211	P1	0.01	97.53	97.55	97.54	1.36	2.6	5
ResNet-50	211	P2	0.01	96.54	96.57	96.54	2.11	2.6	4
ResNet-50	257	P1	0.01	97.53	97.52	97.56	1.65	2.6	0
ResNet-50	257	P2	0.01	98.02	98.02	98.03	1.21	2.6	3
ResNet-50	307	P1	0.01	96.79	96.81	96.78	1.26	2.6	5
ResNet-50	307	P2	0.01	97.53	97.53	97.53	1.19	2.6	4
ResNet-50	349	P1	0.01	97.65	97.67	97.67	1.17	2.6	0
ResNet-50	349	P2	0.01	97.41	97.42	97.42	1.15	2.6	3
ResNet-50	401	P1	0.01	97.04	97.05	97.07	1.61	2.6	5
ResNet-50	401	P2	0.01	97.28	97.29	97.32	1.80	2.6	4
ResNet-50	457	P1	0.01	98.52	98.52	98.53	1.04	2.6	0
ResNet-50	457	P2	0.01	95.56	95.51	95.59	3.58	2.6	3
ResNet-50	503	P1	0.01	97.28	97.29	97.30	1.62	2.6	5
ResNet-50	503	P2	0.01	96.54	96.56	96.53	2.40	2.6	4
ResNet-50	557	P1	0.01	97.16	97.17	97.16	2.31	2.6	0
ResNet-50	557	P2	0.01	96.79	96.79	96.81	2.13	2.6	3
ResNet-50	137	P1	0.03	96.05	96.05	96.09	2.72	2.7	0
ResNet-50	137	P2	0.03	95.56	95.55	95.57	2.81	2.6	4
ResNet-50	173	P1	0.03	97.04	97.04	97.05	1.39	2.7	3
ResNet-50	173	P2	0.03	97.41	97.43	97.42	2.14	2.7	4
ResNet-50	211	P1	0.03	97.41	97.44	97.41	1.61	2.6	0
ResNet-50	211	P2	0.03	95.56	95.60	95.54	2.77	2.6	3
ResNet-50	257	P1	0.03	97.28	97.29	97.30	1.77	2.6	5
ResNet-50	257	P2	0.03	96.42	96.42	96.41	1.96	2.6	4
ResNet-50	307	P1	0.03	95.68	95.74	95.69	2.32	2.6	0
ResNet-50	307	P2	0.03	97.28	97.28	97.28	1.08	2.6	3
ResNet-50	349	P1	0.03	97.41	97.42	97.41	1.35	2.6	5
ResNet-50	349	P2	0.03	96.79	96.82	96.81	1.79	2.6	4
ResNet-50	401	P1	0.03	95.68	95.73	95.68	2.35	2.6	0
ResNet-50	401	P2	0.03	97.65	97.66	97.65	1.61	2.6	3

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	457	P1	0.03	97.53	97.54	97.54	1.76	2.6	5
ResNet-50	457	P2	0.03	95.43	95.41	95.47	3.32	2.6	4
ResNet-50	503	P1	0.03	98.02	98.05	98.03	1.44	2.6	0
ResNet-50	503	P2	0.03	97.16	97.18	97.16	1.83	2.6	3
ResNet-50	557	P1	0.03	97.04	97.06	97.06	2.21	2.6	5
ResNet-50	557	P2	0.03	97.04	97.04	97.06	1.77	2.6	4
ResNet-50	137	P1	0.1	96.91	96.92	96.94	1.87	2.6	5
ResNet-50	137	P2	0.1	94.81	94.83	94.83	3.44	2.6	3
ResNet-50	173	P1	0.1	96.67	96.68	96.68	1.66	2.6	0
ResNet-50	173	P2	0.1	96.67	96.68	96.68	1.61	2.6	5
ResNet-50	211	P1	0.1	96.91	96.91	96.93	1.30	2.6	5
ResNet-50	211	P2	0.1	96.54	96.59	96.53	2.27	2.6	4
ResNet-50	257	P1	0.1	96.67	96.65	96.69	2.06	2.6	0
ResNet-50	257	P2	0.1	97.04	97.04	97.04	2.03	2.6	3
ResNet-50	307	P1	0.1	97.28	97.28	97.30	1.35	2.6	5
ResNet-50	307	P2	0.1	96.67	96.67	96.65	2.19	2.6	4
ResNet-50	349	P1	0.1	97.04	97.05	97.05	1.46	2.6	0
ResNet-50	349	P2	0.1	97.04	97.07	97.03	1.37	2.6	3
ResNet-50	401	P1	0.1	96.91	96.93	96.94	1.71	2.6	5
ResNet-50	401	P2	0.1	97.16	97.18	97.17	2.14	2.6	4
ResNet-50	457	P1	0.1	97.90	97.91	97.90	1.06	2.6	0
ResNet-50	457	P2	0.1	94.32	94.26	94.36	4.31	2.6	3
ResNet-50	503	P1	0.1	96.67	96.71	96.66	2.24	2.6	5
ResNet-50	503	P2	0.1	96.79	96.81	96.81	1.76	2.6	4
ResNet-50	557	P1	0.1	96.79	96.80	96.80	1.43	2.6	0
ResNet-50	557	P2	0.1	97.04	97.04	97.06	1.73	2.6	3
ResNet-50	137	P1	0.3	97.41	97.40	97.44	1.17	2.6	0
ResNet-50	137	P2	0.3	95.56	95.57	95.57	2.59	2.6	4
ResNet-50	173	P1	0.3	95.68	95.69	95.69	1.80	2.6	3
ResNet-50	173	P2	0.3	97.04	97.04	97.05	0.85	2.7	4
ResNet-50	211	P1	0.3	96.79	96.80	96.82	1.45	2.6	0
ResNet-50	211	P2	0.3	96.42	96.45	96.41	2.13	2.6	3
ResNet-50	257	P1	0.3	97.41	97.40	97.43	1.68	2.6	5
ResNet-50	257	P2	0.3	98.02	98.03	98.04	1.65	2.6	4
ResNet-50	307	P1	0.3	98.02	98.03	98.04	1.22	2.6	0
ResNet-50	307	P2	0.3	96.67	96.67	96.65	1.36	2.6	3
ResNet-50	349	P1	0.3	95.93	95.95	95.92	1.88	2.6	5
ResNet-50	349	P2	0.3	97.65	97.67	97.66	0.96	2.6	4
ResNet-50	401	P1	0.3	96.79	96.80	96.81	1.33	2.6	0
ResNet-50	401	P2	0.3	95.93	95.91	95.97	2.46	2.6	3
ResNet-50	457	P1	0.3	98.02	98.03	98.05	1.42	2.6	5
ResNet-50	457	P2	0.3	95.06	95.04	95.10	4.16	2.6	4
ResNet-50	503	P1	0.3	97.04	97.05	97.04	1.00	2.6	0
ResNet-50	503	P2	0.3	96.05	96.05	96.01	2.57	2.6	3
ResNet-50	557	P1	0.3	97.04	97.06	97.04	1.47	2.6	5
ResNet-50	557	P2	0.3	97.04	97.05	97.05	1.52	2.6	4
ResNet-50	137	P1	1	96.67	96.68	96.70	1.17	2.6	5
ResNet-50	137	P2	1	95.56	95.57	95.58	1.61	2.7	3
ResNet-50	173	P1	1	97.16	97.16	97.19	1.38	2.6	0
ResNet-50	173	P2	1	97.41	97.41	97.44	1.54	2.6	3
ResNet-50	211	P1	1	96.79	96.82	96.80	1.64	2.6	5
ResNet-50	211	P2	1	95.56	95.60	95.55	1.84	2.6	4
ResNet-50	257	P1	1	96.54	96.54	96.56	1.96	2.6	0
ResNet-50	257	P2	1	96.42	96.43	96.43	1.27	2.6	3
ResNet-50	307	P1	1	95.68	95.73	95.69	1.70	2.6	5
ResNet-50	307	P2	1	97.65	97.65	97.65	0.85	2.6	4
ResNet-50	349	P1	1	96.91	96.92	96.91	1.44	2.6	0
ResNet-50	349	P2	1	95.80	95.84	95.80	1.87	2.6	3
ResNet-50	401	P1	1	95.93	95.92	95.95	2.13	2.6	5
ResNet-50	401	P2	1	96.67	96.68	96.66	1.38	2.6	4
ResNet-50	457	P1	1	98.40	98.40	98.41	1.28	2.6	0
ResNet-50	457	P2	1	94.94	94.91	94.97	2.62	2.6	3
ResNet-50	503	P1	1	97.41	97.42	97.42	1.21	2.6	5
ResNet-50	503	P2	1	97.41	97.42	97.42	0.77	2.6	4
ResNet-50	557	P1	1	96.79	96.80	96.82	1.60	2.6	0
ResNet-50	557	P2	1	97.65	97.66	97.67	0.83	2.6	3
ResNet-50	137	P1	3	96.30	96.28	96.34	1.67	2.6	0
ResNet-50	137	P2	3	95.80	95.81	95.82	1.13	2.7	4
ResNet-50	173	P1	3	95.93	95.93	95.91	1.23	2.7	5
ResNet-50	173	P2	3	96.17	96.16	96.20	1.32	2.6	4
ResNet-50	211	P1	3	97.41	97.42	97.43	1.13	2.6	0
ResNet-50	211	P2	3	96.17	96.21	96.15	1.04	2.6	3
ResNet-50	257	P1	3	97.04	97.05	97.05	1.23	2.6	5

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	257	P2	3	97.78	97.78	97.79	1.67	2.6	4
ResNet-50	307	P1	3	97.28	97.29	97.30	1.37	2.6	0
ResNet-50	307	P2	3	96.54	96.54	96.54	2.14	2.6	3
ResNet-50	349	P1	3	95.80	95.80	95.80	2.80	2.6	5
ResNet-50	349	P2	3	94.44	94.42	94.50	1.90	2.6	4
ResNet-50	401	P1	3	95.56	95.59	95.57	1.71	2.6	0
ResNet-50	401	P2	3	96.79	96.80	96.81	0.77	2.6	3
ResNet-50	457	P1	3	97.41	97.40	97.43	1.53	2.6	5
ResNet-50	457	P2	3	93.58	93.52	93.59	3.44	2.6	4
ResNet-50	503	P1	3	96.67	96.67	96.69	1.33	2.6	0
ResNet-50	503	P2	3	95.56	95.58	95.54	1.20	2.6	3
ResNet-50	557	P1	3	97.41	97.43	97.42	1.75	2.6	5
ResNet-50	557	P2	3	96.91	96.91	96.93	1.72	2.6	4
ConvNeXt-T	137	P1	0	98.15	98.16	98.16	1.55	5.2	5
ConvNeXt-T	137	P2	0	96.79	96.79	96.81	2.44	5.2	3
ConvNeXt-T	173	P1	0	97.90	97.90	97.89	1.43	5.1	0
ConvNeXt-T	173	P2	0	96.17	96.19	96.17	2.90	5.2	3
ConvNeXt-T	211	P1	0	97.90	97.90	97.91	1.64	5.2	5
ConvNeXt-T	211	P2	0	96.54	96.56	96.54	2.68	5.2	4
ConvNeXt-T	257	P1	0	97.90	97.90	97.91	1.83	5.2	0
ConvNeXt-T	257	P2	0	97.16	97.16	97.16	2.07	5.2	3
ConvNeXt-T	307	P1	0	98.27	98.28	98.28	1.22	5.2	5
ConvNeXt-T	307	P2	0	96.67	96.68	96.66	2.58	5.2	4
ConvNeXt-T	349	P1	0	97.78	97.77	97.80	1.83	5.2	0
ConvNeXt-T	349	P2	0	93.83	93.89	93.83	5.03	5.2	3
ConvNeXt-T	401	P1	0	97.53	97.55	97.56	1.51	5.2	5
ConvNeXt-T	401	P2	0	98.40	98.40	98.41	1.20	5.2	4
ConvNeXt-T	457	P1	0	98.64	98.65	98.66	1.16	5.2	0
ConvNeXt-T	457	P2	0	95.56	95.55	95.57	3.12	5.2	3
ConvNeXt-T	503	P1	0	97.28	97.30	97.31	1.90	5.2	5
ConvNeXt-T	503	P2	0	97.28	97.28	97.29	1.58	5.2	4
ConvNeXt-T	557	P1	0	96.67	96.70	96.67	2.26	5.2	0
ConvNeXt-T	557	P2	0	98.15	98.16	98.14	1.33	5.2	3
ConvNeXt-T	137	P1	0.01	98.02	98.04	98.04	1.83	5.2	0
ConvNeXt-T	137	P2	0.01	96.79	96.79	96.83	2.16	5.2	4
ConvNeXt-T	173	P1	0.01	97.53	97.54	97.53	1.75	5.2	5
ConvNeXt-T	173	P2	0.01	97.41	97.41	97.42	1.88	5.2	4
ConvNeXt-T	211	P1	0.01	96.91	96.92	96.89	2.16	5.2	0
ConvNeXt-T	211	P2	0.01	96.05	96.07	96.06	3.30	5.2	3
ConvNeXt-T	257	P1	0.01	97.16	97.14	97.20	2.04	5.2	5
ConvNeXt-T	257	P2	0.01	97.53	97.51	97.56	1.75	5.2	4
ConvNeXt-T	307	P1	0.01	97.78	97.77	97.81	1.74	5.1	0
ConvNeXt-T	307	P2	0.01	97.53	97.53	97.53	1.74	5.2	3
ConvNeXt-T	349	P1	0.01	96.42	96.43	96.40	2.58	5.2	5
ConvNeXt-T	349	P2	0.01	97.28	97.30	97.29	1.63	5.2	4
ConvNeXt-T	401	P1	0.01	97.53	97.54	97.56	1.83	5.2	0
ConvNeXt-T	401	P2	0.01	96.67	96.66	96.73	2.77	5.2	3
ConvNeXt-T	457	P1	0.01	98.77	98.77	98.78	1.17	5.2	5
ConvNeXt-T	457	P2	0.01	95.31	95.28	95.35	4.46	5.2	4
ConvNeXt-T	503	P1	0.01	97.65	97.66	97.66	1.85	5.2	0
ConvNeXt-T	503	P2	0.01	97.16	97.17	97.19	2.57	5.2	3
ConvNeXt-T	557	P1	0.01	95.31	95.32	95.35	3.35	5.2	5
ConvNeXt-T	557	P2	0.01	96.42	96.43	96.44	2.67	5.2	4
ConvNeXt-T	137	P1	0.03	95.93	95.99	95.99	3.16	5.2	5
ConvNeXt-T	137	P2	0.03	97.04	97.05	97.05	2.52	5.2	3
ConvNeXt-T	173	P1	0.03	97.65	97.66	97.68	2.19	5.2	0
ConvNeXt-T	173	P2	0.03	97.65	97.66	97.66	1.29	5.2	3
ConvNeXt-T	211	P1	0.03	98.15	98.15	98.15	1.43	5.2	5
ConvNeXt-T	211	P2	0.03	96.42	96.45	96.43	2.82	5.2	4
ConvNeXt-T	257	P1	0.03	97.04	97.05	97.06	2.34	5.2	0
ConvNeXt-T	257	P2	0.03	97.16	97.16	97.17	2.35	5.2	3
ConvNeXt-T	307	P1	0.03	97.28	97.29	97.30	1.92	5.2	5
ConvNeXt-T	307	P2	0.03	96.17	96.17	96.17	2.80	5.2	4
ConvNeXt-T	349	P1	0.03	97.16	97.16	97.17	1.70	5.2	0
ConvNeXt-T	349	P2	0.03	96.42	96.42	96.45	3.01	5.2	3
ConvNeXt-T	401	P1	0.03	97.78	97.78	97.78	1.75	5.2	5
ConvNeXt-T	401	P2	0.03	97.90	97.92	97.88	1.09	5.2	4
ConvNeXt-T	457	P1	0.03	97.16	97.15	97.13	1.89	5.2	0
ConvNeXt-T	457	P2	0.03	93.58	93.53	93.56	5.14	5.2	3
ConvNeXt-T	503	P1	0.03	94.69	94.79	94.69	4.44	5.2	5
ConvNeXt-T	503	P2	0.03	98.15	98.16	98.15	1.43	5.2	4
ConvNeXt-T	557	P1	0.03	96.91	96.92	96.93	2.05	5.2	0
ConvNeXt-T	557	P2	0.03	97.28	97.29	97.30	2.22	5.2	3

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	137	P1	0.1	97.28	97.30	97.30	1.50	5.2	0
ConvNeXt-T	137	P2	0.1	96.42	96.42	96.44	2.67	5.2	4
ConvNeXt-T	173	P1	0.1	97.16	97.17	97.16	2.28	5.2	5
ConvNeXt-T	173	P2	0.1	97.78	97.77	97.76	1.66	5.2	4
ConvNeXt-T	211	P1	0.1	98.40	98.40	98.40	1.15	5.1	0
ConvNeXt-T	211	P2	0.1	96.67	96.69	96.66	2.69	5.2	3
ConvNeXt-T	257	P1	0.1	96.91	96.93	96.91	2.05	5.2	5
ConvNeXt-T	257	P2	0.1	97.16	97.16	97.18	2.16	5.2	4
ConvNeXt-T	307	P1	0.1	96.91	96.92	96.95	2.30	5.2	0
ConvNeXt-T	307	P2	0.1	96.91	96.90	96.92	2.59	5.2	3
ConvNeXt-T	349	P1	0.1	95.80	95.79	95.79	2.88	5.2	5
ConvNeXt-T	349	P2	0.1	96.17	96.22	96.18	2.83	5.2	4
ConvNeXt-T	401	P1	0.1	97.90	97.91	97.93	1.52	5.2	0
ConvNeXt-T	401	P2	0.1	97.28	97.29	97.30	1.37	5.2	3
ConvNeXt-T	457	P1	0.1	98.27	98.27	98.27	1.26	5.2	5
ConvNeXt-T	457	P2	0.1	95.43	95.42	95.41	3.24	5.2	4
ConvNeXt-T	503	P1	0.1	96.79	96.78	96.80	2.37	5.2	0
ConvNeXt-T	503	P2	0.1	97.53	97.55	97.54	1.90	5.2	3
ConvNeXt-T	557	P1	0.1	96.91	96.93	96.93	2.45	5.2	5
ConvNeXt-T	557	P2	0.1	98.15	98.16	98.15	1.06	5.2	4
ConvNeXt-T	137	P1	0.3	96.42	96.38	96.47	2.63	5.2	5
ConvNeXt-T	137	P2	0.3	96.91	96.94	96.91	2.29	5.2	3
ConvNeXt-T	173	P1	0.3	98.15	98.16	98.16	1.20	5.2	0
ConvNeXt-T	173	P2	0.3	97.90	97.90	97.90	1.52	5.2	3
ConvNeXt-T	211	P1	0.3	97.78	97.79	97.78	1.68	5.2	5
ConvNeXt-T	211	P2	0.3	97.04	97.07	97.02	2.67	5.2	4
ConvNeXt-T	257	P1	0.3	97.28	97.28	97.31	1.96	5.2	0
ConvNeXt-T	257	P2	0.3	97.04	97.03	97.06	2.62	5.2	3
ConvNeXt-T	307	P1	0.3	95.06	95.09	95.09	2.90	5.2	5
ConvNeXt-T	307	P2	0.3	96.79	96.79	96.78	2.18	5.2	4
ConvNeXt-T	349	P1	0.3	97.53	97.54	97.55	1.78	5.2	0
ConvNeXt-T	349	P2	0.3	98.02	98.03	98.03	1.64	5.2	3
ConvNeXt-T	401	P1	0.3	98.52	98.53	98.52	1.15	5.2	5
ConvNeXt-T	401	P2	0.3	98.77	98.77	98.77	0.67	5.2	4
ConvNeXt-T	457	P1	0.3	98.15	98.17	98.16	1.51	5.2	0
ConvNeXt-T	457	P2	0.3	94.94	94.92	94.93	3.74	5.2	3
ConvNeXt-T	503	P1	0.3	96.91	96.96	96.94	2.06	5.2	5
ConvNeXt-T	503	P2	0.3	97.28	97.32	97.28	1.71	5.2	4
ConvNeXt-T	557	P1	0.3	97.65	97.66	97.67	1.32	5.2	0
ConvNeXt-T	557	P2	0.3	98.15	98.18	98.15	0.96	5.2	3
ConvNeXt-T	137	P1	1	97.53	97.52	97.57	1.28	5.2	0
ConvNeXt-T	137	P2	1	96.42	96.42	96.44	1.91	5.2	4
ConvNeXt-T	173	P1	1	98.15	98.16	98.15	0.68	5.2	5
ConvNeXt-T	173	P2	1	96.67	96.69	96.67	2.28	5.2	4
ConvNeXt-T	211	P1	1	98.64	98.65	98.65	0.83	5.2	0
ConvNeXt-T	211	P2	1	96.30	96.31	96.31	2.02	5.2	3
ConvNeXt-T	257	P1	1	96.79	96.82	96.79	2.02	5.2	5
ConvNeXt-T	257	P2	1	98.02	98.03	98.02	0.70	5.2	4
ConvNeXt-T	307	P1	1	98.52	98.53	98.54	0.45	5.2	0
ConvNeXt-T	307	P2	1	97.53	97.53	97.53	1.22	5.2	3
ConvNeXt-T	349	P1	1	97.65	97.67	97.64	1.06	5.2	5
ConvNeXt-T	349	P2	1	97.53	97.55	97.53	0.97	5.2	4
ConvNeXt-T	401	P1	1	98.27	98.28	98.28	0.47	5.2	0
ConvNeXt-T	401	P2	1	97.41	97.44	97.38	1.49	5.2	3
ConvNeXt-T	457	P1	1	98.89	98.89	98.91	0.60	5.2	5
ConvNeXt-T	457	P2	1	96.17	96.18	96.19	2.62	5.2	4
ConvNeXt-T	503	P1	1	97.16	97.20	97.17	1.23	5.2	0
ConvNeXt-T	503	P2	1	97.78	97.78	97.79	0.98	5.2	3
ConvNeXt-T	557	P1	1	97.53	97.52	97.56	0.99	5.2	5
ConvNeXt-T	557	P2	1	98.64	98.65	98.65	0.64	5.2	4
ConvNeXt-T	137	P1	3	97.41	97.42	97.44	1.21	5.2	5
ConvNeXt-T	137	P2	3	96.42	96.44	96.42	1.07	5.2	3
ConvNeXt-T	173	P1	3	98.27	98.28	98.27	1.97	5.2	0
ConvNeXt-T	173	P2	3	96.91	96.93	96.92	1.60	5.2	3
ConvNeXt-T	211	P1	3	97.90	97.90	97.92	1.24	5.2	5
ConvNeXt-T	211	P2	3	97.78	97.81	97.78	1.20	5.2	4
ConvNeXt-T	257	P1	3	97.53	97.53	97.55	1.34	5.2	0
ConvNeXt-T	257	P2	3	97.53	97.53	97.53	0.73	5.2	3
ConvNeXt-T	307	P1	3	98.77	98.77	98.78	2.14	5.2	5
ConvNeXt-T	307	P2	3	97.16	97.17	97.19	0.75	5.2	4
ConvNeXt-T	349	P1	3	98.27	98.28	98.28	1.46	5.2	0
ConvNeXt-T	349	P2	3	97.04	97.07	97.04	0.64	5.2	3
ConvNeXt-T	401	P1	3	97.90	97.92	97.92	1.69	5.2	5

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	401	P2	3	97.28	97.32	97.28	0.91	5.2	4
ConvNeXt-T	457	P1	3	98.77	98.78	98.78	1.34	5.2	0
ConvNeXt-T	457	P2	3	96.17	96.16	96.21	2.00	5.2	3
ConvNeXt-T	503	P1	3	97.65	97.69	97.66	1.02	5.2	5
ConvNeXt-T	503	P2	3	98.02	98.02	98.02	1.26	5.2	4
ConvNeXt-T	557	P1	3	98.02	98.03	98.04	1.93	5.2	0
ConvNeXt-T	557	P2	3	98.15	98.15	98.16	1.55	5.2	3

M

Amendment 8 run ledgers

Table M.1: Complete Amendment 8 carrier-factorial run ledger (480 runs).

Architecture	Seed	Protocol	condition	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	1103	clean_high	colour	93.05	92.62	92.52	4.32	0.9	7
ResNet-50	1103	clean_low	colour	92.34	91.92	91.82	3.36	0.9	5
ResNet-50	1103	leak_high	colour	94.12	93.66	93.41	3.28	1.0	6
ResNet-50	1103	leak_low	colour	95.37	94.95	94.89	2.38	0.9	4
ResNet-50	1151	clean_high	colour	94.11	94.05	93.81	3.19	0.9	4
ResNet-50	1151	clean_low	colour	94.82	94.83	94.82	2.07	0.9	6
ResNet-50	1151	leak_high	colour	94.82	94.77	94.70	1.95	0.9	6
ResNet-50	1151	leak_low	colour	95.54	95.53	95.57	2.17	0.9	5
ResNet-50	1201	clean_high	colour	96.33	96.09	96.00	1.98	0.9	4
ResNet-50	1201	clean_low	colour	95.52	95.17	95.08	2.22	0.9	6
ResNet-50	1201	leak_high	colour	97.35	97.20	97.24	1.11	0.9	7
ResNet-50	1201	leak_low	colour	96.33	96.14	96.10	1.36	0.9	5
ResNet-50	1249	clean_high	colour	94.28	94.29	94.48	3.32	1.0	4
ResNet-50	1249	clean_low	colour	91.85	91.90	92.03	4.43	1.0	5
ResNet-50	1249	leak_high	colour	95.15	95.18	95.25	2.76	1.0	7
ResNet-50	1249	leak_low	colour	97.40	97.43	97.35	2.02	0.9	6
ResNet-50	1301	clean_high	colour	93.37	93.40	94.08	2.74	0.9	4
ResNet-50	1301	clean_low	colour	96.02	96.12	96.22	2.61	0.9	5
ResNet-50	1301	leak_high	colour	96.97	96.99	97.33	2.42	0.9	7
ResNet-50	1301	leak_low	colour	95.83	95.76	96.16	1.89	0.9	6
ResNet-50	1361	clean_high	colour	94.39	94.17	94.20	3.99	0.9	6
ResNet-50	1361	clean_low	colour	94.21	94.02	94.12	3.44	0.9	5
ResNet-50	1361	leak_high	colour	95.84	95.62	95.78	3.57	0.9	7
ResNet-50	1361	leak_low	colour	95.84	95.69	95.75	2.86	0.9	4
ResNet-50	1409	clean_high	colour	94.16	93.70	93.68	3.63	0.9	6
ResNet-50	1409	clean_low	colour	95.53	95.21	95.33	2.32	0.9	5
ResNet-50	1409	leak_high	colour	97.08	96.87	96.92	1.38	0.9	7
ResNet-50	1409	leak_low	colour	97.67	97.46	97.43	2.02	0.9	4
ResNet-50	1451	clean_high	colour	93.47	93.30	93.14	2.72	0.9	6
ResNet-50	1451	clean_low	colour	93.65	93.50	93.43	3.17	0.9	5
ResNet-50	1451	leak_high	colour	95.83	95.72	95.52	2.50	0.9	7
ResNet-50	1451	leak_low	colour	95.10	94.88	94.68	1.79	1.0	4
ResNet-50	1499	clean_high	colour	93.48	93.14	93.18	4.25	0.9	6
ResNet-50	1499	clean_low	colour	94.47	94.17	94.27	4.36	0.9	5
ResNet-50	1499	leak_high	colour	95.65	95.41	95.44	2.46	0.9	7
ResNet-50	1499	leak_low	colour	94.47	94.26	94.43	3.29	0.9	4
ResNet-50	1553	clean_high	colour	91.22	91.03	91.02	5.51	0.9	7
ResNet-50	1553	clean_low	colour	92.61	92.57	92.35	4.07	0.9	5
ResNet-50	1553	leak_high	colour	93.41	93.24	93.34	2.58	0.9	6
ResNet-50	1553	leak_low	colour	94.81	94.67	94.72	2.76	0.9	4
ResNet-50	1103	clean_high	masked	93.40	93.08	92.99	4.98	1.0	7
ResNet-50	1103	clean_low	masked	93.58	93.14	93.00	3.34	1.0	5
ResNet-50	1103	leak_high	masked	94.12	93.84	93.66	2.71	1.0	6
ResNet-50	1103	leak_low	masked	95.72	95.50	95.34	2.14	1.0	4
ResNet-50	1151	clean_high	masked	93.93	93.93	93.69	4.80	1.0	4
ResNet-50	1151	clean_low	masked	93.93	93.91	93.73	3.32	1.0	7
ResNet-50	1151	leak_high	masked	95.18	95.21	95.10	2.50	1.0	6

Architecture	Seed	Protocol	condition	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	1151	leak_low	masked	96.79	96.79	96.84	1.28	1.0	5
ResNet-50	1201	clean_high	masked	96.33	96.12	96.08	1.54	1.0	4
ResNet-50	1201	clean_low	masked	94.70	94.44	94.37	1.81	1.0	6
ResNet-50	1201	leak_high	masked	97.96	97.82	97.87	1.04	1.0	7
ResNet-50	1201	leak_low	masked	97.96	97.86	97.83	1.56	1.0	5
ResNet-50	1249	clean_high	masked	93.07	93.03	93.36	4.99	1.0	4
ResNet-50	1249	clean_low	masked	93.76	93.78	93.98	4.13	1.0	5
ResNet-50	1249	leak_high	masked	95.15	95.15	95.42	3.24	1.0	7
ResNet-50	1249	leak_low	masked	94.97	95.03	95.02	3.14	1.0	6
ResNet-50	1301	clean_high	masked	93.75	93.70	93.97	3.08	1.0	4
ResNet-50	1301	clean_low	masked	95.08	95.06	95.32	2.48	1.0	5
ResNet-50	1301	leak_high	masked	96.02	96.01	96.34	1.68	1.0	7
ResNet-50	1301	leak_low	masked	95.64	95.66	95.79	2.00	1.0	6
ResNet-50	1361	clean_high	masked	92.59	92.30	92.26	5.37	1.0	6
ResNet-50	1361	clean_low	masked	94.39	94.17	94.28	4.43	1.0	5
ResNet-50	1361	leak_high	masked	95.66	95.49	95.58	2.69	1.0	7
ResNet-50	1361	leak_low	masked	94.21	93.98	93.90	3.93	1.0	4
ResNet-50	1409	clean_high	masked	93.77	93.37	93.53	4.06	1.0	6
ResNet-50	1409	clean_low	masked	96.11	95.82	95.89	2.12	1.0	5
ResNet-50	1409	leak_high	masked	96.50	96.20	96.21	1.85	1.0	7
ResNet-50	1409	leak_low	masked	97.28	97.10	97.16	1.81	1.0	4
ResNet-50	1451	clean_high	masked	93.28	93.13	92.90	3.89	1.0	6
ResNet-50	1451	clean_low	masked	93.28	93.06	93.00	3.54	1.0	5
ResNet-50	1451	leak_high	masked	94.92	94.77	94.65	2.52	1.0	7
ResNet-50	1451	leak_low	masked	96.01	95.93	95.77	0.95	1.0	4
ResNet-50	1499	clean_high	masked	92.69	92.24	92.43	4.04	1.0	6
ResNet-50	1499	clean_low	masked	92.89	92.53	92.63	3.78	1.0	5
ResNet-50	1499	leak_high	masked	95.26	95.11	95.19	2.99	1.0	7
ResNet-50	1499	leak_low	masked	94.47	94.08	94.23	2.83	1.0	4
ResNet-50	1553	clean_high	masked	91.42	91.41	91.34	4.91	1.0	7
ResNet-50	1553	clean_low	masked	91.02	90.87	90.70	4.64	1.0	5
ResNet-50	1553	leak_high	masked	93.81	93.55	93.59	3.65	1.0	6
ResNet-50	1553	leak_low	masked	93.81	93.70	93.46	3.88	1.0	4
ResNet-50	1103	clean_high	standard	94.47	94.06	94.08	3.14	1.0	7
ResNet-50	1103	clean_low	standard	92.69	92.08	91.80	2.96	1.0	5
ResNet-50	1103	leak_high	standard	95.72	95.40	95.32	2.31	1.0	6
ResNet-50	1103	leak_low	standard	95.01	94.64	94.57	2.61	1.0	4
ResNet-50	1151	clean_high	standard	94.46	94.43	94.18	3.08	1.0	4
ResNet-50	1151	clean_low	standard	93.93	93.98	93.72	2.55	1.0	5
ResNet-50	1151	leak_high	standard	96.43	96.43	96.34	1.55	1.0	6
ResNet-50	1151	leak_low	standard	96.07	96.08	95.99	1.83	1.0	7
ResNet-50	1201	clean_high	standard	95.11	94.76	94.77	2.43	1.0	4
ResNet-50	1201	clean_low	standard	95.93	95.73	95.67	1.60	1.0	6
ResNet-50	1201	leak_high	standard	97.15	97.03	96.99	1.44	1.0	7
ResNet-50	1201	leak_low	standard	97.56	97.39	97.41	1.59	1.0	5
ResNet-50	1249	clean_high	standard	94.45	94.51	94.55	3.78	1.0	4
ResNet-50	1249	clean_low	standard	94.63	94.66	94.82	2.75	1.0	5
ResNet-50	1249	leak_high	standard	95.84	95.84	96.01	2.59	1.0	7
ResNet-50	1249	leak_low	standard	96.71	96.71	96.75	1.86	1.0	6
ResNet-50	1301	clean_high	standard	93.75	93.80	94.03	2.90	1.0	4
ResNet-50	1301	clean_low	standard	96.40	96.41	96.77	1.29	1.0	5
ResNet-50	1301	leak_high	standard	96.21	96.25	96.51	1.27	1.0	7
ResNet-50	1301	leak_low	standard	96.40	96.43	96.69	1.96	1.0	6
ResNet-50	1361	clean_high	standard	93.31	93.06	93.06	4.10	1.0	4
ResNet-50	1361	clean_low	standard	93.85	93.52	93.63	4.35	1.0	5
ResNet-50	1361	leak_high	standard	97.29	97.18	97.26	2.21	1.0	7
ResNet-50	1361	leak_low	standard	96.56	96.42	96.54	2.44	1.0	6
ResNet-50	1409	clean_high	standard	96.50	96.21	96.26	2.21	1.0	6
ResNet-50	1409	clean_low	standard	96.89	96.60	96.66	2.49	1.0	5
ResNet-50	1409	leak_high	standard	96.50	96.20	96.25	1.83	1.0	7
ResNet-50	1409	leak_low	standard	97.47	97.31	97.31	1.66	1.0	4
ResNet-50	1451	clean_high	standard	95.10	94.96	94.93	1.89	1.0	6
ResNet-50	1451	clean_low	standard	94.01	93.87	94.09	3.12	1.0	5
ResNet-50	1451	leak_high	standard	97.64	97.55	97.49	1.06	1.0	7
ResNet-50	1451	leak_low	standard	96.73	96.64	96.49	0.94	1.0	4
ResNet-50	1499	clean_high	standard	94.07	93.68	93.94	4.07	1.0	6
ResNet-50	1499	clean_low	standard	94.27	93.99	94.06	3.48	1.0	5
ResNet-50	1499	leak_high	standard	95.26	94.98	94.95	2.93	1.0	7
ResNet-50	1499	leak_low	standard	95.85	95.55	95.57	2.51	1.0	4
ResNet-50	1553	clean_high	standard	88.62	88.16	88.42	7.43	1.0	6
ResNet-50	1553	clean_low	standard	92.22	92.04	91.85	5.06	1.0	5
ResNet-50	1553	leak_high	standard	92.61	92.41	92.47	4.13	1.0	7
ResNet-50	1553	leak_low	standard	95.61	95.56	95.74	3.26	1.0	4

Architecture	Seed	Protocol	condition	Acc.	F1	Bal.	ECE	Min	GPU
DenseNet-121	1103	clean_high	colour	94.12	93.69	93.58	2.82	1.2	7
DenseNet-121	1103	clean_low	colour	94.12	93.86	94.12	1.82	1.2	5
DenseNet-121	1103	leak_high	colour	95.72	95.40	95.34	1.49	1.2	6
DenseNet-121	1103	leak_low	colour	96.08	95.85	95.62	2.96	1.2	4
DenseNet-121	1151	clean_high	colour	92.86	92.80	92.73	3.15	1.2	4
DenseNet-121	1151	clean_low	colour	93.04	92.95	92.75	2.92	1.2	6
DenseNet-121	1151	leak_high	colour	95.00	95.01	94.93	1.53	1.2	7
DenseNet-121	1151	leak_low	colour	95.54	95.55	95.44	1.51	1.2	5
DenseNet-121	1201	clean_high	colour	96.95	96.76	96.73	1.40	1.2	4
DenseNet-121	1201	clean_low	colour	96.74	96.57	96.50	1.59	1.2	6
DenseNet-121	1201	leak_high	colour	97.15	96.97	97.03	0.83	1.2	7
DenseNet-121	1201	leak_low	colour	96.54	96.35	96.29	1.93	1.2	5
DenseNet-121	1249	clean_high	colour	95.32	95.32	95.46	2.19	1.2	4
DenseNet-121	1249	clean_low	colour	93.59	93.60	93.73	2.92	1.2	5
DenseNet-121	1249	leak_high	colour	96.53	96.58	96.65	2.52	1.2	7
DenseNet-121	1249	leak_low	colour	96.71	96.75	96.81	1.41	1.2	6
DenseNet-121	1301	clean_high	colour	95.27	95.29	95.30	2.15	1.2	4
DenseNet-121	1301	clean_low	colour	95.64	95.78	95.90	2.43	1.2	5
DenseNet-121	1301	leak_high	colour	97.54	97.54	97.79	1.30	1.2	7
DenseNet-121	1301	leak_low	colour	96.78	96.82	97.17	2.08	1.2	6
DenseNet-121	1361	clean_high	colour	94.58	94.30	94.42	2.98	1.2	6
DenseNet-121	1361	clean_low	colour	93.67	93.34	93.37	3.20	1.2	5
DenseNet-121	1361	leak_high	colour	95.30	95.17	95.11	2.48	1.2	7
DenseNet-121	1361	leak_low	colour	96.02	95.88	96.01	2.48	1.2	4
DenseNet-121	1409	clean_high	colour	95.91	95.54	95.56	2.69	1.2	6
DenseNet-121	1409	clean_low	colour	96.89	96.66	96.68	2.24	1.2	5
DenseNet-121	1409	leak_high	colour	97.28	97.06	97.16	1.52	1.2	7
DenseNet-121	1409	leak_low	colour	97.47	97.31	97.37	0.82	1.2	4
DenseNet-121	1451	clean_high	colour	94.37	94.18	93.92	1.89	1.2	6
DenseNet-121	1451	clean_low	colour	95.64	95.51	95.49	2.24	1.2	5
DenseNet-121	1451	leak_high	colour	96.19	96.04	95.85	1.78	1.2	7
DenseNet-121	1451	leak_low	colour	97.10	96.94	96.84	1.63	1.2	4
DenseNet-121	1499	clean_high	colour	93.28	92.92	92.95	3.10	1.2	6
DenseNet-121	1499	clean_low	colour	94.47	94.10	94.01	2.89	1.2	5
DenseNet-121	1499	leak_high	colour	96.64	96.34	96.40	1.29	1.2	7
DenseNet-121	1499	leak_low	colour	96.05	95.80	95.86	1.17	1.2	4
DenseNet-121	1553	clean_high	colour	92.02	91.82	91.79	3.24	1.2	7
DenseNet-121	1553	clean_low	colour	91.62	91.41	91.28	3.37	1.2	5
DenseNet-121	1553	leak_high	colour	94.61	94.43	94.19	2.34	1.2	6
DenseNet-121	1553	leak_low	colour	96.21	96.07	96.32	1.52	1.2	4
DenseNet-121	1103	clean_high	masked	93.58	93.30	93.26	3.54	1.2	7
DenseNet-121	1103	clean_low	masked	93.58	93.26	93.40	2.18	1.2	5
DenseNet-121	1103	leak_high	masked	95.19	94.91	94.96	2.37	1.2	6
DenseNet-121	1103	leak_low	masked	95.19	94.88	94.71	2.46	1.2	4
DenseNet-121	1151	clean_high	masked	93.57	93.51	93.60	3.16	1.2	4
DenseNet-121	1151	clean_low	masked	93.39	93.42	93.31	3.30	1.2	7
DenseNet-121	1151	leak_high	masked	95.36	95.35	95.44	1.62	1.2	6
DenseNet-121	1151	leak_low	masked	95.71	95.69	95.88	1.44	1.2	5
DenseNet-121	1201	clean_high	masked	95.52	95.23	95.33	1.40	1.2	4
DenseNet-121	1201	clean_low	masked	97.15	96.98	96.96	0.64	1.2	6
DenseNet-121	1201	leak_high	masked	97.56	97.43	97.42	2.05	1.2	7
DenseNet-121	1201	leak_low	masked	96.95	96.81	96.80	1.18	1.2	5
DenseNet-121	1249	clean_high	masked	94.45	94.48	94.55	3.64	1.2	4
DenseNet-121	1249	clean_low	masked	92.72	92.74	92.97	3.75	1.2	5
DenseNet-121	1249	leak_high	masked	95.49	95.51	95.77	3.26	1.3	7
DenseNet-121	1249	leak_low	masked	96.01	96.06	96.23	2.47	1.2	6
DenseNet-121	1301	clean_high	masked	94.51	94.51	94.83	1.90	1.2	4
DenseNet-121	1301	clean_low	masked	95.08	95.05	95.53	2.14	1.2	5
DenseNet-121	1301	leak_high	masked	96.59	96.61	96.82	1.29	1.2	7
DenseNet-121	1301	leak_low	masked	96.97	97.01	97.27	2.07	1.2	6
DenseNet-121	1361	clean_high	masked	93.31	93.07	93.11	3.69	1.2	6
DenseNet-121	1361	clean_low	masked	93.49	93.25	93.39	3.43	1.2	5
DenseNet-121	1361	leak_high	masked	95.66	95.46	95.51	2.29	1.2	7
DenseNet-121	1361	leak_low	masked	95.66	95.49	95.57	2.20	1.2	4
DenseNet-121	1409	clean_high	masked	95.72	95.41	95.41	2.12	1.2	6
DenseNet-121	1409	clean_low	masked	96.89	96.71	96.83	1.91	1.2	5
DenseNet-121	1409	leak_high	masked	95.72	95.42	95.57	2.48	1.2	7
DenseNet-121	1409	leak_low	masked	96.89	96.71	96.66	1.33	1.2	4
DenseNet-121	1451	clean_high	masked	94.92	94.75	94.52	2.92	1.3	6
DenseNet-121	1451	clean_low	masked	93.47	93.42	93.39	2.69	1.2	5
DenseNet-121	1451	leak_high	masked	95.64	95.47	95.30	1.90	1.2	7
DenseNet-121	1451	leak_low	masked	97.10	97.01	96.96	2.21	1.2	4
DenseNet-121	1499	clean_high	masked	93.87	93.40	93.59	4.37	1.2	6

Architecture	Seed	Protocol	condition	Acc.	F1	Bal.	ECE	Min	GPU
DenseNet-121	1499	clean_low	masked	92.69	92.33	92.46	3.37	1.2	5
DenseNet-121	1499	leak_high	masked	96.25	96.01	96.17	2.19	1.2	7
DenseNet-121	1499	leak_low	masked	94.66	94.38	94.48	2.51	1.2	4
DenseNet-121	1553	clean_high	masked	92.61	92.67	92.64	3.52	1.2	7
DenseNet-121	1553	clean_low	masked	91.42	91.27	91.01	4.64	1.3	5
DenseNet-121	1553	leak_high	masked	92.42	92.30	92.04	3.80	1.2	6
DenseNet-121	1553	leak_low	masked	96.81	96.78	96.87	1.60	1.2	4
DenseNet-121	1103	clean_high	standard	95.01	94.78	94.82	1.48	1.2	7
DenseNet-121	1103	clean_low	standard	94.30	93.94	94.02	2.14	1.2	5
DenseNet-121	1103	leak_high	standard	95.54	95.21	95.05	2.44	1.2	6
DenseNet-121	1103	leak_low	standard	96.43	96.17	95.99	1.89	1.2	4
DenseNet-121	1151	clean_high	standard	93.57	93.52	93.46	2.61	1.2	4
DenseNet-121	1151	clean_low	standard	93.93	93.86	93.61	2.68	1.2	7
DenseNet-121	1151	leak_high	standard	94.46	94.40	94.29	3.21	1.2	6
DenseNet-121	1151	leak_low	standard	96.61	96.58	96.62	1.61	1.2	5
DenseNet-121	1201	clean_high	standard	96.74	96.52	96.50	1.13	1.2	4
DenseNet-121	1201	clean_low	standard	97.56	97.46	97.52	1.19	1.2	6
DenseNet-121	1201	leak_high	standard	98.57	98.50	98.46	1.29	1.2	7
DenseNet-121	1201	leak_low	standard	98.17	98.09	98.07	1.23	1.2	5
DenseNet-121	1249	clean_high	standard	94.97	94.97	95.12	2.85	1.2	4
DenseNet-121	1249	clean_low	standard	94.63	94.65	94.80	2.64	1.2	5
DenseNet-121	1249	leak_high	standard	96.53	96.53	96.81	2.58	1.2	7
DenseNet-121	1249	leak_low	standard	96.36	96.40	96.58	1.93	1.2	6
DenseNet-121	1301	clean_high	standard	96.97	96.95	97.11	1.64	1.2	4
DenseNet-121	1301	clean_low	standard	96.02	96.08	96.46	1.57	1.2	5
DenseNet-121	1301	leak_high	standard	97.54	97.52	97.84	1.30	1.2	7
DenseNet-121	1301	leak_low	standard	97.16	97.19	97.62	1.38	1.2	6
DenseNet-121	1361	clean_high	standard	94.76	94.51	94.59	3.01	1.2	4
DenseNet-121	1361	clean_low	standard	94.03	93.75	93.85	3.61	1.2	5
DenseNet-121	1361	leak_high	standard	96.02	95.82	95.94	1.95	1.2	7
DenseNet-121	1361	leak_low	standard	97.47	97.36	97.48	1.53	1.2	6
DenseNet-121	1409	clean_high	standard	96.69	96.51	96.55	1.90	1.2	6
DenseNet-121	1409	clean_low	standard	95.91	95.57	95.65	1.58	1.2	5
DenseNet-121	1409	leak_high	standard	95.91	95.60	95.62	2.01	1.2	7
DenseNet-121	1409	leak_low	standard	97.86	97.72	97.65	1.46	1.2	4
DenseNet-121	1451	clean_high	standard	95.83	95.66	95.65	1.86	1.2	6
DenseNet-121	1451	clean_low	standard	95.83	95.72	95.69	1.64	1.2	5
DenseNet-121	1451	leak_high	standard	97.10	96.97	96.92	1.02	1.2	7
DenseNet-121	1451	leak_low	standard	96.37	96.24	96.03	1.78	1.2	4
DenseNet-121	1499	clean_high	standard	95.26	94.89	95.02	3.30	1.2	6
DenseNet-121	1499	clean_low	standard	94.07	93.62	93.69	2.56	1.2	5
DenseNet-121	1499	leak_high	standard	95.65	95.31	95.41	2.03	1.2	7
DenseNet-121	1499	leak_low	standard	96.44	96.11	96.15	1.81	1.2	4
DenseNet-121	1553	clean_high	standard	91.62	91.37	91.23	4.40	1.2	7
DenseNet-121	1553	clean_low	standard	92.22	92.11	92.08	4.25	1.2	5
DenseNet-121	1553	leak_high	standard	96.01	95.95	95.92	1.93	1.2	6
DenseNet-121	1553	leak_low	standard	97.21	97.09	97.28	1.81	1.2	4
ViT-B/16	1103	clean_high	colour	92.34	91.98	92.25	5.64	3.4	4
ViT-B/16	1103	clean_low	colour	93.58	93.20	93.59	4.10	3.4	5
ViT-B/16	1103	leak_high	colour	94.12	93.50	93.94	4.26	3.4	6
ViT-B/16	1103	leak_low	colour	95.90	95.70	95.65	3.19	3.4	7
ViT-B/16	1151	clean_high	colour	92.50	92.35	92.29	5.01	3.5	4
ViT-B/16	1151	clean_low	colour	93.57	93.36	93.25	4.50	3.4	6
ViT-B/16	1151	leak_high	colour	95.18	95.14	95.21	2.57	3.5	7
ViT-B/16	1151	leak_low	colour	95.36	95.37	95.21	2.57	3.4	5
ViT-B/16	1201	clean_high	colour	96.13	95.91	95.93	2.53	3.4	4
ViT-B/16	1201	clean_low	colour	94.50	94.31	94.38	3.52	3.3	6
ViT-B/16	1201	leak_high	colour	97.96	97.91	98.00	1.56	3.4	7
ViT-B/16	1201	leak_low	colour	97.15	96.95	96.83	2.08	3.4	5
ViT-B/16	1249	clean_high	colour	94.63	94.66	94.55	3.89	3.5	4
ViT-B/16	1249	clean_low	colour	94.80	94.84	94.86	3.31	3.5	5
ViT-B/16	1249	leak_high	colour	92.20	92.12	92.45	2.97	3.5	7
ViT-B/16	1249	leak_low	colour	96.71	96.79	96.73	1.94	3.4	6
ViT-B/16	1301	clean_high	colour	92.99	92.96	93.65	5.14	3.4	4
ViT-B/16	1301	clean_low	colour	92.42	92.23	91.96	4.66	3.4	5
ViT-B/16	1301	leak_high	colour	93.56	93.70	94.31	3.79	3.4	7
ViT-B/16	1301	leak_low	colour	95.27	95.12	95.06	3.44	3.4	6
ViT-B/16	1361	clean_high	colour	93.49	93.16	93.43	4.66	3.4	6
ViT-B/16	1361	clean_low	colour	92.04	91.80	91.59	5.45	3.4	5
ViT-B/16	1361	leak_high	colour	95.66	95.49	95.43	2.41	3.5	7
ViT-B/16	1361	leak_low	colour	91.86	91.50	91.10	5.26	3.4	4
ViT-B/16	1409	clean_high	colour	94.94	94.56	94.64	3.09	3.4	6
ViT-B/16	1409	clean_low	colour	94.55	94.18	94.27	3.98	3.4	5

Architecture	Seed	Protocol	condition	Acc.	F1	Bal.	ECE	Min	GPU
ViT-B/16	1409	leak_high	colour	95.91	95.61	95.69	2.10	3.4	7
ViT-B/16	1409	leak_low	colour	93.97	93.65	93.89	3.84	3.4	4
ViT-B/16	1451	clean_high	colour	92.74	92.50	92.29	4.03	3.4	6
ViT-B/16	1451	clean_low	colour	95.10	95.03	94.95	2.59	3.4	5
ViT-B/16	1451	leak_high	colour	95.64	95.49	95.36	2.84	3.5	7
ViT-B/16	1451	leak_low	colour	93.10	92.94	93.34	3.72	3.5	4
ViT-B/16	1499	clean_high	colour	92.69	92.18	92.09	5.37	3.4	6
ViT-B/16	1499	clean_low	colour	92.49	92.09	92.18	5.92	3.4	5
ViT-B/16	1499	leak_high	colour	93.68	93.23	93.28	4.87	3.4	7
ViT-B/16	1499	leak_low	colour	94.66	94.45	94.68	3.45	3.4	4
ViT-B/16	1553	clean_high	colour	92.81	92.74	92.37	5.13	3.4	7
ViT-B/16	1553	clean_low	colour	92.02	91.75	91.59	4.80	3.4	5
ViT-B/16	1553	leak_high	colour	96.61	96.50	96.71	2.28	3.4	6
ViT-B/16	1553	leak_low	colour	96.41	96.41	96.42	2.36	3.5	4
ViT-B/16	1103	clean_high	masked	92.16	91.56	91.09	5.02	3.4	7
ViT-B/16	1103	clean_low	masked	90.02	89.46	89.23	6.42	3.4	5
ViT-B/16	1103	leak_high	masked	95.54	95.20	95.19	3.27	3.4	6
ViT-B/16	1103	leak_low	masked	93.76	93.39	93.48	4.48	3.4	4
ViT-B/16	1151	clean_high	masked	89.11	89.48	88.65	8.19	3.4	4
ViT-B/16	1151	clean_low	masked	93.04	93.06	92.79	4.44	3.4	7
ViT-B/16	1151	leak_high	masked	96.61	96.61	96.65	2.74	3.4	6
ViT-B/16	1151	leak_low	masked	94.46	94.48	94.47	2.89	3.4	5
ViT-B/16	1201	clean_high	masked	94.70	94.38	94.24	2.78	3.4	4
ViT-B/16	1201	clean_low	masked	95.11	94.93	94.86	3.35	3.3	6
ViT-B/16	1201	leak_high	masked	98.78	98.69	98.74	1.27	3.4	7
ViT-B/16	1201	leak_low	masked	94.91	94.73	94.76	2.45	3.4	5
ViT-B/16	1249	clean_high	masked	91.85	91.84	91.90	5.66	3.4	4
ViT-B/16	1249	clean_low	masked	93.76	93.73	93.90	4.62	3.4	5
ViT-B/16	1249	leak_high	masked	92.55	92.49	92.99	5.79	3.5	7
ViT-B/16	1249	leak_low	masked	95.84	95.89	95.89	3.46	3.4	6
ViT-B/16	1301	clean_high	masked	95.64	95.61	95.90	3.31	3.4	4
ViT-B/16	1301	clean_low	masked	93.18	93.15	93.46	4.66	3.4	5
ViT-B/16	1301	leak_high	masked	93.75	93.57	93.84	4.39	3.4	7
ViT-B/16	1301	leak_low	masked	96.02	95.99	96.48	2.76	3.4	6
ViT-B/16	1361	clean_high	masked	93.67	93.42	93.54	4.69	3.4	6
ViT-B/16	1361	clean_low	masked	94.03	93.76	93.94	3.96	3.4	5
ViT-B/16	1361	leak_high	masked	91.14	90.65	90.56	6.56	3.4	7
ViT-B/16	1361	leak_low	masked	92.95	92.73	92.43	5.27	3.4	4
ViT-B/16	1409	clean_high	masked	93.58	93.01	92.92	4.08	3.4	6
ViT-B/16	1409	clean_low	masked	93.00	92.76	93.03	4.74	3.4	5
ViT-B/16	1409	leak_high	masked	96.11	95.91	96.05	1.61	3.4	7
ViT-B/16	1409	leak_low	masked	93.97	93.46	93.39	4.27	3.4	4
ViT-B/16	1451	clean_high	masked	94.92	94.84	94.55	3.52	3.4	6
ViT-B/16	1451	clean_low	masked	92.74	92.49	92.64	3.42	3.4	5
ViT-B/16	1451	leak_high	masked	95.64	95.52	95.33	3.14	3.4	7
ViT-B/16	1451	leak_low	masked	96.55	96.48	96.40	2.22	3.4	4
ViT-B/16	1499	clean_high	masked	92.69	92.28	92.25	4.73	3.3	6
ViT-B/16	1499	clean_low	masked	92.09	91.72	92.10	5.32	3.4	5
ViT-B/16	1499	leak_high	masked	95.65	95.28	95.42	2.54	3.4	7
ViT-B/16	1499	leak_low	masked	95.26	94.94	95.13	3.77	3.4	4
ViT-B/16	1553	clean_high	masked	91.22	91.08	91.17	5.95	3.4	7
ViT-B/16	1553	clean_low	masked	90.42	90.17	90.31	6.12	3.4	5
ViT-B/16	1553	leak_high	masked	93.41	93.32	93.22	3.91	3.3	6
ViT-B/16	1553	leak_low	masked	95.41	95.37	95.40	2.58	3.4	4
ViT-B/16	1103	clean_high	standard	93.05	92.68	92.57	5.52	3.4	7
ViT-B/16	1103	clean_low	standard	93.58	93.02	93.44	4.50	3.4	5
ViT-B/16	1103	leak_high	standard	92.87	91.86	91.49	5.26	3.4	6
ViT-B/16	1103	leak_low	standard	93.58	93.00	93.33	5.48	3.4	4
ViT-B/16	1151	clean_high	standard	94.64	94.51	94.66	3.83	3.4	4
ViT-B/16	1151	clean_low	standard	94.82	94.80	94.68	3.39	3.4	7
ViT-B/16	1151	leak_high	standard	96.25	96.16	96.14	1.88	3.4	6
ViT-B/16	1151	leak_low	standard	95.00	94.92	95.33	3.55	3.4	5
ViT-B/16	1201	clean_high	standard	92.87	92.31	92.56	5.63	3.4	4
ViT-B/16	1201	clean_low	standard	96.13	95.82	95.80	2.58	3.3	6
ViT-B/16	1201	leak_high	standard	96.13	96.03	96.08	1.58	3.4	7
ViT-B/16	1201	leak_low	standard	97.76	97.69	97.69	1.41	3.4	5
ViT-B/16	1249	clean_high	standard	94.28	94.29	94.35	4.39	3.5	4
ViT-B/16	1249	clean_low	standard	95.32	95.34	95.49	3.39	3.4	5
ViT-B/16	1249	leak_high	standard	94.63	94.76	94.68	3.81	3.5	7
ViT-B/16	1249	leak_low	standard	95.15	95.17	94.91	3.54	3.4	6
ViT-B/16	1301	clean_high	standard	95.27	95.30	95.10	3.08	3.4	4
ViT-B/16	1301	clean_low	standard	94.89	94.86	94.94	3.72	3.4	5
ViT-B/16	1301	leak_high	standard	94.32	94.26	94.98	4.43	3.4	7

Architecture	Seed	Protocol	condition	Acc.	F1	Bal.	ECE	Min	GPU
ViT-B/16	1301	leak_low	standard	96.59	96.66	96.60	1.93	3.4	6
ViT-B/16	1361	clean_high	standard	95.12	94.86	95.03	4.20	3.4	6
ViT-B/16	1361	clean_low	standard	94.03	93.70	93.88	4.78	3.4	5
ViT-B/16	1361	leak_high	standard	86.44	86.22	85.85	10.16	3.4	7
ViT-B/16	1361	leak_low	standard	96.38	96.22	96.16	2.23	3.4	4
ViT-B/16	1409	clean_high	standard	95.33	95.02	95.09	4.08	3.4	6
ViT-B/16	1409	clean_low	standard	96.69	96.49	96.58	2.79	3.4	5
ViT-B/16	1409	leak_high	standard	87.55	86.87	86.77	7.51	3.4	7
ViT-B/16	1409	leak_low	standard	92.80	91.66	91.76	4.18	3.4	4
ViT-B/16	1451	clean_high	standard	91.29	90.91	91.41	6.29	3.4	6
ViT-B/16	1451	clean_low	standard	92.56	92.22	92.52	4.74	3.4	5
ViT-B/16	1451	leak_high	standard	92.56	92.38	91.88	4.59	3.4	7
ViT-B/16	1451	leak_low	standard	94.37	94.19	93.83	3.26	3.5	4
ViT-B/16	1499	clean_high	standard	90.12	89.75	89.63	6.78	3.3	6
ViT-B/16	1499	clean_low	standard	93.08	92.74	92.94	5.44	3.4	5
ViT-B/16	1499	leak_high	standard	97.23	97.03	97.13	2.61	3.4	7
ViT-B/16	1499	leak_low	standard	92.29	91.67	92.02	6.19	3.4	4
ViT-B/16	1553	clean_high	standard	94.01	93.92	94.05	3.62	3.4	7
ViT-B/16	1553	clean_low	standard	93.81	93.61	93.67	4.38	3.4	5
ViT-B/16	1553	leak_high	standard	96.01	95.85	95.82	2.32	3.3	6
ViT-B/16	1553	leak_low	standard	95.81	95.68	95.53	2.26	3.4	4
ConvNeXt-T	1103	clean_high	colour	94.65	94.42	94.38	3.59	2.0	7
ConvNeXt-T	1103	clean_low	colour	94.30	93.93	93.93	3.21	2.0	5
ConvNeXt-T	1103	leak_high	colour	93.58	93.00	93.42	4.57	2.0	6
ConvNeXt-T	1103	leak_low	colour	96.08	95.91	95.81	3.59	2.0	4
ConvNeXt-T	1151	clean_high	colour	95.18	95.07	94.98	3.07	2.0	4
ConvNeXt-T	1151	clean_low	colour	93.93	93.82	93.60	3.71	2.0	6
ConvNeXt-T	1151	leak_high	colour	93.04	93.02	92.91	5.12	2.0	7
ConvNeXt-T	1151	leak_low	colour	93.93	93.77	93.66	3.27	2.0	5
ConvNeXt-T	1201	clean_high	colour	96.33	96.14	96.10	2.27	2.0	4
ConvNeXt-T	1201	clean_low	colour	96.74	96.61	96.54	2.09	2.0	6
ConvNeXt-T	1201	leak_high	colour	97.76	97.60	97.62	1.42	2.0	7
ConvNeXt-T	1201	leak_low	colour	98.37	98.26	98.30	1.26	2.0	5
ConvNeXt-T	1249	clean_high	colour	94.45	94.47	94.53	3.61	2.0	4
ConvNeXt-T	1249	clean_low	colour	93.41	93.44	93.49	4.68	2.0	5
ConvNeXt-T	1249	leak_high	colour	96.36	96.40	96.46	2.63	2.0	7
ConvNeXt-T	1249	leak_low	colour	94.11	94.03	94.13	3.76	2.0	6
ConvNeXt-T	1301	clean_high	colour	96.40	96.41	96.56	2.92	2.0	4
ConvNeXt-T	1301	clean_low	colour	94.13	93.99	94.38	3.47	2.0	5
ConvNeXt-T	1301	leak_high	colour	96.21	96.16	96.27	2.36	2.0	7
ConvNeXt-T	1301	leak_low	colour	96.78	96.76	96.87	2.25	2.0	6
ConvNeXt-T	1361	clean_high	colour	93.67	93.33	93.34	4.52	2.0	6
ConvNeXt-T	1361	clean_low	colour	93.85	93.65	93.90	4.12	2.0	5
ConvNeXt-T	1361	leak_high	colour	95.48	95.29	95.48	3.26	2.0	7
ConvNeXt-T	1361	leak_low	colour	95.84	95.72	95.65	2.97	2.0	4
ConvNeXt-T	1409	clean_high	colour	95.72	95.30	95.31	2.49	2.0	6
ConvNeXt-T	1409	clean_low	colour	96.50	96.19	96.26	3.00	2.0	5
ConvNeXt-T	1409	leak_high	colour	94.94	94.49	94.60	3.65	2.1	7
ConvNeXt-T	1409	leak_low	colour	95.91	95.74	95.96	2.86	2.0	4
ConvNeXt-T	1451	clean_high	colour	91.65	91.48	91.08	5.98	2.0	6
ConvNeXt-T	1451	clean_low	colour	91.47	91.22	90.99	6.36	2.0	5
ConvNeXt-T	1451	leak_high	colour	96.37	96.27	96.14	3.62	2.0	7
ConvNeXt-T	1451	leak_low	colour	94.37	94.21	94.25	3.46	2.0	4
ConvNeXt-T	1499	clean_high	colour	93.87	93.50	93.55	4.23	2.0	6
ConvNeXt-T	1499	clean_low	colour	92.89	92.49	92.48	5.21	2.0	5
ConvNeXt-T	1499	leak_high	colour	96.84	96.66	96.66	2.65	2.0	7
ConvNeXt-T	1499	leak_low	colour	94.66	94.39	94.67	3.39	2.0	4
ConvNeXt-T	1553	clean_high	colour	90.42	90.18	90.27	6.63	2.0	7
ConvNeXt-T	1553	clean_low	colour	87.23	86.96	86.80	9.65	2.0	5
ConvNeXt-T	1553	leak_high	colour	92.42	92.44	92.33	5.13	2.0	6
ConvNeXt-T	1553	leak_low	colour	97.60	97.57	97.62	0.99	2.0	4
ConvNeXt-T	1103	clean_high	masked	90.20	89.39	89.65	7.48	2.0	7
ConvNeXt-T	1103	clean_low	masked	92.34	91.86	91.72	5.06	2.0	5
ConvNeXt-T	1103	leak_high	masked	95.54	95.28	95.39	2.88	2.0	6
ConvNeXt-T	1103	leak_low	masked	95.90	95.71	95.76	3.19	2.0	4
ConvNeXt-T	1151	clean_high	masked	93.93	93.96	93.72	4.45	2.0	4
ConvNeXt-T	1151	clean_low	masked	93.75	93.79	93.62	4.34	2.0	7
ConvNeXt-T	1151	leak_high	masked	94.82	94.79	94.87	2.51	2.0	6
ConvNeXt-T	1151	leak_low	masked	93.93	93.83	93.70	3.76	2.0	5
ConvNeXt-T	1201	clean_high	masked	96.33	96.10	96.08	2.19	2.0	4
ConvNeXt-T	1201	clean_low	masked	95.52	95.30	95.34	3.60	1.9	6
ConvNeXt-T	1201	leak_high	masked	97.35	97.21	97.31	2.01	2.0	7
ConvNeXt-T	1201	leak_low	masked	97.35	97.25	97.22	1.50	2.0	5

Architecture	Seed	Protocol	condition	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	1249	clean_high	masked	93.76	93.81	93.90	4.57	2.0	4
ConvNeXt-T	1249	clean_low	masked	93.41	93.53	93.38	4.83	2.0	5
ConvNeXt-T	1249	leak_high	masked	95.84	95.87	96.11	3.36	2.0	7
ConvNeXt-T	1249	leak_low	masked	94.97	95.02	95.08	3.96	2.0	6
ConvNeXt-T	1301	clean_high	masked	94.13	94.10	94.70	4.67	2.0	4
ConvNeXt-T	1301	clean_low	masked	94.13	94.13	94.05	3.86	2.0	5
ConvNeXt-T	1301	leak_high	masked	96.59	96.67	96.89	2.67	2.0	7
ConvNeXt-T	1301	leak_low	masked	92.23	92.29	92.29	4.64	2.0	6
ConvNeXt-T	1361	clean_high	masked	93.85	93.64	93.59	4.39	2.0	6
ConvNeXt-T	1361	clean_low	masked	92.95	92.57	92.88	5.45	2.0	5
ConvNeXt-T	1361	leak_high	masked	92.22	91.57	92.21	5.68	2.0	7
ConvNeXt-T	1361	leak_low	masked	95.66	95.50	95.52	2.83	2.0	4
ConvNeXt-T	1409	clean_high	masked	94.75	94.36	94.44	4.18	2.0	6
ConvNeXt-T	1409	clean_low	masked	96.50	96.26	96.31	2.66	2.0	5
ConvNeXt-T	1409	leak_high	masked	94.75	94.47	94.60	3.57	2.0	7
ConvNeXt-T	1409	leak_low	masked	96.89	96.74	96.87	2.53	2.0	4
ConvNeXt-T	1451	clean_high	masked	91.29	91.16	91.54	5.41	2.0	6
ConvNeXt-T	1451	clean_low	masked	93.65	93.54	93.55	4.00	2.0	5
ConvNeXt-T	1451	leak_high	masked	91.83	91.90	91.91	6.38	2.0	7
ConvNeXt-T	1451	leak_low	masked	95.10	94.93	95.02	2.26	2.0	4
ConvNeXt-T	1499	clean_high	masked	93.08	92.71	92.81	4.90	2.0	6
ConvNeXt-T	1499	clean_low	masked	93.87	93.46	93.59	4.74	2.0	5
ConvNeXt-T	1499	leak_high	masked	94.66	94.31	94.58	3.97	2.0	7
ConvNeXt-T	1499	leak_low	masked	96.05	95.69	95.75	2.89	2.0	4
ConvNeXt-T	1553	clean_high	masked	90.22	90.10	90.13	7.40	2.0	7
ConvNeXt-T	1553	clean_low	masked	90.02	90.01	89.52	7.78	2.0	5
ConvNeXt-T	1553	leak_high	masked	94.41	94.37	94.45	3.46	2.0	6
ConvNeXt-T	1553	leak_low	masked	94.81	94.67	94.77	2.76	2.0	4
ConvNeXt-T	1103	clean_high	standard	94.65	94.30	94.38	4.18	2.0	7
ConvNeXt-T	1103	clean_low	standard	93.58	93.37	93.87	5.09	2.0	5
ConvNeXt-T	1103	leak_high	standard	95.54	95.17	95.34	3.16	2.0	6
ConvNeXt-T	1103	leak_low	standard	95.90	95.63	95.66	2.77	2.0	4
ConvNeXt-T	1151	clean_high	standard	94.64	94.64	94.45	4.04	2.0	4
ConvNeXt-T	1151	clean_low	standard	90.18	90.30	89.97	7.78	2.2	7
ConvNeXt-T	1151	leak_high	standard	97.14	97.16	97.25	2.36	2.0	6
ConvNeXt-T	1151	leak_low	standard	95.54	95.49	95.33	2.14	2.2	5
ConvNeXt-T	1201	clean_high	standard	97.15	96.99	96.90	2.20	2.0	4
ConvNeXt-T	1201	clean_low	standard	96.74	96.48	96.43	2.72	2.0	6
ConvNeXt-T	1201	leak_high	standard	97.96	97.83	97.87	1.52	2.0	7
ConvNeXt-T	1201	leak_low	standard	97.35	97.14	97.05	2.28	2.0	5
ConvNeXt-T	1249	clean_high	standard	95.84	95.89	95.91	3.18	2.0	4
ConvNeXt-T	1249	clean_low	standard	94.97	95.00	94.97	3.70	2.0	5
ConvNeXt-T	1249	leak_high	standard	97.05	97.10	97.14	2.26	2.0	7
ConvNeXt-T	1249	leak_low	standard	96.88	96.88	97.03	2.34	2.0	6
ConvNeXt-T	1301	clean_high	standard	93.94	93.83	94.62	4.34	2.0	4
ConvNeXt-T	1301	clean_low	standard	95.83	95.78	96.15	3.01	2.0	5
ConvNeXt-T	1301	leak_high	standard	96.97	96.96	97.36	2.13	2.0	7
ConvNeXt-T	1301	leak_low	standard	96.78	96.76	96.93	2.30	2.0	6
ConvNeXt-T	1361	clean_high	standard	94.58	94.32	94.47	4.09	2.0	4
ConvNeXt-T	1361	clean_low	standard	95.48	95.29	95.30	3.43	2.0	5
ConvNeXt-T	1361	leak_high	standard	95.48	95.30	95.28	3.03	2.0	7
ConvNeXt-T	1361	leak_low	standard	96.38	96.27	96.38	2.27	2.0	6
ConvNeXt-T	1409	clean_high	standard	96.89	96.70	96.81	2.69	2.0	6
ConvNeXt-T	1409	clean_low	standard	94.94	94.58	94.73	3.90	2.0	5
ConvNeXt-T	1409	leak_high	standard	97.67	97.47	97.49	1.86	2.0	7
ConvNeXt-T	1409	leak_low	standard	97.47	97.30	97.39	1.96	2.0	4
ConvNeXt-T	1451	clean_high	standard	94.92	94.80	94.86	3.57	2.0	6
ConvNeXt-T	1451	clean_low	standard	92.20	91.97	91.76	6.05	2.0	5
ConvNeXt-T	1451	leak_high	standard	95.28	95.19	95.22	3.22	2.0	7
ConvNeXt-T	1451	leak_low	standard	97.28	97.16	97.20	2.06	2.0	4
ConvNeXt-T	1499	clean_high	standard	94.66	94.31	94.53	4.48	2.0	6
ConvNeXt-T	1499	clean_low	standard	95.65	95.33	95.35	3.30	2.0	5
ConvNeXt-T	1499	leak_high	standard	96.05	95.74	95.83	2.86	2.0	7
ConvNeXt-T	1499	leak_low	standard	97.04	96.74	96.82	2.46	2.0	4
ConvNeXt-T	1553	clean_high	standard	93.21	93.09	93.21	5.16	2.0	7
ConvNeXt-T	1553	clean_low	standard	90.42	90.47	90.54	7.39	2.0	5
ConvNeXt-T	1553	leak_high	standard	96.41	96.36	96.27	2.76	2.0	6
ConvNeXt-T	1553	leak_low	standard	96.01	95.88	96.04	2.50	2.0	4

Table M.2: Complete Amendment 8 linear-adversary run ledger (300 runs).

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	601	P1	0	97.16	97.18	97.19	1.88	2.6	6
ResNet-50	601	P2	0	97.41	97.42	97.42	1.68	2.6	7
ResNet-50	653	P1	0	97.28	97.31	97.29	2.18	2.6	6
ResNet-50	653	P2	0	96.67	96.69	96.68	2.23	2.6	4
ResNet-50	701	P1	0	97.53	97.54	97.53	1.47	2.6	6
ResNet-50	701	P2	0	96.79	96.81	96.81	2.25	2.6	7
ResNet-50	751	P1	0	97.53	97.56	97.55	1.31	2.6	6
ResNet-50	751	P2	0	95.56	95.55	95.55	3.75	2.6	4
ResNet-50	809	P1	0	97.78	97.77	97.80	1.37	2.6	5
ResNet-50	809	P2	0	97.04	97.05	97.02	1.90	2.6	6
ResNet-50	853	P1	0	97.41	97.41	97.44	2.18	2.6	7
ResNet-50	853	P2	0	96.17	96.17	96.18	2.40	2.6	4
ResNet-50	907	P1	0	98.64	98.65	98.65	0.84	2.6	5
ResNet-50	907	P2	0	97.41	97.41	97.43	1.60	2.6	6
ResNet-50	953	P1	0	97.53	97.55	97.53	1.28	2.6	5
ResNet-50	953	P2	0	95.80	95.87	95.79	2.85	2.6	7
ResNet-50	1009	P1	0	97.16	97.18	97.16	1.72	2.6	4
ResNet-50	1009	P2	0	97.41	97.42	97.41	1.58	2.6	6
ResNet-50	1051	P1	0	98.27	98.29	98.28	0.81	2.6	5
ResNet-50	1051	P2	0	97.41	97.40	97.43	0.99	2.6	4
ResNet-50	601	P1	0.1	97.28	97.31	97.29	1.27	2.6	4
ResNet-50	601	P2	0.1	95.93	95.94	95.96	2.60	2.6	5
ResNet-50	653	P1	0.1	97.53	97.55	97.55	1.11	2.6	5
ResNet-50	653	P2	0.1	96.17	96.16	96.17	2.08	2.6	7
ResNet-50	701	P1	0.1	96.91	96.90	96.93	1.62	2.6	4
ResNet-50	701	P2	0.1	95.56	95.56	95.58	2.70	2.6	5
ResNet-50	751	P1	0.1	97.16	97.18	97.18	1.50	2.6	5
ResNet-50	751	P2	0.1	95.80	95.79	95.79	3.07	2.6	7
ResNet-50	809	P1	0.1	96.67	96.68	96.72	2.09	2.6	7
ResNet-50	809	P2	0.1	96.42	96.45	96.40	2.51	2.6	4
ResNet-50	853	P1	0.1	96.67	96.67	96.68	1.30	2.6	5
ResNet-50	853	P2	0.1	96.30	96.29	96.30	2.59	2.6	6
ResNet-50	907	P1	0.1	97.41	97.42	97.42	1.07	2.6	7
ResNet-50	907	P2	0.1	96.42	96.41	96.39	2.13	2.6	4
ResNet-50	953	P1	0.1	97.16	97.20	97.17	1.30	2.6	4
ResNet-50	953	P2	0.1	96.79	96.82	96.79	2.21	2.6	6
ResNet-50	1009	P1	0.1	97.16	97.18	97.16	1.63	2.6	5
ResNet-50	1009	P2	0.1	97.04	97.05	97.05	1.75	2.6	7
ResNet-50	1051	P1	0.1	97.28	97.29	97.30	1.00	2.6	6
ResNet-50	1051	P2	0.1	96.42	96.43	96.42	1.76	2.6	7
ResNet-50	601	P1	0.3	96.79	96.81	96.81	1.57	2.6	6
ResNet-50	601	P2	0.3	96.67	96.67	96.68	0.96	2.6	7
ResNet-50	653	P1	0.3	97.16	97.17	97.18	1.23	2.6	4
ResNet-50	653	P2	0.3	96.05	96.03	96.05	1.57	2.6	5
ResNet-50	701	P1	0.3	96.91	96.91	96.94	1.17	2.6	6
ResNet-50	701	P2	0.3	95.43	95.48	95.45	1.36	2.6	7
ResNet-50	751	P1	0.3	96.17	96.15	96.22	1.30	2.6	6
ResNet-50	751	P2	0.3	96.17	96.19	96.19	2.31	2.6	4
ResNet-50	809	P1	0.3	96.42	96.43	96.46	1.91	2.6	5
ResNet-50	809	P2	0.3	95.80	95.79	95.79	2.39	2.6	6
ResNet-50	853	P1	0.3	95.06	95.05	95.04	2.75	2.6	7
ResNet-50	853	P2	0.3	96.17	96.14	96.19	1.12	2.6	4
ResNet-50	907	P1	0.3	96.91	96.95	96.90	2.31	2.6	5
ResNet-50	907	P2	0.3	95.80	95.82	95.77	1.89	2.6	6
ResNet-50	953	P1	0.3	97.78	97.79	97.78	1.12	2.6	5
ResNet-50	953	P2	0.3	96.17	96.24	96.17	1.75	2.6	7
ResNet-50	1009	P1	0.3	97.53	97.55	97.53	0.84	2.6	4
ResNet-50	1009	P2	0.3	94.07	94.07	94.05	3.09	2.6	6
ResNet-50	1051	P1	0.3	97.28	97.28	97.31	0.89	2.6	5
ResNet-50	1051	P2	0.3	96.54	96.56	96.56	1.75	2.6	4
ResNet-50	601	P1	1	96.54	96.57	96.58	1.73	2.6	4
ResNet-50	601	P2	1	94.94	94.97	94.96	3.16	2.6	5
ResNet-50	653	P1	1	92.59	92.64	92.62	2.65	2.6	6
ResNet-50	653	P2	1	93.83	93.87	93.80	3.92	2.6	7
ResNet-50	701	P1	1	90.12	89.73	90.05	3.74	2.6	4
ResNet-50	701	P2	1	92.96	92.99	92.98	3.22	2.6	5
ResNet-50	751	P1	1	93.46	93.51	93.52	4.36	2.6	5
ResNet-50	751	P2	1	91.73	91.81	91.71	3.58	2.6	7
ResNet-50	809	P1	1	93.83	93.85	93.86	1.57	2.6	7
ResNet-50	809	P2	1	94.44	94.43	94.41	1.53	2.6	4

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	853	P1	1	94.81	94.79	94.81	3.11	2.6	5
ResNet-50	853	P2	1	93.95	93.88	93.92	2.14	2.6	6
ResNet-50	907	P1	1	94.57	94.56	94.60	3.46	2.6	7
ResNet-50	907	P2	1	91.48	91.53	91.43	1.86	2.6	4
ResNet-50	953	P1	1	95.19	95.17	95.22	1.58	2.6	4
ResNet-50	953	P2	1	95.80	95.84	95.79	2.78	2.6	6
ResNet-50	1009	P1	1	96.42	96.43	96.43	4.57	2.6	5
ResNet-50	1009	P2	1	91.36	91.34	91.34	4.86	2.6	7
ResNet-50	1051	P1	1	94.44	94.48	94.46	2.70	2.6	6
ResNet-50	1051	P2	1	91.60	91.60	91.56	4.29	2.6	7
ResNet-50	601	P1	3	84.20	84.37	84.20	11.37	2.6	6
ResNet-50	601	P2	3	65.68	60.65	65.87	15.25	2.6	7
ResNet-50	653	P1	3	88.89	89.00	88.94	15.06	2.6	4
ResNet-50	653	P2	3	91.85	91.87	91.87	6.43	2.6	5
ResNet-50	701	P1	3	88.27	88.27	88.30	10.16	2.6	6
ResNet-50	701	P2	3	85.19	85.20	85.28	16.72	2.6	7
ResNet-50	751	P1	3	89.38	89.36	89.43	8.57	2.6	6
ResNet-50	751	P2	3	71.36	71.59	71.29	18.14	2.6	4
ResNet-50	809	P1	3	87.41	87.44	87.41	14.36	2.6	5
ResNet-50	809	P2	3	84.81	84.78	84.89	8.87	2.6	6
ResNet-50	853	P1	3	79.88	79.31	79.98	15.51	2.6	7
ResNet-50	853	P2	3	88.77	88.59	88.70	15.66	2.6	4
ResNet-50	907	P1	3	92.10	92.07	92.15	13.84	2.6	5
ResNet-50	907	P2	3	74.94	72.53	75.11	17.50	2.6	6
ResNet-50	953	P1	3	78.02	78.75	78.00	15.11	2.6	5
ResNet-50	953	P2	3	85.19	84.98	85.18	13.01	2.6	7
ResNet-50	1009	P1	3	63.09	59.40	62.60	7.80	2.6	4
ResNet-50	1009	P2	3	80.37	80.19	80.30	23.57	2.6	6
ResNet-50	1051	P1	3	85.43	85.48	85.41	12.46	2.6	5
ResNet-50	1051	P2	3	90.74	90.82	90.75	12.48	2.6	4
DenseNet-121	601	P1	0	97.04	97.05	97.06	1.45	3.3	6
DenseNet-121	601	P2	0	97.16	97.17	97.18	1.39	3.3	7
DenseNet-121	653	P1	0	96.67	96.70	96.67	1.99	3.3	4
DenseNet-121	653	P2	0	97.16	97.18	97.15	1.27	3.3	5
DenseNet-121	701	P1	0	97.90	97.90	97.91	1.39	3.3	6
DenseNet-121	701	P2	0	96.79	96.80	96.83	2.02	3.3	7
DenseNet-121	751	P1	0	98.27	98.29	98.30	0.91	3.3	6
DenseNet-121	751	P2	0	96.79	96.78	96.81	1.87	3.3	4
DenseNet-121	809	P1	0	98.40	98.40	98.41	0.96	3.3	5
DenseNet-121	809	P2	0	97.41	97.39	97.39	1.62	3.3	6
DenseNet-121	853	P1	0	97.41	97.41	97.43	1.44	3.3	7
DenseNet-121	853	P2	0	96.17	96.19	96.17	2.47	3.3	4
DenseNet-121	907	P1	0	97.53	97.54	97.55	1.59	3.3	5
DenseNet-121	907	P2	0	96.91	96.91	96.91	1.25	3.3	6
DenseNet-121	953	P1	0	98.02	98.03	98.04	1.33	3.3	5
DenseNet-121	953	P2	0	97.28	97.31	97.26	1.54	3.3	7
DenseNet-121	1009	P1	0	98.02	98.03	98.03	1.47	3.3	4
DenseNet-121	1009	P2	0	96.79	96.80	96.80	1.75	3.3	6
DenseNet-121	1051	P1	0	97.53	97.55	97.55	1.06	3.3	5
DenseNet-121	1051	P2	0	97.04	97.03	97.06	1.77	3.3	4
DenseNet-121	601	P1	0.1	96.79	96.79	96.81	1.42	3.3	6
DenseNet-121	601	P2	0.1	97.41	97.42	97.42	1.04	3.3	4
DenseNet-121	653	P1	0.1	95.93	95.96	95.95	2.80	3.3	6
DenseNet-121	653	P2	0.1	97.28	97.30	97.29	1.96	3.3	7
DenseNet-121	701	P1	0.1	97.65	97.66	97.67	1.55	3.3	6
DenseNet-121	701	P2	0.1	95.06	95.10	95.08	2.50	3.3	4
DenseNet-121	751	P1	0.1	97.04	97.06	97.04	1.74	3.3	5
DenseNet-121	751	P2	0.1	97.65	97.66	97.67	1.70	3.3	6
DenseNet-121	809	P1	0.1	96.54	96.53	96.57	2.04	3.3	7
DenseNet-121	809	P2	0.1	97.04	97.05	97.02	1.61	3.3	4
DenseNet-121	853	P1	0.1	97.65	97.66	97.68	1.32	3.3	5
DenseNet-121	853	P2	0.1	96.91	96.93	96.91	2.40	3.3	6
DenseNet-121	907	P1	0.1	97.65	97.66	97.65	1.25	3.3	5
DenseNet-121	907	P2	0.1	96.91	96.94	96.90	2.23	3.3	7
DenseNet-121	953	P1	0.1	97.28	97.29	97.31	1.98	3.3	4
DenseNet-121	953	P2	0.1	96.91	96.94	96.90	2.26	3.3	6
DenseNet-121	1009	P1	0.1	97.41	97.41	97.42	1.24	3.3	5
DenseNet-121	1009	P2	0.1	95.93	95.91	95.96	2.82	3.3	4
DenseNet-121	1051	P1	0.1	96.42	96.44	96.43	1.35	3.3	6
DenseNet-121	1051	P2	0.1	97.04	97.03	97.04	0.97	3.3	7
DenseNet-121	601	P1	0.3	93.70	93.70	93.71	2.71	3.3	5
DenseNet-121	601	P2	0.3	96.67	96.67	96.70	1.37	3.3	7
DenseNet-121	653	P1	0.3	97.41	97.42	97.43	0.97	3.3	4

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
DenseNet-121	653	P2	0.3	95.80	95.82	95.86	1.74	3.3	5
DenseNet-121	701	P1	0.3	97.53	97.53	97.55	2.07	3.3	5
DenseNet-121	701	P2	0.3	94.69	94.72	94.74	2.66	3.3	7
DenseNet-121	751	P1	0.3	96.91	96.95	96.93	1.32	3.3	7
DenseNet-121	751	P2	0.3	97.78	97.78	97.80	1.64	3.3	4
DenseNet-121	809	P1	0.3	96.17	96.19	96.19	1.65	3.3	5
DenseNet-121	809	P2	0.3	97.41	97.41	97.42	1.87	3.3	6
DenseNet-121	853	P1	0.3	97.04	97.04	97.04	1.81	3.3	7
DenseNet-121	853	P2	0.3	96.67	96.65	96.68	1.38	3.3	4
DenseNet-121	907	P1	0.3	96.79	96.79	96.79	1.00	3.3	4
DenseNet-121	907	P2	0.3	95.56	95.58	95.53	2.00	3.3	6
DenseNet-121	953	P1	0.3	97.16	97.17	97.17	0.92	3.3	5
DenseNet-121	953	P2	0.3	96.42	96.44	96.41	1.72	3.3	7
DenseNet-121	1009	P1	0.3	96.67	96.70	96.67	1.45	3.3	6
DenseNet-121	1009	P2	0.3	93.33	93.29	93.36	2.37	3.3	7
DenseNet-121	1051	P1	0.3	96.91	96.92	96.94	0.96	3.3	5
DenseNet-121	1051	P2	0.3	96.17	96.19	96.17	1.06	3.3	4
DenseNet-121	601	P1	1	92.59	92.56	92.58	4.23	3.3	6
DenseNet-121	601	P2	1	90.86	90.96	90.95	5.81	3.3	4
DenseNet-121	653	P1	1	93.21	93.28	93.16	2.32	3.3	6
DenseNet-121	653	P2	1	92.72	92.72	92.72	4.76	3.3	7
DenseNet-121	701	P1	1	86.54	86.52	86.60	5.01	3.3	6
DenseNet-121	701	P2	1	84.69	84.64	84.58	3.33	3.3	4
DenseNet-121	751	P1	1	90.25	90.08	90.26	2.81	3.3	5
DenseNet-121	751	P2	1	88.52	88.40	88.42	3.68	3.3	6
DenseNet-121	809	P1	1	94.57	94.51	94.62	4.61	3.3	7
DenseNet-121	809	P2	1	89.63	89.53	89.54	3.40	3.3	4
DenseNet-121	853	P1	1	92.96	92.89	92.91	4.25	3.3	5
DenseNet-121	853	P2	1	73.58	70.89	73.44	4.99	3.3	6
DenseNet-121	907	P1	1	89.63	89.70	89.50	2.95	3.3	5
DenseNet-121	907	P2	1	91.73	91.76	91.73	4.61	3.3	7
DenseNet-121	953	P1	1	92.59	92.61	92.57	1.10	3.3	4
DenseNet-121	953	P2	1	91.85	91.84	91.91	3.29	3.3	6
DenseNet-121	1009	P1	1	93.09	93.17	93.12	3.88	3.3	5
DenseNet-121	1009	P2	1	88.27	88.23	88.22	2.18	3.3	4
DenseNet-121	1051	P1	1	89.38	89.45	89.37	3.03	3.3	6
DenseNet-121	1051	P2	1	87.78	87.77	87.84	4.71	3.3	7
DenseNet-121	601	P1	3	63.09	62.82	63.10	4.84	3.3	5
DenseNet-121	601	P2	3	69.01	68.02	68.75	17.64	3.3	7
DenseNet-121	653	P1	3	64.44	63.31	64.49	10.23	3.3	4
DenseNet-121	653	P2	3	74.07	74.04	73.74	15.13	3.3	5
DenseNet-121	701	P1	3	63.70	62.00	63.77	9.14	3.3	5
DenseNet-121	701	P2	3	59.26	60.38	59.26	7.80	3.3	7
DenseNet-121	751	P1	3	76.54	75.83	76.58	18.67	3.3	7
DenseNet-121	751	P2	3	68.89	67.64	68.90	11.27	3.3	4
DenseNet-121	809	P1	3	64.57	63.39	64.30	9.81	3.3	5
DenseNet-121	809	P2	3	76.67	76.37	76.48	16.50	3.3	6
DenseNet-121	853	P1	3	59.26	56.41	59.36	6.20	3.3	7
DenseNet-121	853	P2	3	52.35	49.76	52.02	8.03	3.3	4
DenseNet-121	907	P1	3	69.88	69.68	69.68	7.50	3.3	4
DenseNet-121	907	P2	3	81.23	81.01	81.16	15.90	3.3	6
DenseNet-121	953	P1	3	52.96	52.57	53.28	6.27	3.3	5
DenseNet-121	953	P2	3	53.70	53.90	53.64	8.01	3.3	7
DenseNet-121	1009	P1	3	65.68	60.71	65.40	7.08	3.3	6
DenseNet-121	1009	P2	3	64.44	64.16	64.54	16.23	3.3	7
DenseNet-121	1051	P1	3	79.14	78.85	79.00	20.84	3.3	5
DenseNet-121	1051	P2	3	61.48	57.65	61.19	10.02	3.3	4
ConvNeXt-T	601	P1	0	98.40	98.40	98.41	1.60	5.2	4
ConvNeXt-T	601	P2	0	97.16	97.17	97.18	1.93	5.2	5
ConvNeXt-T	653	P1	0	97.65	97.68	97.64	1.82	5.2	6
ConvNeXt-T	653	P2	0	97.53	97.53	97.54	1.75	5.2	7
ConvNeXt-T	701	P1	0	98.52	98.52	98.53	1.12	5.2	4
ConvNeXt-T	701	P2	0	94.94	94.94	94.98	4.14	5.2	5
ConvNeXt-T	751	P1	0	98.40	98.40	98.42	1.28	5.2	5
ConvNeXt-T	751	P2	0	96.67	96.67	96.66	2.46	5.2	7
ConvNeXt-T	809	P1	0	97.16	97.16	97.19	2.57	5.2	7
ConvNeXt-T	809	P2	0	96.42	96.42	96.39	2.49	5.2	4
ConvNeXt-T	853	P1	0	97.53	97.53	97.55	1.67	5.2	5
ConvNeXt-T	853	P2	0	97.16	97.18	97.13	2.17	5.2	6
ConvNeXt-T	907	P1	0	98.02	98.04	98.03	1.56	5.2	7
ConvNeXt-T	907	P2	0	95.93	95.91	95.96	3.26	5.2	4
ConvNeXt-T	953	P1	0	96.42	96.35	96.47	2.74	5.2	4
ConvNeXt-T	953	P2	0	96.42	96.46	96.43	2.76	5.1	6

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	1009	P1	0	97.04	97.08	97.02	1.87	5.2	5
ConvNeXt-T	1009	P2	0	97.04	97.03	97.08	2.44	5.2	7
ConvNeXt-T	1051	P1	0	98.52	98.53	98.53	0.97	5.1	6
ConvNeXt-T	1051	P2	0	96.17	96.19	96.15	2.38	5.2	7
ConvNeXt-T	601	P1	0.1	97.41	97.44	97.42	2.27	5.2	6
ConvNeXt-T	601	P2	0.1	98.15	98.15	98.15	1.15	5.3	7
ConvNeXt-T	653	P1	0.1	98.02	98.04	98.01	1.51	5.2	4
ConvNeXt-T	653	P2	0.1	96.54	96.55	96.53	2.97	5.2	5
ConvNeXt-T	701	P1	0.1	96.67	96.70	96.66	3.02	5.2	6
ConvNeXt-T	701	P2	0.1	95.93	95.93	95.96	2.95	5.2	7
ConvNeXt-T	751	P1	0.1	98.27	98.28	98.29	1.50	5.2	6
ConvNeXt-T	751	P2	0.1	95.93	95.96	95.90	3.19	5.2	4
ConvNeXt-T	809	P1	0.1	97.41	97.41	97.43	1.88	5.2	5
ConvNeXt-T	809	P2	0.1	95.31	95.30	95.26	3.63	5.1	6
ConvNeXt-T	853	P1	0.1	97.41	97.42	97.42	1.86	5.2	7
ConvNeXt-T	853	P2	0.1	96.42	96.40	96.42	2.89	5.2	4
ConvNeXt-T	907	P1	0.1	99.01	99.02	99.01	0.81	5.2	5
ConvNeXt-T	907	P2	0.1	96.54	96.55	96.54	2.88	5.1	6
ConvNeXt-T	953	P1	0.1	96.42	96.38	96.48	2.60	5.2	5
ConvNeXt-T	953	P2	0.1	97.04	97.07	97.04	2.38	5.2	7
ConvNeXt-T	1009	P1	0.1	97.41	97.42	97.44	2.41	5.2	4
ConvNeXt-T	1009	P2	0.1	97.16	97.16	97.17	2.09	5.1	6
ConvNeXt-T	1051	P1	0.1	97.53	97.52	97.57	1.91	5.2	5
ConvNeXt-T	1051	P2	0.1	96.54	96.57	96.52	2.56	5.2	4
ConvNeXt-T	601	P1	0.3	96.91	96.95	96.92	2.43	5.2	4
ConvNeXt-T	601	P2	0.3	96.91	96.93	96.91	1.51	5.2	5
ConvNeXt-T	653	P1	0.3	97.90	97.91	97.90	1.31	5.2	6
ConvNeXt-T	653	P2	0.3	96.54	96.57	96.54	2.31	5.2	7
ConvNeXt-T	701	P1	0.3	97.65	97.66	97.66	1.69	5.2	4
ConvNeXt-T	701	P2	0.3	96.30	96.29	96.35	2.82	5.2	5
ConvNeXt-T	751	P1	0.3	98.64	98.65	98.65	0.61	5.2	5
ConvNeXt-T	751	P2	0.3	91.11	90.63	90.99	6.76	5.2	7
ConvNeXt-T	809	P1	0.3	97.41	97.43	97.43	1.52	5.2	7
ConvNeXt-T	809	P2	0.3	95.93	95.94	95.91	3.09	5.2	4
ConvNeXt-T	853	P1	0.3	97.16	97.15	97.20	1.66	5.2	5
ConvNeXt-T	853	P2	0.3	96.05	96.01	95.98	2.84	5.1	6
ConvNeXt-T	907	P1	0.3	98.27	98.28	98.28	1.17	5.2	7
ConvNeXt-T	907	P2	0.3	96.54	96.55	96.54	1.84	5.2	4
ConvNeXt-T	953	P1	0.3	96.42	96.40	96.46	2.68	5.2	4
ConvNeXt-T	953	P2	0.3	96.54	96.59	96.54	2.67	5.2	6
ConvNeXt-T	1009	P1	0.3	97.90	97.91	97.91	1.79	5.2	5
ConvNeXt-T	1009	P2	0.3	96.54	96.54	96.56	3.03	5.2	7
ConvNeXt-T	1051	P1	0.3	97.53	97.56	97.54	1.65	5.1	6
ConvNeXt-T	1051	P2	0.3	97.78	97.80	97.76	1.37	5.2	7
ConvNeXt-T	601	P1	1	97.16	97.18	97.18	0.86	5.2	6
ConvNeXt-T	601	P2	1	96.17	96.18	96.16	4.22	5.3	7
ConvNeXt-T	653	P1	1	97.16	97.15	97.18	3.60	5.2	4
ConvNeXt-T	653	P2	1	96.67	96.68	96.65	2.62	5.2	5
ConvNeXt-T	701	P1	1	97.28	97.29	97.29	1.58	5.2	6
ConvNeXt-T	701	P2	1	93.21	93.19	93.29	2.63	5.2	7
ConvNeXt-T	751	P1	1	96.79	96.78	96.82	2.46	5.1	6
ConvNeXt-T	751	P2	1	95.19	95.20	95.14	4.02	5.2	4
ConvNeXt-T	809	P1	1	96.17	96.14	96.22	1.65	5.2	5
ConvNeXt-T	809	P2	1	94.44	94.50	94.45	1.87	5.1	6
ConvNeXt-T	853	P1	1	97.90	97.91	97.91	0.90	5.2	7
ConvNeXt-T	853	P2	1	95.56	95.56	95.53	2.03	5.2	4
ConvNeXt-T	907	P1	1	97.41	97.42	97.41	1.05	5.2	5
ConvNeXt-T	907	P2	1	96.54	96.55	96.55	3.32	5.1	6
ConvNeXt-T	953	P1	1	96.05	96.00	96.11	3.54	5.2	5
ConvNeXt-T	953	P2	1	95.31	95.35	95.29	3.00	5.2	7
ConvNeXt-T	1009	P1	1	97.90	97.91	97.92	1.94	5.2	4
ConvNeXt-T	1009	P2	1	95.93	95.98	95.94	4.03	5.2	6
ConvNeXt-T	1051	P1	1	96.17	96.21	96.18	2.23	5.2	5
ConvNeXt-T	1051	P2	1	97.28	97.28	97.27	1.92	5.2	4
ConvNeXt-T	601	P1	3	87.53	87.78	87.58	4.66	5.2	4
ConvNeXt-T	601	P2	3	82.35	83.05	82.19	6.82	5.2	5
ConvNeXt-T	653	P1	3	82.84	82.65	82.79	4.79	5.2	6
ConvNeXt-T	653	P2	3	90.49	90.36	90.58	3.55	5.2	7
ConvNeXt-T	701	P1	3	87.90	88.02	87.97	5.80	5.2	4
ConvNeXt-T	701	P2	3	91.11	91.24	91.06	12.60	5.2	5
ConvNeXt-T	751	P1	3	88.89	89.13	88.94	6.94	5.2	5
ConvNeXt-T	751	P2	3	19.63	6.77	20.12	14.58	5.2	7
ConvNeXt-T	809	P1	3	92.10	92.14	92.18	6.37	5.2	7

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	809	P2	3	93.33	93.28	93.33	9.11	5.2	4
ConvNeXt-T	853	P1	3	86.17	85.38	86.34	11.14	5.2	5
ConvNeXt-T	853	P2	3	87.28	87.02	87.11	4.75	5.1	6
ConvNeXt-T	907	P1	3	89.63	89.46	89.64	8.63	5.2	7
ConvNeXt-T	907	P2	3	84.44	84.44	84.40	4.86	5.2	4
ConvNeXt-T	953	P1	3	93.58	93.57	93.61	8.92	5.2	4
ConvNeXt-T	953	P2	3	64.94	62.05	64.93	12.03	5.1	6
ConvNeXt-T	1009	P1	3	86.42	86.10	86.52	5.14	5.2	5
ConvNeXt-T	1009	P2	3	85.80	85.78	85.60	6.35	5.2	7
ConvNeXt-T	1051	P1	3	88.52	88.76	88.51	4.31	5.1	6
ConvNeXt-T	1051	P2	3	20.12	6.70	20.00	17.86	5.2	7

Table M.3: Complete Amendment 8 normalisation run ledger (160 runs).

Architecture	Seed	Protocol	cache_name	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	601	P1	sipakmed_reinhard100	95.31	95.38	95.29	3.19	2.6	5
ResNet-50	601	P2	sipakmed_reinhard100	94.32	94.33	94.32	3.64	2.6	4
ResNet-50	653	P1	sipakmed_reinhard100	95.68	95.70	95.68	3.46	2.6	5
ResNet-50	653	P2	sipakmed_reinhard100	96.30	96.32	96.29	2.82	2.6	4
ResNet-50	701	P1	sipakmed_reinhard100	95.31	95.31	95.34	2.70	2.6	5
ResNet-50	701	P2	sipakmed_reinhard100	93.46	93.48	93.46	4.74	2.6	4
ResNet-50	751	P1	sipakmed_reinhard100	96.30	96.30	96.34	2.39	2.6	5
ResNet-50	751	P2	sipakmed_reinhard100	94.57	94.55	94.57	3.49	2.6	4
ResNet-50	809	P1	sipakmed_reinhard100	95.68	95.69	95.72	2.27	2.7	5
ResNet-50	809	P2	sipakmed_reinhard100	97.28	97.27	97.31	2.03	2.7	6
ResNet-50	853	P1	sipakmed_reinhard100	96.42	96.44	96.42	1.74	2.7	6
ResNet-50	853	P2	sipakmed_reinhard100	95.31	95.30	95.31	3.15	2.7	7
ResNet-50	907	P1	sipakmed_reinhard100	96.05	96.06	96.05	2.52	2.7	7
ResNet-50	907	P2	sipakmed_reinhard100	92.96	92.94	92.94	4.61	2.7	5
ResNet-50	953	P1	sipakmed_reinhard100	96.17	96.19	96.21	2.49	2.7	5
ResNet-50	953	P2	sipakmed_reinhard100	96.67	96.70	96.66	3.02	2.7	6
ResNet-50	1009	P1	sipakmed_reinhard100	95.80	95.83	95.81	2.00	2.7	6
ResNet-50	1009	P2	sipakmed_reinhard100	95.06	95.07	95.05	2.87	2.7	7
ResNet-50	1051	P1	sipakmed_reinhard100	95.56	95.55	95.58	2.60	2.7	7
ResNet-50	1051	P2	sipakmed_reinhard100	94.20	94.26	94.15	4.07	2.7	5
ResNet-50	601	P1	sipakmed_reinhard50	96.91	96.93	96.92	2.07	2.6	6
ResNet-50	601	P2	sipakmed_reinhard50	96.67	96.69	96.68	1.71	2.6	7
ResNet-50	653	P1	sipakmed_reinhard50	97.41	97.42	97.41	1.49	2.6	6
ResNet-50	653	P2	sipakmed_reinhard50	95.43	95.47	95.42	3.06	2.6	7
ResNet-50	701	P1	sipakmed_reinhard50	97.53	97.52	97.55	1.32	2.6	6
ResNet-50	701	P2	sipakmed_reinhard50	94.69	94.71	94.70	3.19	2.6	7
ResNet-50	751	P1	sipakmed_reinhard50	97.90	97.92	97.92	1.47	2.6	6
ResNet-50	751	P2	sipakmed_reinhard50	96.30	96.29	96.30	2.27	2.6	7
ResNet-50	809	P1	sipakmed_reinhard50	96.42	96.44	96.44	2.22	3.5	4
ResNet-50	809	P2	sipakmed_reinhard50	96.05	96.07	96.04	2.83	2.7	7
ResNet-50	853	P1	sipakmed_reinhard50	97.65	97.66	97.66	1.36	3.6	4
ResNet-50	853	P2	sipakmed_reinhard50	96.67	96.64	96.67	2.04	2.7	5
ResNet-50	907	P1	sipakmed_reinhard50	98.15	98.17	98.16	1.23	3.6	4
ResNet-50	907	P2	sipakmed_reinhard50	96.17	96.19	96.15	2.63	2.7	6
ResNet-50	953	P1	sipakmed_reinhard50	96.79	96.80	96.79	1.83	3.6	4
ResNet-50	953	P2	sipakmed_reinhard50	97.04	97.06	97.04	2.57	2.8	7
ResNet-50	1009	P1	sipakmed_reinhard50	96.67	96.70	96.65	2.25	3.6	4
ResNet-50	1009	P2	sipakmed_reinhard50	96.42	96.40	96.47	2.62	2.7	5
ResNet-50	1051	P1	sipakmed_reinhard50	97.65	97.66	97.66	1.56	3.6	4
ResNet-50	1051	P2	sipakmed_reinhard50	95.80	95.81	95.76	2.38	2.7	6
DenseNet-121	601	P1	sipakmed_reinhard100	94.69	94.71	94.73	2.19	3.3	5
DenseNet-121	601	P2	sipakmed_reinhard100	94.94	94.96	94.94	3.11	3.3	4
DenseNet-121	653	P1	sipakmed_reinhard100	97.16	97.18	97.17	1.29	3.3	5
DenseNet-121	653	P2	sipakmed_reinhard100	95.31	95.37	95.29	2.23	3.3	4
DenseNet-121	701	P1	sipakmed_reinhard100	95.43	95.39	95.47	2.76	3.3	5
DenseNet-121	701	P2	sipakmed_reinhard100	93.83	93.89	93.81	3.80	3.3	4
DenseNet-121	751	P1	sipakmed_reinhard100	96.79	96.80	96.83	2.08	3.3	5
DenseNet-121	751	P2	sipakmed_reinhard100	95.56	95.53	95.55	2.67	4.2	4
DenseNet-121	809	P1	sipakmed_reinhard100	97.28	97.29	97.29	1.76	4.2	4
DenseNet-121	809	P2	sipakmed_reinhard100	96.17	96.15	96.19	2.38	3.3	6
DenseNet-121	853	P1	sipakmed_reinhard100	96.30	96.28	96.32	1.83	4.2	4
DenseNet-121	853	P2	sipakmed_reinhard100	95.43	95.44	95.41	2.13	3.3	7
DenseNet-121	907	P1	sipakmed_reinhard100	96.67	96.69	96.66	1.49	3.3	5
DenseNet-121	907	P2	sipakmed_reinhard100	94.20	94.24	94.17	4.11	4.2	4

Architecture	Seed	Protocol	cache_name	Acc.	F1	Bal.	ECE	Min	GPU
DenseNet-121	953	P1	sipakmed_reinhard100	95.43	95.45	95.47	2.17	3.3	6
DenseNet-121	953	P2	sipakmed_reinhard100	97.04	97.05	97.02	1.92	4.2	4
DenseNet-121	1009	P1	sipakmed_reinhard100	96.05	96.10	96.05	2.35	4.3	4
DenseNet-121	1009	P2	sipakmed_reinhard100	94.94	94.96	94.92	3.36	3.3	7
DenseNet-121	1051	P1	sipakmed_reinhard100	96.17	96.18	96.19	1.86	4.3	4
DenseNet-121	1051	P2	sipakmed_reinhard100	95.56	95.56	95.52	1.84	3.3	5
DenseNet-121	601	P1	sipakmed_reinhard50	97.04	97.05	97.06	1.83	3.3	6
DenseNet-121	601	P2	sipakmed_reinhard50	96.54	96.54	96.56	1.60	3.3	7
DenseNet-121	653	P1	sipakmed_reinhard50	98.27	98.29	98.27	1.06	3.3	6
DenseNet-121	653	P2	sipakmed_reinhard50	96.54	96.59	96.54	2.22	3.3	7
DenseNet-121	701	P1	sipakmed_reinhard50	97.04	97.02	97.06	1.56	3.3	6
DenseNet-121	701	P2	sipakmed_reinhard50	95.43	95.46	95.45	2.90	3.3	7
DenseNet-121	751	P1	sipakmed_reinhard50	97.90	97.92	97.90	1.20	3.3	6
DenseNet-121	751	P2	sipakmed_reinhard50	96.54	96.55	96.57	2.34	3.3	7
DenseNet-121	809	P1	sipakmed_reinhard50	96.91	96.91	96.95	1.87	3.3	7
DenseNet-121	809	P2	sipakmed_reinhard50	96.54	96.56	96.52	2.18	3.3	5
DenseNet-121	853	P1	sipakmed_reinhard50	97.41	97.40	97.44	1.12	3.3	5
DenseNet-121	853	P2	sipakmed_reinhard50	96.42	96.41	96.39	1.96	3.3	6
DenseNet-121	907	P1	sipakmed_reinhard50	97.90	97.91	97.92	1.48	3.3	6
DenseNet-121	907	P2	sipakmed_reinhard50	95.31	95.34	95.28	3.12	3.3	7
DenseNet-121	953	P1	sipakmed_reinhard50	97.16	97.16	97.17	1.39	3.3	7
DenseNet-121	953	P2	sipakmed_reinhard50	96.79	96.84	96.78	2.34	3.3	5
DenseNet-121	1009	P1	sipakmed_reinhard50	97.53	97.55	97.54	1.48	3.3	5
DenseNet-121	1009	P2	sipakmed_reinhard50	95.68	95.71	95.67	2.84	3.3	6
DenseNet-121	1051	P1	sipakmed_reinhard50	97.90	97.91	97.90	1.36	3.3	6
DenseNet-121	1051	P2	sipakmed_reinhard50	97.16	97.16	97.15	0.98	3.3	7
EfficientNet-B0	601	P1	sipakmed_reinhard100	95.93	95.96	95.94	3.67	1.8	5
EfficientNet-B0	601	P2	sipakmed_reinhard100	94.44	94.48	94.43	3.04	1.8	4
EfficientNet-B0	653	P1	sipakmed_reinhard100	96.30	96.33	96.28	2.15	1.8	5
EfficientNet-B0	653	P2	sipakmed_reinhard100	95.93	95.95	95.92	2.51	1.8	4
EfficientNet-B0	701	P1	sipakmed_reinhard100	95.68	95.66	95.70	2.93	1.8	5
EfficientNet-B0	701	P2	sipakmed_reinhard100	95.06	95.10	95.06	3.59	1.8	4
EfficientNet-B0	751	P1	sipakmed_reinhard100	95.93	95.94	95.95	3.01	1.8	5
EfficientNet-B0	751	P2	sipakmed_reinhard100	97.04	97.03	97.05	1.76	1.8	6
EfficientNet-B0	809	P1	sipakmed_reinhard100	96.67	96.66	96.70	2.18	1.8	6
EfficientNet-B0	809	P2	sipakmed_reinhard100	96.05	96.04	96.06	2.35	1.8	7
EfficientNet-B0	853	P1	sipakmed_reinhard100	95.31	95.29	95.33	2.51	1.8	7
EfficientNet-B0	853	P2	sipakmed_reinhard100	96.42	96.41	96.41	2.12	1.8	5
EfficientNet-B0	907	P1	sipakmed_reinhard100	97.04	97.05	97.04	1.92	1.8	5
EfficientNet-B0	907	P2	sipakmed_reinhard100	93.95	94.00	93.93	4.61	1.8	6
EfficientNet-B0	953	P1	sipakmed_reinhard100	96.17	96.16	96.19	2.98	1.8	6
EfficientNet-B0	953	P2	sipakmed_reinhard100	97.28	97.31	97.28	1.69	1.8	7
EfficientNet-B0	1009	P1	sipakmed_reinhard100	96.05	96.09	96.05	3.49	1.8	7
EfficientNet-B0	1009	P2	sipakmed_reinhard100	94.57	94.53	94.57	3.49	1.8	5
EfficientNet-B0	1051	P1	sipakmed_reinhard100	97.04	97.04	97.04	1.29	1.8	5
EfficientNet-B0	1051	P2	sipakmed_reinhard100	95.19	95.18	95.13	3.17	1.8	6
EfficientNet-B0	601	P1	sipakmed_reinhard50	97.16	97.17	97.18	2.23	1.8	6
EfficientNet-B0	601	P2	sipakmed_reinhard50	96.54	96.55	96.54	1.39	1.8	7
EfficientNet-B0	653	P1	sipakmed_reinhard50	98.52	98.53	98.52	0.78	1.8	6
EfficientNet-B0	653	P2	sipakmed_reinhard50	96.67	96.68	96.67	1.74	1.8	7
EfficientNet-B0	701	P1	sipakmed_reinhard50	97.41	97.40	97.41	1.77	1.8	6
EfficientNet-B0	701	P2	sipakmed_reinhard50	95.31	95.34	95.34	3.31	1.8	7
EfficientNet-B0	751	P1	sipakmed_reinhard50	97.78	97.79	97.80	1.37	1.8	6
EfficientNet-B0	751	P2	sipakmed_reinhard50	96.79	96.80	96.79	2.20	1.8	7
EfficientNet-B0	809	P1	sipakmed_reinhard50	97.41	97.41	97.44	1.75	1.8	7
EfficientNet-B0	809	P2	sipakmed_reinhard50	96.05	96.05	96.03	2.68	1.8	5
EfficientNet-B0	853	P1	sipakmed_reinhard50	96.67	96.65	96.69	2.12	1.8	5
EfficientNet-B0	853	P2	sipakmed_reinhard50	96.30	96.28	96.28	2.03	1.8	6
EfficientNet-B0	907	P1	sipakmed_reinhard50	97.53	97.55	97.54	1.35	1.8	6
EfficientNet-B0	907	P2	sipakmed_reinhard50	96.30	96.31	96.28	2.67	1.8	7
EfficientNet-B0	953	P1	sipakmed_reinhard50	96.67	96.68	96.67	2.18	1.8	7
EfficientNet-B0	953	P2	sipakmed_reinhard50	97.41	97.44	97.40	1.38	1.8	5
EfficientNet-B0	1009	P1	sipakmed_reinhard50	96.91	96.95	96.92	1.99	1.8	5
EfficientNet-B0	1009	P2	sipakmed_reinhard50	97.04	97.04	97.04	1.93	1.8	6
EfficientNet-B0	1051	P1	sipakmed_reinhard50	97.65	97.65	97.67	1.50	1.8	6
EfficientNet-B0	1051	P2	sipakmed_reinhard50	97.04	97.05	97.03	1.55	1.8	7
ConvNeXt-T	601	P1	sipakmed_reinhard100	96.05	96.07	96.05	3.46	5.2	5
ConvNeXt-T	601	P2	sipakmed_reinhard100	95.56	95.56	95.58	3.10	5.2	4
ConvNeXt-T	653	P1	sipakmed_reinhard100	97.90	97.92	97.91	1.67	5.2	5
ConvNeXt-T	653	P2	sipakmed_reinhard100	95.06	95.00	95.08	3.33	5.2	4
ConvNeXt-T	701	P1	sipakmed_reinhard100	96.30	96.31	96.31	2.67	5.2	5
ConvNeXt-T	701	P2	sipakmed_reinhard100	93.21	93.22	93.26	5.78	5.2	4
ConvNeXt-T	751	P1	sipakmed_reinhard100	96.42	96.44	96.46	2.76	5.3	5

Architecture	Seed	Protocol	cache_name	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	751	P2	sipakmed_reinhard100	96.17	96.16	96.16	2.65	5.9	4
ConvNeXt-T	809	P1	sipakmed_reinhard100	97.41	97.39	97.43	2.02	6.3	4
ConvNeXt-T	809	P2	sipakmed_reinhard100	96.30	96.30	96.27	2.19	5.2	6
ConvNeXt-T	853	P1	sipakmed_reinhard100	96.30	96.29	96.32	2.41	5.3	6
ConvNeXt-T	853	P2	sipakmed_reinhard100	96.54	96.51	96.55	2.70	5.5	7
ConvNeXt-T	907	P1	sipakmed_reinhard100	96.67	96.63	96.71	2.24	5.5	7
ConvNeXt-T	907	P2	sipakmed_reinhard100	94.57	94.65	94.53	3.86	5.4	5
ConvNeXt-T	953	P1	sipakmed_reinhard100	94.69	94.66	94.75	3.77	5.4	5
ConvNeXt-T	953	P2	sipakmed_reinhard100	96.05	96.06	96.08	3.19	5.2	6
ConvNeXt-T	1009	P1	sipakmed_reinhard100	93.58	93.63	93.57	4.53	6.4	4
ConvNeXt-T	1009	P2	sipakmed_reinhard100	94.81	94.82	94.81	3.60	5.5	7
ConvNeXt-T	1051	P1	sipakmed_reinhard100	95.80	95.80	95.85	3.45	5.5	7
ConvNeXt-T	1051	P2	sipakmed_reinhard100	96.42	96.41	96.43	2.74	5.4	5
ConvNeXt-T	601	P1	sipakmed_reinhard50	97.53	97.54	97.54	1.93	5.1	6
ConvNeXt-T	601	P2	sipakmed_reinhard50	97.16	97.17	97.17	2.18	5.2	7
ConvNeXt-T	653	P1	sipakmed_reinhard50	97.78	97.80	97.79	1.70	5.2	6
ConvNeXt-T	653	P2	sipakmed_reinhard50	92.84	92.79	92.78	5.21	5.2	7
ConvNeXt-T	701	P1	sipakmed_reinhard50	96.54	96.55	96.56	2.15	5.1	6
ConvNeXt-T	701	P2	sipakmed_reinhard50	93.70	93.68	93.77	4.95	5.2	7
ConvNeXt-T	751	P1	sipakmed_reinhard50	97.78	97.80	97.80	1.42	5.2	6
ConvNeXt-T	751	P2	sipakmed_reinhard50	95.06	94.96	94.99	3.64	5.4	7
ConvNeXt-T	809	P1	sipakmed_reinhard50	97.16	97.18	97.16	1.98	5.5	7
ConvNeXt-T	809	P2	sipakmed_reinhard50	95.80	95.77	95.76	3.40	5.4	5
ConvNeXt-T	853	P1	sipakmed_reinhard50	97.28	97.28	97.28	1.36	5.4	5
ConvNeXt-T	853	P2	sipakmed_reinhard50	97.04	97.02	97.01	1.67	6.3	4
ConvNeXt-T	907	P1	sipakmed_reinhard50	97.78	97.78	97.80	1.41	5.2	6
ConvNeXt-T	907	P2	sipakmed_reinhard50	94.94	94.94	94.96	3.97	6.3	4
ConvNeXt-T	953	P1	sipakmed_reinhard50	95.68	95.63	95.72	3.09	5.5	7
ConvNeXt-T	953	P2	sipakmed_reinhard50	96.30	96.32	96.32	2.95	6.3	4
ConvNeXt-T	1009	P1	sipakmed_reinhard50	96.42	96.46	96.44	2.40	5.4	5
ConvNeXt-T	1009	P2	sipakmed_reinhard50	95.68	95.66	95.71	3.15	5.3	6
ConvNeXt-T	1051	P1	sipakmed_reinhard50	96.17	96.23	96.17	2.66	5.3	6
ConvNeXt-T	1051	P2	sipakmed_reinhard50	96.91	96.91	96.92	2.42	6.4	4

Table M.4: Complete LEACE ledger (480 models): refitted-head accuracy and recoverability at rank 0 and rank 256 (percent).

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
ResNet-18	137	P1	97.78	97.53	22.8	1.8
ResNet-18	137	P2	96.54	96.42	23.8	2.5
ResNet-18	173	P1	97.90	97.53	25.0	2.0
ResNet-18	173	P2	97.78	97.53	22.9	2.0
ResNet-18	211	P1	97.65	98.02	24.2	2.6
ResNet-18	211	P2	96.91	96.54	22.3	2.8
ResNet-18	257	P1	96.79	97.28	24.8	1.8
ResNet-18	257	P2	98.02	98.40	24.2	2.4
ResNet-18	307	P1	98.52	98.40	22.1	2.5
ResNet-18	307	P2	97.78	97.28	22.9	2.9
ResNet-18	349	P1	97.41	97.28	22.3	1.9
ResNet-18	349	P2	97.16	95.80	22.4	2.5
ResNet-18	401	P1	98.15	97.90	21.9	1.7
ResNet-18	401	P2	98.52	98.77	24.1	1.9
ResNet-18	457	P1	98.40	97.90	25.2	1.8
ResNet-18	457	P2	96.05	94.44	21.8	2.0
ResNet-18	503	P1	98.15	98.27	24.4	2.1
ResNet-18	503	P2	96.91	96.54	23.2	2.4
ResNet-18	557	P1	98.27	97.53	23.3	2.0
ResNet-18	557	P2	97.65	97.78	25.2	2.6
ResNet-18	601	P1	98.02	97.78	23.2	2.0
ResNet-18	601	P2	97.41	96.91	22.9	2.9
ResNet-18	653	P1	97.78	98.02	21.9	2.2
ResNet-18	653	P2	97.28	97.16	24.1	2.2
ResNet-18	701	P1	97.78	97.41	22.8	1.5
ResNet-18	701	P2	96.67	95.80	21.6	1.9
ResNet-18	751	P1	97.78	97.90	21.0	1.9
ResNet-18	751	P2	96.54	96.17	23.8	1.8
ResNet-18	809	P1	97.78	97.65	24.8	1.7
ResNet-18	809	P2	96.91	96.42	23.6	2.9
ResNet-18	853	P1	98.02	97.53	23.2	2.7

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
ResNet-18	853	P2	97.28	97.41	21.9	3.1
ResNet-18	907	P1	97.78	97.90	23.1	1.4
ResNet-18	907	P2	97.78	97.90	22.5	2.6
ResNet-18	953	P1	98.02	97.53	21.5	1.9
ResNet-18	953	P2	97.65	97.90	22.9	0.8
ResNet-18	1009	P1	98.15	98.27	22.4	1.1
ResNet-18	1009	P2	97.28	96.91	25.5	2.2
ResNet-18	1051	P1	97.78	97.65	23.0	2.6
ResNet-18	1051	P2	98.02	97.41	23.5	2.4
ResNet-50	137	P1	98.15	97.90	25.7	10.4
ResNet-50	137	P2	96.30	96.30	26.2	11.2
ResNet-50	173	P1	98.15	98.52	23.6	10.8
ResNet-50	173	P2	97.78	98.27	26.9	10.3
ResNet-50	211	P1	97.78	98.15	24.6	10.6
ResNet-50	211	P2	97.16	97.04	24.5	10.1
ResNet-50	257	P1	98.27	98.02	23.8	9.4
ResNet-50	257	P2	97.04	97.53	25.5	11.2
ResNet-50	307	P1	98.89	98.15	23.0	10.1
ResNet-50	307	P2	97.28	97.41	22.4	9.8
ResNet-50	349	P1	97.41	97.41	25.5	10.0
ResNet-50	349	P2	96.67	96.91	25.1	10.0
ResNet-50	401	P1	98.64	97.90	22.5	9.7
ResNet-50	401	P2	97.41	97.53	24.5	9.1
ResNet-50	457	P1	98.40	98.40	25.2	10.8
ResNet-50	457	P2	95.56	95.06	25.1	10.6
ResNet-50	503	P1	98.15	98.15	24.8	12.3
ResNet-50	503	P2	97.28	97.28	23.2	11.6
ResNet-50	557	P1	97.90	97.65	25.4	11.1
ResNet-50	557	P2	98.27	98.02	25.2	10.1
ResNet-50	601	P1	98.15	98.40	23.8	9.8
ResNet-50	601	P2	98.52	97.65	25.1	9.7
ResNet-50	653	P1	98.15	98.40	26.1	12.1
ResNet-50	653	P2	97.53	97.28	25.9	11.4
ResNet-50	701	P1	98.02	97.53	23.5	10.4
ResNet-50	701	P2	97.04	96.91	25.6	10.7
ResNet-50	751	P1	98.40	98.27	25.5	11.8
ResNet-50	751	P2	96.17	96.05	23.7	9.7
ResNet-50	809	P1	97.90	97.41	25.9	11.4
ResNet-50	809	P2	97.28	97.41	23.9	10.0
ResNet-50	853	P1	97.16	96.79	25.5	11.2
ResNet-50	853	P2	96.91	96.79	26.4	10.4
ResNet-50	907	P1	98.77	98.15	23.0	9.0
ResNet-50	907	P2	97.16	97.78	25.2	8.6
ResNet-50	953	P1	97.78	97.53	24.8	10.6
ResNet-50	953	P2	97.04	96.91	24.5	9.6
ResNet-50	1009	P1	98.40	97.78	23.6	10.6
ResNet-50	1009	P2	97.53	97.16	23.6	9.9
ResNet-50	1051	P1	97.90	97.78	25.7	11.7
ResNet-50	1051	P2	97.28	97.53	24.5	8.6
DenseNet-121	137	P1	98.52	97.65	29.2	3.4
DenseNet-121	137	P2	97.16	97.04	29.1	4.4
DenseNet-121	173	P1	98.15	98.27	30.8	4.5
DenseNet-121	173	P2	98.52	98.52	29.6	4.7
DenseNet-121	211	P1	98.64	98.40	29.7	4.4
DenseNet-121	211	P2	97.41	97.65	28.5	3.4
DenseNet-121	257	P1	97.41	97.28	30.2	3.7
DenseNet-121	257	P2	97.65	96.79	29.1	4.4
DenseNet-121	307	P1	98.77	98.64	29.6	3.2
DenseNet-121	307	P2	98.40	98.27	30.4	5.0
DenseNet-121	349	P1	98.15	97.90	27.9	5.0
DenseNet-121	349	P2	97.65	97.04	27.7	4.0
DenseNet-121	401	P1	98.02	97.90	30.3	4.2
DenseNet-121	401	P2	97.53	97.65	29.8	3.7
DenseNet-121	457	P1	98.27	98.02	27.6	3.9
DenseNet-121	457	P2	96.17	95.80	29.2	4.7
DenseNet-121	503	P1	98.15	97.78	30.8	4.4
DenseNet-121	503	P2	97.90	98.02	27.5	3.4
DenseNet-121	557	P1	98.15	97.65	31.1	2.9
DenseNet-121	557	P2	98.77	98.64	30.3	5.3
DenseNet-121	601	P1	98.64	97.65	28.9	3.1
DenseNet-121	601	P2	98.40	97.53	31.6	4.1
DenseNet-121	653	P1	98.15	98.40	28.7	3.9
DenseNet-121	653	P2	97.53	96.54	28.9	3.8

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
DenseNet-121	701	P1	98.02	97.53	30.2	3.3
DenseNet-121	701	P2	97.16	97.53	29.1	5.2
DenseNet-121	751	P1	98.15	98.27	27.4	4.1
DenseNet-121	751	P2	96.91	96.91	27.9	4.4
DenseNet-121	809	P1	98.27	98.02	29.4	3.9
DenseNet-121	809	P2	97.78	97.53	29.5	5.1
DenseNet-121	853	P1	98.02	97.78	32.5	4.0
DenseNet-121	853	P2	96.54	96.05	30.5	5.7
DenseNet-121	907	P1	98.27	98.15	27.8	2.8
DenseNet-121	907	P2	96.91	96.91	26.3	4.1
DenseNet-121	953	P1	97.90	97.90	30.7	3.3
DenseNet-121	953	P2	97.78	97.41	29.1	4.4
DenseNet-121	1009	P1	98.52	98.02	29.2	2.7
DenseNet-121	1009	P2	97.16	97.28	29.0	4.6
DenseNet-121	1051	P1	98.64	98.52	28.9	3.4
DenseNet-121	1051	P2	97.16	97.65	31.5	3.9
EfficientNet-B0	137	P1	98.15	97.41	21.5	4.7
EfficientNet-B0	137	P2	97.41	97.41	23.5	5.3
EfficientNet-B0	173	P1	98.02	97.78	22.9	6.8
EfficientNet-B0	173	P2	98.15	98.02	23.9	4.1
EfficientNet-B0	211	P1	98.64	98.27	23.3	5.9
EfficientNet-B0	211	P2	97.53	97.16	23.8	7.0
EfficientNet-B0	257	P1	97.90	98.15	24.2	7.0
EfficientNet-B0	257	P2	96.42	97.04	22.2	5.3
EfficientNet-B0	307	P1	98.40	98.15	21.6	4.8
EfficientNet-B0	307	P2	96.79	96.79	21.7	5.8
EfficientNet-B0	349	P1	97.28	97.16	23.6	5.1
EfficientNet-B0	349	P2	97.90	98.15	24.8	5.8
EfficientNet-B0	401	P1	98.27	98.40	23.0	4.4
EfficientNet-B0	401	P2	96.91	96.79	21.3	5.0
EfficientNet-B0	457	P1	97.65	98.15	22.3	6.2
EfficientNet-B0	457	P2	94.57	95.19	22.6	5.1
EfficientNet-B0	503	P1	96.67	96.67	21.1	5.0
EfficientNet-B0	503	P2	97.28	97.78	19.9	5.0
EfficientNet-B0	557	P1	98.02	97.90	22.3	5.1
EfficientNet-B0	557	P2	98.40	98.15	24.3	6.0
EfficientNet-B0	601	P1	98.27	98.52	20.6	5.7
EfficientNet-B0	601	P2	97.65	97.41	23.0	5.2
EfficientNet-B0	653	P1	98.64	98.64	23.0	4.7
EfficientNet-B0	653	P2	97.04	97.53	24.9	6.6
EfficientNet-B0	701	P1	97.53	97.78	21.3	4.8
EfficientNet-B0	701	P2	97.16	97.16	22.2	5.4
EfficientNet-B0	751	P1	98.77	98.27	22.4	6.4
EfficientNet-B0	751	P2	97.41	97.16	21.1	6.8
EfficientNet-B0	809	P1	98.52	98.15	22.6	6.8
EfficientNet-B0	809	P2	96.91	97.16	23.0	5.4
EfficientNet-B0	853	P1	97.65	97.78	23.5	6.2
EfficientNet-B0	853	P2	96.54	96.05	22.8	7.0
EfficientNet-B0	907	P1	98.77	98.40	21.0	4.5
EfficientNet-B0	907	P2	97.53	97.53	21.9	4.6
EfficientNet-B0	953	P1	97.90	98.52	21.0	5.3
EfficientNet-B0	953	P2	98.40	98.02	21.7	3.9
EfficientNet-B0	1009	P1	98.15	98.15	21.3	5.1
EfficientNet-B0	1009	P2	97.53	97.04	22.8	4.8
EfficientNet-B0	1051	P1	97.65	97.65	23.7	5.8
EfficientNet-B0	1051	P2	98.52	98.27	23.8	5.9
MobileNet-V2	137	P1	97.04	97.16	19.0	4.0
MobileNet-V2	137	P2	97.16	96.42	23.3	4.0
MobileNet-V2	173	P1	97.90	97.78	20.0	3.7
MobileNet-V2	173	P2	97.65	98.02	20.3	3.3
MobileNet-V2	211	P1	97.41	97.41	19.7	3.1
MobileNet-V2	211	P2	95.56	95.43	20.8	3.9
MobileNet-V2	257	P1	97.04	96.67	22.3	4.7
MobileNet-V2	257	P2	96.17	96.79	20.6	4.4
MobileNet-V2	307	P1	98.52	98.40	22.2	4.4
MobileNet-V2	307	P2	95.80	96.05	21.3	4.1
MobileNet-V2	349	P1	97.41	96.67	19.2	4.6
MobileNet-V2	349	P2	96.42	96.05	21.1	3.9
MobileNet-V2	401	P1	97.65	97.41	21.5	3.9
MobileNet-V2	401	P2	97.04	96.79	19.7	4.1
MobileNet-V2	457	P1	97.16	97.16	19.2	3.7
MobileNet-V2	457	P2	94.57	94.81	20.0	4.0
MobileNet-V2	503	P1	97.16	97.16	19.9	5.4

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
MobileNet-V2	503	P2	96.79	96.17	20.8	3.9
MobileNet-V2	557	P1	96.54	96.67	20.9	3.8
MobileNet-V2	557	P2	98.02	98.02	20.8	4.8
MobileNet-V2	601	P1	97.78	97.41	19.9	4.6
MobileNet-V2	601	P2	97.28	96.79	20.5	4.1
MobileNet-V2	653	P1	97.78	97.28	21.7	4.7
MobileNet-V2	653	P2	96.05	96.30	22.5	3.3
MobileNet-V2	701	P1	96.91	97.04	19.3	3.7
MobileNet-V2	701	P2	96.05	96.17	20.0	4.7
MobileNet-V2	751	P1	96.79	96.54	21.1	4.6
MobileNet-V2	751	P2	95.68	95.31	21.5	5.2
MobileNet-V2	809	P1	97.04	96.67	21.8	4.8
MobileNet-V2	809	P2	96.91	96.91	21.6	4.2
MobileNet-V2	853	P1	97.53	97.65	21.7	5.9
MobileNet-V2	853	P2	96.05	96.30	20.0	3.8
MobileNet-V2	907	P1	98.15	97.28	19.3	2.9
MobileNet-V2	907	P2	97.41	97.28	19.8	3.7
MobileNet-V2	953	P1	97.41	96.79	21.8	4.7
MobileNet-V2	953	P2	97.90	96.54	20.8	4.7
MobileNet-V2	1009	P1	97.65	97.90	21.8	3.8
MobileNet-V2	1009	P2	96.54	96.17	22.2	5.4
MobileNet-V2	1051	P1	97.28	96.79	19.9	3.3
MobileNet-V2	1051	P2	97.16	97.04	21.5	3.5
ViT-B/16	137	P1	97.53	97.16	30.4	6.8
ViT-B/16	137	P2	97.28	96.30	30.7	5.5
ViT-B/16	173	P1	98.64	98.52	27.4	7.0
ViT-B/16	173	P2	97.53	97.53	26.5	4.5
ViT-B/16	211	P1	98.40	98.52	29.0	7.0
ViT-B/16	211	P2	96.42	96.42	29.6	7.1
ViT-B/16	257	P1	96.91	96.67	29.6	6.6
ViT-B/16	257	P2	97.28	96.79	28.8	6.4
ViT-B/16	307	P1	99.14	99.38	30.2	7.0
ViT-B/16	307	P2	96.17	96.54	30.0	5.9
ViT-B/16	349	P1	97.16	97.16	30.7	5.5
ViT-B/16	349	P2	96.79	96.79	29.6	7.4
ViT-B/16	401	P1	97.78	97.65	28.8	5.5
ViT-B/16	401	P2	97.53	97.16	30.2	6.8
ViT-B/16	457	P1	98.77	98.64	28.7	6.5
ViT-B/16	457	P2	94.69	95.31	29.6	5.1
ViT-B/16	503	P1	96.79	96.91	29.7	6.2
ViT-B/16	503	P2	96.67	97.04	29.5	5.9
ViT-B/16	557	P1	97.53	97.28	27.2	7.2
ViT-B/16	557	P2	97.53	97.04	29.6	8.0
ViT-B/16	601	P1	98.15	98.02	28.5	6.1
ViT-B/16	601	P2	97.90	97.53	30.7	5.0
ViT-B/16	653	P1	98.15	97.90	26.2	4.8
ViT-B/16	653	P2	97.65	97.53	26.8	6.5
ViT-B/16	701	P1	97.41	97.65	30.3	5.3
ViT-B/16	701	P2	96.54	96.54	30.8	7.1
ViT-B/16	751	P1	98.15	98.15	29.1	5.9
ViT-B/16	751	P2	95.93	95.80	28.7	7.0
ViT-B/16	809	P1	97.04	97.16	29.0	5.7
ViT-B/16	809	P2	95.68	95.31	30.2	7.2
ViT-B/16	853	P1	97.28	97.78	28.3	6.1
ViT-B/16	853	P2	96.79	96.67	28.5	6.2
ViT-B/16	907	P1	99.01	98.64	28.3	5.5
ViT-B/16	907	P2	96.42	96.42	29.1	6.1
ViT-B/16	953	P1	96.54	96.79	27.8	7.0
ViT-B/16	953	P2	97.53	97.78	29.1	4.5
ViT-B/16	1009	P1	98.02	97.90	28.4	5.0
ViT-B/16	1009	P2	96.91	96.91	29.1	6.4
ViT-B/16	1051	P1	98.15	98.15	28.3	5.9
ViT-B/16	1051	P2	97.28	97.41	30.9	6.7
ConvNeXt-T	137	P1	98.27	98.27	30.3	5.5
ConvNeXt-T	137	P2	97.28	96.67	32.3	6.7
ConvNeXt-T	173	P1	98.52	98.40	32.0	5.4
ConvNeXt-T	173	P2	98.15	98.27	28.4	4.7
ConvNeXt-T	211	P1	98.27	97.78	28.2	4.8
ConvNeXt-T	211	P2	97.04	97.16	28.9	6.0
ConvNeXt-T	257	P1	98.40	98.27	30.2	5.7
ConvNeXt-T	257	P2	97.16	97.16	28.9	4.2
ConvNeXt-T	307	P1	98.77	98.89	30.3	5.3
ConvNeXt-T	307	P2	96.79	97.41	30.7	6.0

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
ConvNeXt-T	349	P1	97.90	97.90	29.8	5.3
ConvNeXt-T	349	P2	97.04	97.04	27.6	4.8
ConvNeXt-T	401	P1	98.52	98.52	30.0	5.2
ConvNeXt-T	401	P2	98.15	98.15	29.6	6.4
ConvNeXt-T	457	P1	98.64	98.64	29.7	4.8
ConvNeXt-T	457	P2	95.93	95.68	27.2	5.3
ConvNeXt-T	503	P1	97.65	97.78	30.8	6.4
ConvNeXt-T	503	P2	97.90	97.78	29.5	5.4
ConvNeXt-T	557	P1	97.78	97.53	28.1	4.7
ConvNeXt-T	557	P2	99.01	98.89	31.8	5.5
ConvNeXt-T	601	P1	97.90	97.78	29.2	5.5
ConvNeXt-T	601	P2	97.53	97.28	30.8	6.5
ConvNeXt-T	653	P1	98.02	98.40	27.8	4.7
ConvNeXt-T	653	P2	97.53	97.41	31.1	4.8
ConvNeXt-T	701	P1	97.65	97.78	27.0	4.0
ConvNeXt-T	701	P2	96.05	96.05	26.4	4.4
ConvNeXt-T	751	P1	98.64	98.15	28.5	4.8
ConvNeXt-T	751	P2	96.67	96.79	31.4	5.0
ConvNeXt-T	809	P1	97.90	97.65	29.2	6.1
ConvNeXt-T	809	P2	96.91	96.91	30.7	5.0
ConvNeXt-T	853	P1	98.02	97.90	26.8	3.7
ConvNeXt-T	853	P2	96.67	96.79	28.9	4.6
ConvNeXt-T	907	P1	98.64	98.40	29.0	4.5
ConvNeXt-T	907	P2	97.41	97.41	30.5	5.1
ConvNeXt-T	953	P1	97.78	97.65	29.6	5.0
ConvNeXt-T	953	P2	97.78	98.02	30.1	6.0
ConvNeXt-T	1009	P1	98.27	97.90	29.8	5.3
ConvNeXt-T	1009	P2	97.41	97.41	29.0	4.6
ConvNeXt-T	1051	P1	98.40	98.40	30.1	4.8
ConvNeXt-T	1051	P2	97.53	97.41	30.0	4.7
Swin-T	137	P1	97.04	97.41	31.0	6.1
Swin-T	137	P2	96.54	96.67	28.9	6.2
Swin-T	173	P1	98.64	98.52	28.7	6.7
Swin-T	173	P2	97.41	97.65	26.5	7.3
Swin-T	211	P1	98.27	98.27	29.7	7.1
Swin-T	211	P2	96.79	96.54	30.3	5.8
Swin-T	257	P1	97.78	97.90	28.4	6.5
Swin-T	257	P2	97.90	97.41	27.7	7.1
Swin-T	307	P1	98.02	98.02	27.5	6.6
Swin-T	307	P2	97.16	97.04	30.5	7.5
Swin-T	349	P1	98.02	98.02	30.3	6.5
Swin-T	349	P2	96.91	96.67	29.7	6.1
Swin-T	401	P1	97.90	97.78	28.9	5.8
Swin-T	401	P2	97.16	97.53	27.4	6.7
Swin-T	457	P1	98.89	98.52	29.8	7.1
Swin-T	457	P2	95.31	96.17	31.2	7.0
Swin-T	503	P1	98.02	97.78	30.1	7.4
Swin-T	503	P2	98.02	97.78	30.1	7.4
Swin-T	557	P1	98.40	98.27	30.8	7.0
Swin-T	557	P2	98.40	97.78	30.3	8.4
Swin-T	601	P1	97.90	97.78	27.1	8.1
Swin-T	601	P2	98.02	98.40	27.8	6.7
Swin-T	653	P1	98.27	97.90	29.5	7.1
Swin-T	653	P2	97.28	97.28	28.7	5.9
Swin-T	701	P1	97.90	98.15	29.8	5.3
Swin-T	701	P2	96.67	96.54	29.2	7.4
Swin-T	751	P1	98.89	98.89	27.2	6.6
Swin-T	751	P2	96.42	96.79	29.8	7.9
Swin-T	809	P1	97.65	97.53	30.8	5.9
Swin-T	809	P2	95.80	96.42	27.8	5.8
Swin-T	853	P1	98.64	98.52	27.4	6.1
Swin-T	853	P2	97.41	97.04	26.8	5.9
Swin-T	907	P1	98.89	98.77	26.9	5.9
Swin-T	907	P2	96.91	96.91	29.6	6.4
Swin-T	953	P1	97.65	97.78	28.2	6.1
Swin-T	953	P2	98.15	97.90	26.4	5.8
Swin-T	1009	P1	98.77	98.64	30.7	6.4
Swin-T	1009	P2	97.65	96.91	27.6	6.4
Swin-T	1051	P1	97.90	98.27	27.0	4.7
Swin-T	1051	P2	98.15	98.02	29.6	6.6
RegNetY-1.6GF	137	P1	97.04	97.41	21.5	2.1
RegNetY-1.6GF	137	P2	96.17	95.93	22.4	4.4
RegNetY-1.6GF	173	P1	98.27	97.90	21.7	4.1

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
RegNetY-1.6GF	173	P2	98.15	97.78	20.6	2.9
RegNetY-1.6GF	211	P1	98.40	97.65	23.8	3.3
RegNetY-1.6GF	211	P2	96.17	95.80	21.2	3.1
RegNetY-1.6GF	257	P1	97.04	97.41	19.2	2.5
RegNetY-1.6GF	257	P2	96.79	96.91	21.0	4.7
RegNetY-1.6GF	307	P1	98.27	98.15	20.0	4.2
RegNetY-1.6GF	307	P2	96.54	96.30	20.0	3.3
RegNetY-1.6GF	349	P1	97.90	98.02	19.9	2.9
RegNetY-1.6GF	349	P2	97.28	96.54	20.2	2.2
RegNetY-1.6GF	401	P1	98.27	98.02	21.1	3.1
RegNetY-1.6GF	401	P2	97.65	97.41	20.6	2.7
RegNetY-1.6GF	457	P1	98.77	98.15	22.8	4.4
RegNetY-1.6GF	457	P2	95.43	95.06	19.0	3.5
RegNetY-1.6GF	503	P1	97.78	97.28	18.4	3.4
RegNetY-1.6GF	503	P2	97.28	96.67	21.3	3.3
RegNetY-1.6GF	557	P1	97.65	97.28	21.7	2.8
RegNetY-1.6GF	557	P2	97.90	97.65	21.9	3.2
RegNetY-1.6GF	601	P1	97.16	97.04	20.3	3.5
RegNetY-1.6GF	601	P2	98.02	97.53	19.7	2.8
RegNetY-1.6GF	653	P1	97.78	97.28	20.4	3.8
RegNetY-1.6GF	653	P2	97.04	97.41	22.2	3.8
RegNetY-1.6GF	701	P1	97.53	97.16	21.8	3.3
RegNetY-1.6GF	701	P2	95.68	95.43	23.1	4.4
RegNetY-1.6GF	751	P1	98.02	97.90	21.6	3.1
RegNetY-1.6GF	751	P2	96.54	96.30	21.2	2.6
RegNetY-1.6GF	809	P1	97.65	98.15	21.2	3.4
RegNetY-1.6GF	809	P2	97.28	96.54	22.5	3.3
RegNetY-1.6GF	853	P1	98.15	98.27	18.9	2.8
RegNetY-1.6GF	853	P2	96.42	96.54	23.0	3.7
RegNetY-1.6GF	907	P1	98.27	98.15	20.3	4.7
RegNetY-1.6GF	907	P2	96.17	95.93	19.8	3.3
RegNetY-1.6GF	953	P1	97.90	98.02	22.2	3.9
RegNetY-1.6GF	953	P2	98.27	98.27	20.3	3.3
RegNetY-1.6GF	1009	P1	97.90	98.02	23.1	2.4
RegNetY-1.6GF	1009	P2	96.17	95.68	20.6	3.1
RegNetY-1.6GF	1051	P1	97.65	97.16	20.2	2.7
RegNetY-1.6GF	1051	P2	97.16	96.67	21.3	4.0
EfficientNetV2-S	137	P1	96.79	96.79	19.2	1.7
EfficientNetV2-S	137	P2	96.30	95.80	20.3	5.2
EfficientNetV2-S	173	P1	98.02	97.78	18.8	3.5
EfficientNetV2-S	173	P2	98.02	98.15	20.9	5.1
EfficientNetV2-S	211	P1	98.02	98.15	19.1	3.8
EfficientNetV2-S	211	P2	96.79	96.54	19.7	1.8
EfficientNetV2-S	257	P1	96.91	97.04	20.9	3.5
EfficientNetV2-S	257	P2	97.16	97.16	20.8	4.4
EfficientNetV2-S	307	P1	98.40	98.27	20.3	1.2
EfficientNetV2-S	307	P2	96.05	96.67	18.3	4.5
EfficientNetV2-S	349	P1	97.53	97.78	18.4	3.8
EfficientNetV2-S	349	P2	97.41	97.04	21.1	4.1
EfficientNetV2-S	401	P1	97.90	97.90	17.9	5.0
EfficientNetV2-S	401	P2	97.53	97.16	22.1	5.1
EfficientNetV2-S	457	P1	98.15	98.27	21.9	2.6
EfficientNetV2-S	457	P2	95.31	95.93	21.0	4.1
EfficientNetV2-S	503	P1	97.41	97.04	18.8	2.8
EfficientNetV2-S	503	P2	95.93	96.05	21.3	5.4
EfficientNetV2-S	557	P1	97.78	97.90	19.8	4.8
EfficientNetV2-S	557	P2	99.26	98.77	18.5	4.6
EfficientNetV2-S	601	P1	97.28	97.53	19.9	4.1
EfficientNetV2-S	601	P2	97.78	96.91	21.6	5.1
EfficientNetV2-S	653	P1	97.65	97.65	19.7	2.7
EfficientNetV2-S	653	P2	96.17	96.42	21.6	4.7
EfficientNetV2-S	701	P1	97.04	96.79	19.2	1.9
EfficientNetV2-S	701	P2	95.43	95.31	21.0	5.3
EfficientNetV2-S	751	P1	97.90	97.90	21.7	4.7
EfficientNetV2-S	751	P2	96.05	96.17	20.0	4.8
EfficientNetV2-S	809	P1	98.02	97.78	20.3	4.8
EfficientNetV2-S	809	P2	96.91	96.67	22.6	5.1
EfficientNetV2-S	853	P1	97.28	97.04	20.4	1.7
EfficientNetV2-S	853	P2	96.67	95.93	20.9	4.5
EfficientNetV2-S	907	P1	98.52	98.27	17.2	3.9
EfficientNetV2-S	907	P2	96.67	96.30	20.6	5.2
EfficientNetV2-S	953	P1	97.90	97.78	20.2	5.3
EfficientNetV2-S	953	P2	98.52	98.52	19.8	4.8

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
EfficientNetV2-S	1009	P1	97.16	97.16	20.2	3.8
EfficientNetV2-S	1009	P2	96.17	96.30	19.1	2.4
EfficientNetV2-S	1051	P1	98.27	98.27	15.8	1.9
EfficientNetV2-S	1051	P2	97.04	97.28	16.3	1.5
MaxViT-T	137	P1	98.02	97.78	21.2	3.3
MaxViT-T	137	P2	96.42	96.67	24.4	5.4
MaxViT-T	173	P1	98.89	98.27	22.4	3.8
MaxViT-T	173	P2	96.79	96.91	22.3	4.6
MaxViT-T	211	P1	97.90	98.15	22.4	3.3
MaxViT-T	211	P2	96.30	96.54	22.6	2.8
MaxViT-T	257	P1	97.28	97.28	21.7	3.7
MaxViT-T	257	P2	97.53	97.65	21.3	2.9
MaxViT-T	307	P1	98.02	98.02	21.0	3.8
MaxViT-T	307	P2	97.04	96.67	21.1	4.1
MaxViT-T	349	P1	97.90	97.78	23.0	4.1
MaxViT-T	349	P2	97.04	97.28	21.1	5.1
MaxViT-T	401	P1	97.65	97.53	20.3	2.9
MaxViT-T	401	P2	97.78	97.78	22.1	3.8
MaxViT-T	457	P1	98.02	98.02	19.2	3.1
MaxViT-T	457	P2	95.43	95.68	21.3	3.4
MaxViT-T	503	P1	97.04	96.91	21.0	3.8
MaxViT-T	503	P2	97.53	97.65	23.7	4.5
MaxViT-T	557	P1	98.02	98.27	24.1	4.5
MaxViT-T	557	P2	98.52	98.89	25.1	5.4
MaxViT-T	601	P1	98.02	97.78	21.8	4.7
MaxViT-T	601	P2	97.90	97.53	24.8	3.2
MaxViT-T	653	P1	98.52	98.15	19.6	4.1
MaxViT-T	653	P2	97.78	97.90	20.2	2.1
MaxViT-T	701	P1	98.15	98.15	21.3	4.0
MaxViT-T	701	P2	95.93	96.05	22.8	3.5
MaxViT-T	751	P1	97.41	97.53	19.9	3.2
MaxViT-T	751	P2	96.79	96.91	20.8	3.5
MaxViT-T	809	P1	97.16	97.16	22.4	3.7
MaxViT-T	809	P2	96.91	96.54	23.1	4.5
MaxViT-T	853	P1	97.78	97.78	21.9	3.5
MaxViT-T	853	P2	96.67	96.54	25.7	5.0
MaxViT-T	907	P1	98.40	98.27	22.4	3.9
MaxViT-T	907	P2	96.05	96.17	22.6	3.9
MaxViT-T	953	P1	97.41	97.16	21.5	3.5
MaxViT-T	953	P2	98.15	98.02	23.0	4.2
MaxViT-T	1009	P1	98.02	97.90	21.8	4.0
MaxViT-T	1009	P2	97.28	97.16	22.5	4.2
MaxViT-T	1051	P1	97.90	98.15	22.8	4.8
MaxViT-T	1051	P2	98.15	97.90	21.6	3.5
DINOv2 ViT-S/14	137	P1	95.68	93.70	29.6	8.8
DINOv2 ViT-S/14	137	P2	93.21	92.47	32.9	12.5
DINOv2 ViT-S/14	173	P1	95.31	95.80	30.8	10.3
DINOv2 ViT-S/14	173	P2	93.83	91.73	30.9	9.2
DINOv2 ViT-S/14	211	P1	95.93	95.68	31.2	11.0
DINOv2 ViT-S/14	211	P2	93.95	93.33	29.6	10.5
DINOv2 ViT-S/14	257	P1	95.19	93.95	31.1	9.7
DINOv2 ViT-S/14	257	P2	93.46	93.09	30.0	9.9
DINOv2 ViT-S/14	307	P1	97.28	96.17	29.2	9.4
DINOv2 ViT-S/14	307	P2	95.31	93.70	31.0	9.4
DINOv2 ViT-S/14	349	P1	96.54	95.06	32.1	8.5
DINOv2 ViT-S/14	349	P2	93.70	92.84	28.3	7.4
DINOv2 ViT-S/14	401	P1	95.68	93.95	29.7	9.8
DINOv2 ViT-S/14	401	P2	95.43	94.69	28.1	9.8
DINOv2 ViT-S/14	457	P1	96.79	96.91	30.2	9.8
DINOv2 ViT-S/14	457	P2	92.84	92.96	30.4	8.7
DINOv2 ViT-S/14	503	P1	95.19	94.81	28.8	9.7
DINOv2 ViT-S/14	503	P2	95.06	94.57	29.5	10.3
DINOv2 ViT-S/14	557	P1	96.17	96.91	30.9	7.8
DINOv2 ViT-S/14	557	P2	94.94	94.20	33.1	10.0
DINOv2 ViT-S/14	601	P1	96.05	95.19	31.8	9.8
DINOv2 ViT-S/14	601	P2	93.21	90.86	30.2	9.9
DINOv2 ViT-S/14	653	P1	95.06	95.68	30.4	10.1
DINOv2 ViT-S/14	653	P2	94.32	93.09	28.9	11.0
DINOv2 ViT-S/14	701	P1	94.81	95.56	32.0	9.1
DINOv2 ViT-S/14	701	P2	91.48	91.11	29.4	8.7
DINOv2 ViT-S/14	751	P1	97.04	95.93	30.8	8.4
DINOv2 ViT-S/14	751	P2	93.33	93.83	30.7	9.8
DINOv2 ViT-S/14	809	P1	95.93	95.06	30.5	9.4

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
DINOv2 ViT-S/14	809	P2	94.20	93.83	30.8	8.3
DINOv2 ViT-S/14	853	P1	94.94	94.07	31.5	9.1
DINOv2 ViT-S/14	853	P2	94.44	93.09	28.9	10.6
DINOv2 ViT-S/14	907	P1	96.67	96.17	31.5	7.7
DINOv2 ViT-S/14	907	P2	92.84	91.73	30.3	11.1
DINOv2 ViT-S/14	953	P1	95.19	94.20	31.7	9.4
DINOv2 ViT-S/14	953	P2	93.95	94.81	29.0	8.6
DINOv2 ViT-S/14	1009	P1	96.79	96.42	31.6	9.8
DINOv2 ViT-S/14	1009	P2	95.43	94.32	32.5	8.8
DINOv2 ViT-S/14	1051	P1	96.05	95.31	30.8	8.3
DINOv2 ViT-S/14	1051	P2	94.20	92.10	31.2	10.5

Table M.5: Complete test-time ablation ledger (160 models; accuracy in percent).

Architecture	Seed	Cell	Orig.	Grey	Stain	Low-pass	Flat	No bg
ResNet-50	137	clean_high	97.32	64.75	90.04	70.11	92.72	88.31
ResNet-50	137	clean_low	96.36	56.70	90.80	51.92	93.87	91.00
ResNet-50	137	leak_high	97.13	63.22	89.85	59.77	93.68	89.46
ResNet-50	137	leak_low	97.89	68.20	90.80	80.65	95.40	82.38
ResNet-50	173	clean_high	95.55	49.32	88.20	39.85	92.26	89.36
ResNet-50	173	clean_low	94.97	70.79	89.17	63.25	91.68	90.52
ResNet-50	173	leak_high	96.71	61.32	88.01	49.13	92.07	90.14
ResNet-50	173	leak_low	96.71	53.77	84.91	74.27	91.49	91.30
ResNet-50	211	clean_high	94.80	57.03	84.01	42.58	92.49	91.71
ResNet-50	211	clean_low	94.22	54.14	85.93	56.26	87.28	89.98
ResNet-50	211	leak_high	96.15	59.73	87.67	40.27	92.29	91.33
ResNet-50	211	leak_low	97.50	64.16	86.71	61.66	94.03	93.45
ResNet-50	257	clean_high	94.07	42.45	88.34	59.46	91.20	91.59
ResNet-50	257	clean_low	95.03	59.85	89.29	68.64	90.44	92.54
ResNet-50	257	leak_high	96.56	52.77	90.06	53.15	94.84	91.20
ResNet-50	257	leak_low	96.75	59.27	90.82	61.57	93.50	91.97
ResNet-50	307	clean_high	94.92	58.59	90.82	59.57	90.04	90.43
ResNet-50	307	clean_low	93.95	65.23	91.60	60.35	88.28	87.89
ResNet-50	307	leak_high	95.51	64.84	91.80	66.60	87.11	90.82
ResNet-50	307	leak_low	97.85	52.34	85.35	54.88	87.50	93.16
ResNet-50	349	clean_high	92.29	61.09	87.97	33.83	88.16	85.90
ResNet-50	349	clean_low	94.74	66.73	93.42	47.18	89.29	88.35
ResNet-50	349	leak_high	94.92	57.52	87.41	51.88	87.78	87.59
ResNet-50	349	leak_low	97.18	65.41	91.92	68.23	93.61	91.92
ResNet-50	401	clean_high	92.07	66.15	86.46	55.32	89.17	85.30
ResNet-50	401	clean_low	93.23	57.83	88.39	41.59	88.20	88.97
ResNet-50	401	leak_high	95.16	69.05	89.17	66.92	89.94	90.14
ResNet-50	401	leak_low	94.78	56.48	88.59	53.19	92.07	89.17
ResNet-50	457	clean_high	96.34	66.18	85.92	45.52	91.96	89.58
ResNet-50	457	clean_low	95.80	63.62	85.37	54.66	87.75	88.85
ResNet-50	457	leak_high	96.89	58.68	85.74	69.47	90.68	89.58
ResNet-50	457	leak_low	96.16	66.54	89.03	52.83	91.59	93.05
ResNet-50	503	clean_high	95.47	55.85	82.83	62.45	92.26	88.49
ResNet-50	503	clean_low	95.85	64.91	84.91	71.89	90.94	91.13
ResNet-50	503	leak_high	96.42	57.17	81.70	51.89	93.77	89.25
ResNet-50	503	leak_low	97.55	54.91	82.26	39.81	93.40	90.75
ResNet-50	557	clean_high	93.64	63.09	86.73	64.55	85.64	88.36
ResNet-50	557	clean_low	94.73	46.91	84.91	40.18	89.27	88.91
ResNet-50	557	leak_high	95.64	69.09	87.82	67.82	85.82	86.18
ResNet-50	557	leak_low	95.64	53.64	86.55	38.18	87.27	89.09
DenseNet-121	137	clean_high	96.93	56.70	92.72	71.26	93.49	70.88
DenseNet-121	137	clean_low	94.25	53.83	86.40	68.39	89.46	77.59
DenseNet-121	137	leak_high	97.51	53.64	90.23	71.26	94.25	76.44
DenseNet-121	137	leak_low	96.36	60.73	87.93	61.30	91.57	75.67
DenseNet-121	173	clean_high	94.58	66.92	91.30	62.48	93.23	81.24
DenseNet-121	173	clean_low	96.32	72.92	89.75	76.02	91.49	77.37
DenseNet-121	173	leak_high	96.52	62.86	92.07	82.40	93.42	85.88
DenseNet-121	173	leak_low	97.68	61.12	91.88	86.07	89.17	81.04
DenseNet-121	211	clean_high	95.95	49.52	89.79	77.46	89.60	79.00
DenseNet-121	211	clean_low	95.18	63.78	87.67	70.33	91.91	91.71
DenseNet-121	211	leak_high	97.69	61.08	89.79	75.53	95.76	84.01
DenseNet-121	211	leak_low	97.69	65.32	85.36	84.59	90.75	92.10
DenseNet-121	257	clean_high	94.84	56.02	89.87	73.04	92.16	90.06
DenseNet-121	257	clean_low	96.75	52.77	91.20	79.35	94.07	87.95

Architecture	Seed	Cell	Orig.	Grey	Stain	Low-pass	Flat	No bg
DenseNet-121	257	leak_high	95.98	47.99	88.15	82.03	92.16	87.19
DenseNet-121	257	leak_low	97.51	58.32	89.87	67.11	93.69	83.56
DenseNet-121	307	clean_high	94.34	57.03	90.43	59.57	90.04	73.83
DenseNet-121	307	clean_low	94.14	66.99	89.84	63.09	91.80	81.84
DenseNet-121	307	leak_high	98.05	73.24	92.97	79.88	93.55	83.98
DenseNet-121	307	leak_low	97.85	62.11	91.80	69.34	93.55	80.66
DenseNet-121	349	clean_high	92.86	46.62	93.42	69.74	88.91	83.83
DenseNet-121	349	clean_low	90.41	44.17	88.91	69.17	81.95	85.15
DenseNet-121	349	leak_high	96.80	61.84	93.42	75.38	86.28	84.96
DenseNet-121	349	leak_low	96.80	63.91	93.05	79.51	90.23	85.71
DenseNet-121	401	clean_high	93.04	62.67	89.56	72.15	92.46	84.91
DenseNet-121	401	clean_low	94.58	67.50	90.14	63.44	91.88	88.01
DenseNet-121	401	leak_high	95.94	74.66	91.88	65.96	93.42	88.01
DenseNet-121	401	leak_low	95.94	58.22	89.56	72.34	93.23	85.49
DenseNet-121	457	clean_high	95.61	56.12	88.48	68.01	91.41	83.91
DenseNet-121	457	clean_low	94.33	49.73	89.03	73.49	91.22	83.91
DenseNet-121	457	leak_high	97.62	58.87	90.31	82.45	93.05	86.11
DenseNet-121	457	leak_low	97.07	54.84	91.22	78.43	91.41	85.01
DenseNet-121	503	clean_high	95.66	70.94	85.85	74.72	94.15	79.62
DenseNet-121	503	clean_low	95.85	60.75	87.17	75.47	92.26	80.38
DenseNet-121	503	leak_high	96.04	61.13	87.74	76.42	94.72	84.34
DenseNet-121	503	leak_low	97.17	71.70	86.42	84.53	93.77	82.83
DenseNet-121	557	clean_high	94.00	56.55	91.27	71.45	88.91	80.36
DenseNet-121	557	clean_low	94.91	49.45	90.00	83.27	83.27	83.82
DenseNet-121	557	leak_high	95.45	62.55	90.55	75.27	90.91	85.09
DenseNet-121	557	leak_low	97.09	61.09	90.36	75.45	91.64	81.45
ViT-B/16	137	clean_high	95.79	54.02	90.04	94.64	92.53	90.61
ViT-B/16	137	clean_low	96.36	50.38	86.97	96.36	89.46	90.61
ViT-B/16	137	leak_high	98.28	54.21	91.38	96.93	93.68	92.72
ViT-B/16	137	leak_low	94.83	40.80	86.78	92.91	89.85	91.57
ViT-B/16	173	clean_high	88.97	70.60	83.95	86.07	84.33	86.85
ViT-B/16	173	clean_low	90.52	68.28	84.53	85.69	79.30	81.43
ViT-B/16	173	leak_high	95.94	59.57	88.59	94.97	90.72	92.65
ViT-B/16	173	leak_low	95.16	59.57	89.56	94.00	86.07	91.10
ViT-B/16	211	clean_high	96.15	61.27	88.63	95.57	87.48	91.52
ViT-B/16	211	clean_low	95.38	57.23	88.05	95.18	86.71	93.45
ViT-B/16	211	leak_high	97.30	61.46	91.52	97.50	89.21	90.37
ViT-B/16	211	leak_low	98.46	61.85	89.79	97.88	91.33	94.61
ViT-B/16	257	clean_high	93.88	51.24	92.16	92.35	91.20	82.79
ViT-B/16	257	clean_low	93.69	70.36	92.35	91.97	90.82	86.81
ViT-B/16	257	leak_high	94.84	56.21	92.93	93.31	91.97	88.72
ViT-B/16	257	leak_low	97.32	55.83	91.78	96.56	90.63	92.93
ViT-B/16	307	clean_high	95.90	56.64	92.58	95.70	88.09	93.95
ViT-B/16	307	clean_low	93.36	52.15	90.62	94.34	88.48	86.72
ViT-B/16	307	leak_high	95.90	53.12	91.80	93.75	90.04	91.80
ViT-B/16	307	leak_low	95.51	55.27	89.65	93.55	86.72	88.87
ViT-B/16	349	clean_high	92.29	49.25	91.35	89.66	90.60	86.84
ViT-B/16	349	clean_low	94.92	57.14	92.29	93.05	92.11	87.59
ViT-B/16	349	leak_high	95.11	60.90	90.04	90.98	88.53	87.03
ViT-B/16	349	leak_low	96.80	62.78	93.23	96.05	94.92	91.17
ViT-B/16	401	clean_high	93.62	64.41	90.91	93.04	91.10	86.65
ViT-B/16	401	clean_low	92.84	47.20	91.49	91.88	92.46	89.94
ViT-B/16	401	leak_high	95.16	54.93	88.39	93.23	89.56	91.30
ViT-B/16	401	leak_low	96.13	53.58	93.23	94.78	90.72	92.26
ViT-B/16	457	clean_high	95.61	52.83	93.42	94.52	89.95	90.86
ViT-B/16	457	clean_low	95.80	58.50	92.69	95.80	91.22	88.67
ViT-B/16	457	leak_high	94.88	51.55	85.01	93.24	89.03	85.37
ViT-B/16	457	leak_low	95.98	52.83	89.95	94.88	91.41	86.47
ViT-B/16	503	clean_high	93.21	63.96	84.15	92.64	83.02	84.53
ViT-B/16	503	clean_low	92.26	64.15	83.40	91.13	89.81	84.72
ViT-B/16	503	leak_high	96.98	58.11	88.11	95.85	90.38	92.45
ViT-B/16	503	leak_low	95.47	56.60	85.47	94.34	92.83	88.49
ViT-B/16	557	clean_high	95.09	65.64	90.00	93.82	83.64	90.55
ViT-B/16	557	clean_low	96.00	53.09	88.36	94.00	88.91	93.64
ViT-B/16	557	leak_high	95.64	52.18	90.00	93.64	85.64	86.73
ViT-B/16	557	leak_low	94.55	49.45	92.18	95.64	84.73	85.82
ConvNeXt-T	137	clean_high	96.93	64.75	93.68	85.82	96.17	89.08
ConvNeXt-T	137	clean_low	94.64	61.49	92.34	81.61	91.19	88.89
ConvNeXt-T	137	leak_high	97.13	71.46	94.64	91.38	94.83	91.19
ConvNeXt-T	137	leak_low	95.98	55.36	92.91	83.91	90.80	91.38
ConvNeXt-T	173	clean_high	95.36	64.41	90.72	79.50	94.58	88.39
ConvNeXt-T	173	clean_low	94.20	66.15	86.65	87.23	92.65	85.49
ConvNeXt-T	173	leak_high	95.36	67.12	89.75	89.17	91.10	92.07

Architecture	Seed	Cell	Orig.	Grey	Stain	Low-pass	Flat	No bg
ConvNeXt-T	173	leak_low	95.16	70.41	88.01	83.75	94.58	87.43
ConvNeXt-T	211	clean_high	96.34	64.93	89.98	83.04	92.68	89.98
ConvNeXt-T	211	clean_low	97.88	69.36	92.68	91.14	93.45	91.71
ConvNeXt-T	211	leak_high	99.04	64.93	93.45	86.32	94.61	92.68
ConvNeXt-T	211	leak_low	95.95	62.43	90.56	89.60	93.83	92.49
ConvNeXt-T	257	clean_high	94.07	69.02	90.63	77.25	92.16	89.48
ConvNeXt-T	257	clean_low	94.26	56.02	92.16	83.56	89.10	89.67
ConvNeXt-T	257	leak_high	95.98	63.10	96.18	79.54	93.50	90.06
ConvNeXt-T	257	leak_low	97.13	65.97	94.46	86.81	93.50	94.07
ConvNeXt-T	307	clean_high	97.07	71.48	95.51	92.38	92.38	93.36
ConvNeXt-T	307	clean_low	96.68	71.88	94.73	86.72	89.84	92.58
ConvNeXt-T	307	leak_high	98.63	69.92	95.90	88.28	92.77	92.19
ConvNeXt-T	307	leak_low	98.24	73.83	95.12	84.77	94.53	92.77
ConvNeXt-T	349	clean_high	93.98	64.66	93.80	71.24	93.42	87.59
ConvNeXt-T	349	clean_low	96.05	63.16	94.17	91.73	93.42	88.72
ConvNeXt-T	349	leak_high	95.30	66.17	94.17	89.66	93.05	83.46
ConvNeXt-T	349	leak_low	95.11	63.91	93.80	88.53	92.11	91.73
ConvNeXt-T	401	clean_high	93.42	66.34	91.30	83.56	89.56	89.36
ConvNeXt-T	401	clean_low	93.62	65.18	93.04	64.60	91.88	89.36
ConvNeXt-T	401	leak_high	95.74	70.60	93.04	70.60	94.58	89.94
ConvNeXt-T	401	leak_low	93.62	50.68	90.52	62.67	91.49	88.20
ConvNeXt-T	457	clean_high	95.80	67.09	94.52	87.39	94.52	90.49
ConvNeXt-T	457	clean_low	91.77	51.01	88.12	81.54	89.40	89.03
ConvNeXt-T	457	leak_high	95.98	65.81	92.69	83.73	94.52	89.21
ConvNeXt-T	457	leak_low	96.53	69.29	94.52	75.87	94.88	88.67
ConvNeXt-T	503	clean_high	95.09	51.32	89.25	84.72	92.64	93.40
ConvNeXt-T	503	clean_low	95.66	67.36	87.36	88.49	93.02	91.70
ConvNeXt-T	503	leak_high	95.66	68.87	87.55	89.81	94.34	92.64
ConvNeXt-T	503	leak_low	96.23	64.15	89.06	88.11	94.34	92.26
ConvNeXt-T	557	clean_high	94.00	62.18	90.36	89.64	91.27	92.00
ConvNeXt-T	557	clean_low	94.73	57.64	92.91	84.00	90.91	90.36
ConvNeXt-T	557	leak_high	94.00	67.09	88.91	74.18	90.55	88.73
ConvNeXt-T	557	leak_low	96.00	67.64	90.91	84.36	90.00	90.73

Table M.6: Every smoke test (Study 2 before registration; Amendment 8 before registration). Only crop-level, leak-cell, or single-protocol runs were smoke-tested, so no registered contrast could be observed.

Run	Kind	Architecture	Protocol	Epochs	Seconds
cric_p1_resnet18	train	resnet18	P1	1.0	8
leak_high_resnet50	train	resnet50	leak_high	1.0	7
p1_convnext_tiny	train	convnext_tiny	P1	1.0	27
p1_densenet121	train	densenet121	P1	1.0	12
p1_dinov2_vits14	train	dinov2_vits14	P1	1.0	13
p1_dinov2_vits14_math	train	dinov2_vits14	P1	1.0	15
p1_efficientnet_b0	train	efficientnet_b0	P1	1.0	7
p1_efficientnet_v2_s	train	efficientnet_v2_s	P1	1.0	13
p1_grl_resnet50	train	resnet50	P1	1.0	8
p1_maxvit_t	train	maxvit_t	P1	1.0	28
p1_mobilenet_v2	train	mobilenet_v2	P1	1.0	6
p1_regnet_y_1_6gf	train	regnet_y_1_6gf	P1	1.0	9
p1_resnet18	train	resnet18	P1	1.0	4
p1_resnet18_repeat	train	resnet18	P1	1.0	4
p1_resnet50	train	resnet50	P1	1.0	10
p1_swin_t	train	swin_t	P1	1.0	17
p1_vit_b_16	train	vit_b_16	P1	1.0	35
p1_vit_b_16_math	train	vit_b_16	P1	1.0	32
p1_vit_b_16_math_repeat	train	vit_b_16	P1	1.0	32
sipakmed_resnet50_leak_high_seed1103_colour	train	resnet50	leak_high	1.0	6
sipakmed_vit_b_16_leak_high_seed1103_masked	train	vit_b_16	leak_high	1.0	15
sipakmed_convnext_tiny_leak_high_seed137	factorial_followup	convnext_tiny	leak_high	nan	15
sipakmed_densenet121_leak_high_seed137	factorial_followup	densenet121	leak_high	nan	8
sipakmed_resnet50_leak_high_seed137	factorial_followup	resnet50	leak_high	nan	8

Run	Kind	Architecture	Protocol	Epochs	Seconds
sipakmed__vit__b__16__leak__high__seed137	factorial_followup	vit__b__16	leak__high	nan	11
sipakmed__resnet18__p1__seed137	leace	resnet18	P1	nan	25
sipakmed__vit__b__16__p1__seed137	leace	vit__b__16	P1	nan	26
sipakmed__convnext__tiny__p1__seed601__linadv1	train	convnext__tiny	P1	2.0	41
sipakmed__densenet121__p1__seed601__linadv3	train	densenet121	P1	1.0	12
sipakmed__efficientnet__b0__p1__seed601__reinhard100	train	efficientnet__b0	P1	1.0	7

N

Variance components of every registered model

Table N.1: Variance components (REML) of every registered mixed model. Correlations of random intercepts and slopes are listed as 'corr.'.

Model	Component	Variance	SD / corr.
H1	split: (Intercept)	0.2271	0.477
	seed: (Intercept)	0.0000	0.000
	architecture: (Intercept)	1.2769	1.130
	architecture: x	0.0104	0.102
	architecture: (Intercept) × x	–	−1.000 (corr.)
	Residual: residual	0.5915	0.769
P2 − P3	split: (Intercept)	0.4033	0.635
	seed: (Intercept)	0.1955	0.442
	architecture: (Intercept)	2.0081	1.417
	architecture: x	0.2253	0.475
	architecture: (Intercept) × x	–	−1.000 (corr.)
	Residual: residual	0.6948	0.834
H2	split: (Intercept)	0.6662	0.816
	seed: (Intercept)	0.0295	0.172
	architecture: (Intercept)	0.2783	0.528
	architecture: x	0.0220	0.148
	architecture: (Intercept) × x	–	−0.639 (corr.)
	Residual: residual	0.7960	0.892
H3	training_set: (Intercept)	0.0000	0.000
	seed: (Intercept)	0.4778	0.691
	architecture: (Intercept)	0.0776	0.279
	architecture: leak	0.0963	0.310
	architecture: (Intercept) × leak	–	0.001 (corr.)
	Residual: residual	1.4223	1.193
H4	training_set: (Intercept)	0.1932	0.440
	seed: (Intercept)	0.7472	0.864
	architecture: (Intercept)	0.1369	0.370
	architecture: leak	0.1412	0.376
	architecture: (Intercept) × leak	–	1.000 (corr.)
	Residual: residual	1.6183	1.272
H5	seed: (Intercept)	1.0619	1.031
	Residual: residual	0.6736	0.821
F2	run: (Intercept)	0.0000	0.000
	training_set: (Intercept)	0.0000	0.000
	seed: (Intercept)	0.7919	0.890
	architecture: (Intercept)	0.7394	0.860
	architecture: leak	0.0004	0.021
	architecture: (Intercept) × leak	–	−1.000 (corr.)
	Residual: residual	4.0889	2.022
F3	training_set: (Intercept)	0.1245	0.353
	seed: (Intercept)	0.9803	0.990
	architecture: (Intercept)	0.2018	0.449
	architecture: leak	0.0042	0.065

Model	Component	Variance	SD / corr.
	architecture: (Intercept) × leak	–	−1.000 (corr.)
	Residual: residual	1.2390	1.113
F4	training_set: (Intercept)	0.2028	0.450
	seed: (Intercept)	0.7043	0.839
	architecture: (Intercept)	0.3280	0.573
	architecture: leak	0.1868	0.432
	architecture: (Intercept) × leak	–	1.000 (corr.)
	Residual: residual	1.7444	1.321
F5	seed: (Intercept)	0.0000	0.000
	architecture: (Intercept)	0.0017	0.042
	Residual: residual	0.3183	0.564
F6	seed: (Intercept)	0.3559	0.597
	Residual: residual	0.9056	0.952
F7	seed: (Intercept)	0.2447	0.495
	Residual: residual	1.4134	1.189
F8	block: (Intercept)	0.4937	0.703
	Residual: residual	0.4997	0.707

O

Complete commit chronology

Table O.1: Complete commit chronology of the project (all branches; IST).

Commit	Date	Message
67e8ded	2026-08-22 04:54	preregister confirmatory analysis
8ebc6d5	2026-08-22 05:27	build leakage verification pipeline
12c9de7	2026-08-22 05:31	add parent identity probe and complete source checksums
4ba6a96	2026-08-22 05:34	make image provenance recovery independent
1dbd6ac	2026-08-22 05:35	separate provenance recovery stages
c551f2d	2026-08-22 05:38	add matched classification harness
14a1569	2026-08-22 05:39	add near duplicate audit and P3 grouping
79b798a	2026-08-22 05:40	amend SIPaKMeD class order before analysis
70b5ffa	2026-08-22 05:44	add bounded GPU probe launcher
4aa3e60	2026-08-22 05:49	fail closed on image provenance confidence
c18d393	2026-08-22 05:54	validate Cx22 against declared source images
f7ca331	2026-08-22 05:54	add bounded matched evaluation launcher
318120f	2026-08-22 05:55	summarize matched runs in one correction family
8f6f76e	2026-08-22 05:57	complete preregistered probe ablations
680fa53	2026-08-22 05:58	record reproducible model run metadata
0c5166e	2026-08-22 06:00	checkpoint exhaustive provenance matching
ea053e9	2026-08-22 06:01	verify structural protocol ordering
65b0230	2026-08-22 06:03	export fixed protocol split assets
3261dc0	2026-08-22 06:06	render fail-closed verification contract
79e5b31	2026-08-22 06:07	require named provenance evidence in report
cded044	2026-08-22 06:09	measure literal cross-split pixel overlap
55708d4	2026-08-22 06:09	keep training commands free of analysis-only imports
6369698	2026-08-22 06:11	generate self-contained thesis figure
da548a4	2026-08-22 06:12	bound clean reproduction matcher to three workers
83d56af	2026-08-22 06:14	verify published class crop counts
03e7442	2026-08-22 06:15	document fail-closed release workflow
be46a53	2026-08-22 06:17	separate canonical and audit provenance maps
c617374	2026-08-22 06:23	bound provenance image cache to class working set
4c6282e	2026-08-22 07:00	use multiscale provenance matching only as fallback
67892b7	2026-08-22 07:19	align provenance validation with registered criteria
268e56a	2026-08-22 07:34	build explicit section 11 acceptance ledger
438fa3f	2026-08-22 07:36	measure duplicate detector precision and recall
62a51ef	2026-08-22 07:39	record virgin-host reproduction evidence
dd0a56e	2026-08-22 07:41	generate five-seed parent probe figure
b90805a	2026-08-22 07:45	reproduce P3 splits and pixel overlap end to end
82e541d	2026-08-22 07:56	add matched protocol results figure
adf5539	2026-08-22 08:04	document full matched classifier matrix
bf8a4ec	2026-08-22 08:06	enforce paper table reporting contract
ab40681	2026-08-22 09:32	separate Figure 2 control labels
a72fb5f	2026-08-22 09:55	record split and image library versions
4da9f62	2026-08-22 10:05	pin CUDA 12.8 training environment
f936802	2026-08-22 10:44	verify classification matrix integrity
157b882	2026-08-22 10:46	report secondary ordering violations
10fb2a3	2026-08-22 10:48	pin reproducible split runtime
970c74e	2026-08-22 10:58	gate acceptance on matrix integrity
8ebb2b9	2026-08-22 11:15	simulate exact P1 allocation rule
525df1d	2026-08-22 12:14	report architecture ranking stability

Commit	Date	Message
1c2da7f	2026-08-22 13:22	record preregistered provenance and five-seed results
2c5d5c3	2026-08-22 13:23	record post-result acceptance evidence
d347b07	2026-08-24 03:15	preregister Cx22 segmentation rescore
a58f72f	2026-08-24 03:24	add resumable Cx22 segmentation extension
635e660	2026-08-24 03:33	decode Cx22 instance labels with bounded memory
2fdb136	2026-08-24 03:46	refresh Cx22 controller heartbeat during epochs
d66dba4	2026-08-24 03:55	add memory-mapped Cx22 training cache
a18ddef	2026-08-24 04:01	preregister mixed-precision Cx22 execution
f10c9c5	2026-08-24 04:02	enable preregistered mixed precision for Cx22
48c9b65	2026-08-28 05:53	stabilize Cx22 MAnet training in FP32
373276a	2026-08-28 06:00	enforce deterministic CUDA workspace for Cx22
a9594b3	2026-09-06 23:32	remove redundant Cx22 instance runtime overhead
01c4c31	2026-09-06 23:38	stage Cx22 checkpoints off network storage
6179cfa	2026-10-18 05:01	add GPU-resident Study 2 training stack
6c12dd2	2026-10-18 05:01	add leakage-by-diversity factorial and transplant designs
d97ea01	2026-10-18 05:01	fetch CRIC slide images with checksums
8bdfd47	2026-10-18 05:01	add Study 2 power simulation and analysis code
5f15787	2026-10-18 05:07	export registered Study 2 splits and run matrix
a62e8e7	2026-10-18 05:07	preregister Study 2
24d372d	2026-10-18 05:12	add structural leakage-risk report and Cx22 literature check
f6f6ab3	2026-10-18 05:21	start single-column Study 2 manuscript and analysis driver
ee057a8	2026-10-18 05:24	measure literal source-pixel overlap for CRIC and Cx22 crop splits
d3dd9f4	2026-10-18 06:18	add Study 2 result figures
66b4091	2026-10-18 06:18	add Study 2 archive-and-vacate script
582327a	2026-10-18 06:19	document Study 2 in the README
7fe2e44	2026-10-18 06:25	dry-run the full Study 2 analysis on synthetic run outputs
ffd9d5a	2026-10-18 06:26	record post hoc pooled reanalyses of Study 1
7b8c96f	2026-10-18 06:29	amend Study 2 failure handling before any 2A run
035ef4d	2026-10-18 06:30	draft outcome-independent manuscript sections
965f20a	2026-10-19 13:42	implement Amendment 8 follow-up experiments (code and designs, not yet registered)
80171fe	2026-10-19 14:02	register Amendment 8: follow-up experiments on how leakage is used
458817d	2026-10-19 14:13	record the registered Study 2 results and draft their manuscript sections
43b6973	2026-10-20 01:25	Amendment 9: fix carrier condition labelling before any F3/F4 estimate
b958b8b	2026-10-20 03:26	record the registered Amendment 8 follow-up results and complete the manuscript
8f4e2f3	2026-10-20 03:27	summarise Study 2 and Amendment 8 results in the README

P

Registration hashes

SHA-256 hashes of every file and cache fixed by the Study 2 registration and by Amendments 7, 8, and 9, as recorded in `reports/study2/registration_hashes.txt`.

```
7b0cafa1d6b305c5c176cbd450bc0e081b515b417c4a901285488ffaaecad79d PREREGISTRATION_STUDY2.md
7822d4a344b06d70629bc7e4419a3a08da9e6c3da44db4f4e163588a0e3f8e8e analysis/study2/protocol_lmm.R
99ac4ebc60c20da9e08004d69d7c5e6a39c3747d6fd984ca7b9a46857a7f4f58 analysis/study2/mechanism_lmm.R
1dd75a0ec895ef93eebe5da8b1948e357ed92b9cb72b033be0af66a595114ada analysis/study2/power_simulation.R
39d2c989c1801cc045f96f17e79509b788d079c744a5e325d2973211172fa6c5 analysis/study2/collect_runs.py
842af403e275229df60241b1f06b9d5ca8af3ef4705e42ef0b42ea50e9e5f722 analysis/study2/prepare_mechanism.py
52ca6a9a01657ce596a13dac28500979dea19a269318b8b665bf909307c0ee8c analysis/study2/secondary.py
c7872775fb85bedc2f2702b3b3a33dcb3d4df93f3692e9baa48f5e5839bcd632 src/cytofair/train_study2.py
aff5ffd89cd66f3c033491d946ace916914c3d177b1e5a10370a5b036378c3ad src/cytofair/gpu_augment.py
39f4485e76bab3a8bdbc9d2234ffd11427c2ad8596430e27e95786385819fce9 src/cytofair/backbones.py
81777104aafabf76ec339ed5a771241d0e0026192d0b798c1828f08bbe3f6e33 src/cytofair/factorial.py
35b6c0b455de6a0d71d8e38309f098763c9ee0ed8ee6f7d4487357baa36993f4 src/cytofair/transplant.py
3f218331e3f78eecc63c105efa831e03138d491ec3f66aed3be02a18319055df src/cytofair/recoverability.py
4e88e1f9ed2b992e51b1f0cb9b55cbd0112438ac67a39510ddea13852a958be4 src/cytofair/study2_design.py
1a2203e7f7b59fae61c4eb9c5181e11ae6a34dd9601045df9e36f5687b2d4d8d src/cytofair/image_cache.py
a952b5c92b945201adbf3102ba16ad4f4c4b1dcb5df6c9c0452ae182800c5101 src/cytofair/fingerprint.py
65a5b4726b730b6919b7c7923f57a9254841033b1fa8d3d44794545c1e46c105 scripts/study2/build_tasks.py
77692934e3cc194e638b6872320f56da46c2baa602d2e91820b1db617e0c8b59 scripts/study2/launch_matrix.py
4cb34f07e9640e632eddaecb128896ae1d9014fa5a82164c7381a6cf381310c1 splits/study2/study2_split_manifest.csv
722d4979be3b63476941deb214d22e0fadbd94669ef2af69a128af3368b18154 splits/study2/study2_transplant_manifest.csv
9e5fac3a0a27b48e4a49184f4d82f8d53c532ff20807911a2853a99682259058 splits/study2/sipakmed_probe_split.csv
d02b7d02519869b417dab81a99d26d2d4a97b0a403439011eb6416e86bca2c37 data_manifests/cric_images_verified.json
06fe16103eae26f970db1c651b8c86660b3eb32c8ff3e56219933e9563e75863 splits/study2/sipakmed_2b/
    background_fraction_seed137.csv
1dd2db6e7d4a06c3995281a7876ce1971ffd45ae455ef3e655e43e2c672f28b3 splits/study2/sipakmed_2b/
    background_fraction_seed173.csv
5063538684a60f94747905d777ee6ba79174c84bd5813da2de237f87cf833cd2 splits/study2/sipakmed_2b/
    background_fraction_seed211.csv
3ef5674738a3901867e2d9744f45ea2d186ac8d6d02674e6f23fbb811f2a6100 splits/study2/sipakmed_2b/
    background_fraction_seed257.csv
914c82d53259068bb7e79e51d7425e2010726c55685221e2e3a820689e62e833 splits/study2/sipakmed_2b/
    background_fraction_seed307.csv
637911ee4155f2593fe472d2be8e7fad606b16525d778bcea5201afcd34b07c0 splits/study2/sipakmed_2b/
    background_fraction_seed349.csv
69a0e0a88ad9de3befa5dfe7ec175612ff8ca5a66087055084a2c351101cb6a3 splits/study2/sipakmed_2b/
    background_fraction_seed401.csv
d87520a7b40aa4a2b43411f80573951ad111faabcfd7f48ab75c2b82ccf258f7 splits/study2/sipakmed_2b/
    background_fraction_seed457.csv
c5e40327ff7a52f86c6a231a73dc52783f31cb93c59245415bfaa7ee37dc3d61 splits/study2/sipakmed_2b/
    background_fraction_seed503.csv
05f936a70d9b5329628cfdfa3ecf9f2f035e0e445d7faaf3d1e5eb95788c37a2 splits/study2/sipakmed_2b/
    background_fraction_seed557.csv
df1d723eba270fdeaac9e42b3663350cdd95503b601d64eb0e31c1e05438ba30 splits/study2/tasks/cric.jsonl
c611865c7919bd329744fa5a03e1e3d27cc70e20f7ac9d768cdfc73d645585fa splits/study2/tasks/factorial.jsonl
ae478ddb914a75c057a07bded9f9edf17ae41ddf216db6a4753d7b1d0e0ada96 splits/study2/tasks/fingerprint.jsonl
c4d96582a084370c448b7b46ec2cbbcde5ae89cd0df8d86476cf5bbbdb412f54 splits/study2/tasks/grl.jsonl
b57b43f34058cad7b6871dfb665e3028adf9609abad3b625004c341170454954 splits/study2/tasks/phase1.jsonl
a52b656719d79879ece36a8cb260789bb2c49ca8567be76799e01c397876ad2a cache:cric:pixels (11534 crops)
b8d905e39e31af99d760039a03c99dc138ef27fa703889729a83a066553c8e1f cache:cric:index
```

```
d5bbaf57f176cf6cdf838a3b4773e2d5fb8c7bf019570344fa6a8b7dcd30c748 cache:sipakmed:pixels (4049 crops)
d29078c8e211819e8fa6ea5f681c5106121bd8f15c0db66769075ea30842d66f cache:sipakmed:index
afe9d620a1de88ab0fdaed3f1abfb5e96d34425819db48be566aa95ea3ea3eca cache:transplant_seed137:pixels (4176 crops)
994672d672c448a1b51f7ae34caca1b21e7d524e06af2b62ec1fcc0e48f0eac6 cache:transplant_seed137:index
87e0e17ab774d40a8f7d4546676f5b281bf4774bf9c8f7a1a62540c25378e35e cache:transplant_seed173:pixels (4136 crops)
ac22a79f280a7f7a3bf735b80d66e9db1d9063d5a4382b4c7327e6b3fbc10c4e cache:transplant_seed173:index
5baadc6106666d174bd7927ad049fcb426ce40bd8cb7a01030ad213141514937 cache:transplant_seed211:pixels (4152 crops)
5f141d0430020d89f35ae080691dc6c5b98a38bc177914f8e9ccb519ff7efe0e cache:transplant_seed211:index
e4ad01639c33080b427f2964922854eebbb44b05c8da85f4ef63bf8351cd6853 cache:transplant_seed257:pixels (4184 crops)
3a07633ca021e8a058a779de4485bdb89ab10c27bc06a37e213457213eaaa1f9 cache:transplant_seed257:index
df36fe07cb47edee3bceeb35943fa1e99bcd2118f7395fa8cf34cca510dd3784 cache:transplant_seed307:pixels (4096 crops)
2f066cbd45475e8eba414c7675d67e607abc92423ba2f23f25452fcacaf2ce16 cache:transplant_seed307:index
eb7ca1021c78dad6e8b20e6728d819359e9498426acd4a2d996a5308620f1e3e cache:transplant_seed349:pixels (4256 crops)
91442b22f10454434cd789b0a8190a01eb59d9321c3f88510327fe0e3ec94ab4 cache:transplant_seed349:index
ce34f911fd43a1c7f5702d9a92acd87ee59444375b47f67291552e64376e071f cache:transplant_seed401:pixels (4136 crops)
4086823828c93e5f7025a57187d11566806bafa2396e16c0284048f5ac00371a cache:transplant_seed401:index
c0fb31f795f94bfd5a5f5f6fe5582d9140a6e0c03d5e5efd025dda89a1164491 cache:transplant_seed457:pixels (4376 crops)
707d08d4f41741aea2bfc8b1a1705f8e52b4b47bb6359eba152296d2a47e0cc6 cache:transplant_seed457:index
9c177a52ce8519b1c1a171c914362db1f91ef5933e8d78df15bef338c998b235 cache:transplant_seed503:pixels (4240 crops)
d6e35027b4894467d4945bee514e6d44689d8ca7ed26c943643604303a2f0f19 cache:transplant_seed503:index
e0b87b9e3d0b0ff04804503b64c4c9deb13987813fe7bcbe02899f8247636189 cache:transplant_seed557:pixels (4400 crops)
065c83b86cbb54d6b05495156cefd3a4e154d75209c19f94922fbb400f0e8383 cache:transplant_seed557:index

# Amendment 7 (2026-10-18): files changed or added after registration, before any 2A run
42bf439fbff44ae9a2d2638ac7fb3f3eb92d47c0fba5acd8d4076272a75e9b86 PREREGISTRATION_STUDY2.md
8d6eb5674bcb6c19b135dc8085e5171dca7a5dfe3f74caee9a306bf1f93f6076 analysis/study2/collect_runs.py
b1cec6d36aaac901dbae55164bcbdccc32835492dab96b9ee89e86d688f6eabf analysis/study2/mechanism_lmm.R
54488ce3f68b0707dbea92da138b3afdffc2d5c5335b0b072f5624e721707d97 analysis/study2/run_all.sh

# Amendment 8 (2026-10-19): follow-up experiments, registered before any follow-up run or analysis
6877eb1ec9598958590e861194098ad0f60a68c64980c88445d118d25d406525 PREREGISTRATION_STUDY2.md
ac286bfed400c57e5f8de377102633132dcbecc6fbe0b2b5fd288da12eb0d68f PREREGISTRATION_STUDY2_AMENDMENT8.md
2d0e79997085d536a02a1d19c31c90103271979d9bf80d7fbfdee4f8ac001639 src/cytofair/carriers.py
689f4ea37617bf13f16d6a067e3ad61bd2cfb58775945e9c2aa550c0a2fc3553 src/cytofair/followup_eval.py
3b181ee8bf9b6824a69a1b1b54ccf7dc3d5424cd232c97dcb242467568a4395b src/cytofair/train_study2.py
e93fe9fa699284d25045abc1d065ef936c37c7fd8e706005a72185c341087847 src/cytofair/gpu_augment.py
1653a502581fa1322d8a412e811ac3d4d9fb8dc7ee9483e717f50bef766acfe8 src/cytofair/backbones.py
e31ef811e9639d4ff78dc6edf4b7dce5c1c713662f1713c36a834f09011588ff src/cytofair/cli.py
4645afa59daa826ec2e14696ae66abc46d52e230d3151e84e2e95f7e43cbf181 src/cytofair/study2_design.py
7afccc8affa20d164c288b851529b05008099a9843280a5fb8b8d8879a0bc900 scripts/study2/build_followup_tasks.py
77692934e3cc194e638b6872320f56da46c2baa602d2e91820b1db617e0c8b59 scripts/study2/launch_matrix.py
6108cbb436b6c14af6925c038bcc7d05ab6725ecb7ed6182f3a3c6c45a199c81 splits/study2/followup/
    followup_split_manifest.csv
560fbb4b81d0faad66433fb6854c2e9f23ecf35e00f353af8cc46de3f22f41c1 splits/study2/followup/
    followup_ablation_manifest.csv
bae4d51de34c37cab5bd1c384b01ce7b3638788089c2c05308163ebd8c4b547c analysis/study2/followup/cric_context.py
8211d214b8476967eedfc59275a8f8731e187707090ded1edd948273060a8179 analysis/study2/followup/followup_lmm.R
8edf79e706a82cb82572422f85ce05d777a9eda6a6ca5482e2bbb40cea9b8103 analysis/study2/followup/prepare_followup.py
f41638914033f1a8895a6ffa89fbf5ba034d8fccc37a8b8aaf8cd6344ebe4094 analysis/study2/followup/run_followup.sh
3f800acc4ac742032d3988df807a0096bfcbe2dc10ac12df3be2e6a4affa79ae analysis/study2/followup/
    simulate_followup_inputs.py
5d53f3715e6432107ab7686d3b2b31c4ee00b3402fcce75fda681af2ae8a8edc splits/study2/followup/ablations/
    ablation_plan_seed137.csv
4bafa40d2251f5837c03cc6e06911f9cd5850ffd501da260bc21b4c0471f46bc splits/study2/followup/ablations/
    ablation_plan_seed173.csv
fbf92dc4c3370a00e8b38c17325c01a7d0b509e9343f70f306a4f1dc94c5cb23 splits/study2/followup/ablations/
    ablation_plan_seed211.csv
29d29e52d9f54a6bb08728a7e577e8674e40cb51a317c4651c82a787f857729d splits/study2/followup/ablations/
    ablation_plan_seed257.csv
a348bee178a67d29497453221f45c7a30f6a1d1a9e31645ac52750ccab8ad9bd splits/study2/followup/ablations/
    ablation_plan_seed307.csv
643b8b42ce1d485d63c511ad77b84dbbe2dc75a228fd7f663d47e021efb0862d splits/study2/followup/ablations/
    ablation_plan_seed349.csv
ed7787d88b9b9e5adf19353027027e1bbed76366eda8da576470cd25bc1b442e splits/study2/followup/ablations/
    ablation_plan_seed401.csv
26cb7c492e9c84050444bf7be8a6cedf9de5d5ef11a39e9fd53d625a329ddab6 splits/study2/followup/ablations/
    ablation_plan_seed457.csv
bd8a91179515b8c30fc437a17ff787faadd1bebd33b27d03687f129b49a1ed8e splits/study2/followup/ablations/
    ablation_plan_seed503.csv
448d5c2bef95c4bc9882234e5516d9098a9f848fbba6112078eb2467541e764e splits/study2/followup/ablations/
    ablation_plan_seed557.csv
9bff878f72a63fc89d1fb374182964174665a0f01c2a2e4b3214f07c94bc45d9 splits/study2/followup/tasks/carrier.jsonl
05a615f1d19c92d4095cd9909b108e4b2dacdb064859c9d3a535e64e7f949d6d splits/study2/followup/tasks/ffact.jsonl
ca8d6db070b0da00457759f905f85f59ccba39383cb05f0f2c04a679d65cb3e2 splits/study2/followup/tasks/leace.jsonl
4f683b6e7a0a90c9fb631df68b2f61777aedf472b6a5ce477a155f6107891642 splits/study2/followup/tasks/linadv.jsonl
```

```
e7d301e0ffa9397cca00af0b60d5675ef7953d6beecfbc76f37aa6d7034a23f3 splits/study2/followup/tasks/normalization.
    jsonl
1117f1219a1c4e7b4f04e3256c1a441ce1d007e1e66b6a2ecfad1de8b9bf862a cache:ablation_seed137:pixels (3132 crops)
879097586ed1b7942e976e5ebccf58fca3a440c8e3ed946b8ee1f8aa88966caa cache:ablation_seed137:index
3d2f43f345a703465888513f1636243d126df06da00e3acd490337d0465ae718 cache:ablation_seed173:pixels (3102 crops)
67e6344f98c824a89729202970ba9f2008a1648144c5e4e26f0315e15e3ab761 cache:ablation_seed173:index
58d57a0b460829781f7510df5048e3604f1542b20e89f4ef57caa99ba61d6a11 cache:ablation_seed211:pixels (3114 crops)
703ac1d02282a9427c49ebc80e9da189547c79bdc4f62ef24b3a25b194d11f37 cache:ablation_seed211:index
96fb0e6729372b4702acb9492b1c35d676c05584423baf00f6449825aef88d4a cache:ablation_seed257:pixels (3138 crops)
483c8612c71cdead087c1387daf655bd3cb1b390b64d60f0928a3e39b5259a2d cache:ablation_seed257:index
850a11d7d3f7ac9ef4507658a087f28ed08f9dfe03062ed0c23ce490fa8a67b9 cache:ablation_seed307:pixels (3072 crops)
3bf6ed06b3c4b1067bd3fe093111a53f847f873e1c60dd275a6b8dd8d565bc8d cache:ablation_seed307:index
7c9f1e2c85cdb4362a6efffbaf36827f774e07ad27edc32a3f19d72673e2e2a0 cache:ablation_seed349:pixels (3192 crops)
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c8e9a22fdb2ab9abe33083a6c847c1814b367bd140d67b535be2fd7b8c2865e4 cache:ablation_seed401:index
1c44558cfda145ef285a366d038861d49f1978c0d8d372591ac70de35666718c cache:ablation_seed457:pixels (3282 crops)
11b162ca4f2cc7fd0edd26bc2d987470eaf6485da44137f810ee38dc104e5893 cache:ablation_seed457:index
c316def50ed2e82d750d8b83627602d6bd1d7ad1671c13f923f478fb6a43d57f cache:ablation_seed503:pixels (3180 crops)
2973973f0c623f03da92d70ef37d552843636203655ed4a8ef3df7e9b1b7000d cache:ablation_seed503:index
c868502d5e670ec86040c5aa63d9b9fce3928e65472b2eb1f4928b77b724de26 cache:ablation_seed557:pixels (3300 crops)
f33d0d36634066ef1e7aa6890f500e3a9408928db3e399742ea24724a2e35776 cache:ablation_seed557:index
a0d069a9324563f36fe31ad80003d81ea2efb120e9bcf53ae513efdae9400efe cache:sipakmed_masked:pixels (4049 crops)
d45211370d5e5eee02ab1afc234e2f48572d6aa76e8692e3f3e818dd269d3b99 cache:sipakmed_masked:index
10974f8fe420e4850294f0b959beafc08feb53aa514c272bdc778e4f47366428 cache:sipakmed_reinhard100:pixels (4049
    crops)
017675d683b8f50007c61a0a0305eef7fed8c29ee1e5b0932c68924326c706a8 cache:sipakmed_reinhard100:index
a7265a0f8162cc862c38fab2bedfee783aa19f25260aaf89b5b777eab48b5c05 cache:sipakmed_reinhard50:pixels (4049 crops
    )
e1faf04759b407365319da6852422952818c8d2b72681a25bd09992e0ef34734 cache:sipakmed_reinhard50:index

# Amendment 9 (2026-10-20): preparer bug fix, before any F3/F4 estimate was seen
fd8e2e1f17291592eb3df50cb6ea83fea5910f729852beb15eac6c56f772e22b PREREGISTRATION_STUDY2.md
b58e212b92ce5c2e06b8f85554332bd70f3108aff575bde8d01c8e240fe27f0c PREREGISTRATION_STUDY2_AMENDMENT8.md
63f03ca9082a3a6b59dabb68e8f7550fd28536391aa41f8721ecd9be939a5c79 analysis/study2/followup/prepare_followup.py
94d1fc5cfe1aad4ed0f632d03c6ad6d057b792f31c4f61646eb0ffbd452cec94 tests/test_followup_prepare.py
```

Q

The Study 2 preregistration, verbatim

```
# CytoFair Study 2 preregistration: powered replication and causal decomposition

**Registered:** 2026-10-18 (Asia/Kolkata), in the commit that adds this file.
**Relationship to the original plan:** registered as Amendment 6 of ‘PREREGISTRATION.md‘.
**Status at registration:** no Study 2 model has been trained for any registered cell.
The only Study 2 GPU executions before this commit are one-epoch pipeline smoke tests on
P1 splits (listed in ‘reports/study2/excluded_attempts.md‘); none produced a protocol
contrast, and their metrics were checked only for finiteness.

## 0. Why a second study

Study 1 (frozen 7 September 2026) established structural non-independence, literal pixel
sharing, and a learnable parent fingerprint, but left four limitations that this study
addresses:

1. **Specification.** One question ("does crop-level splitting inflate accuracy?") was tested
   six times, once per architecture, inside a 12-test BH family. Study 2 replaces this with
   one mixed-model test of the protocol effect (H1), powered by simulation.
2. **Availability versus use.** Study 1 showed a fingerprint is available, not that
   classifiers route predictions through it. Study 2 adds a counterfactual background
   transplant (2B, H4) and adversarial fingerprint suppression (2A, H5).
3. **Leakage versus diversity.** At equal crop counts, P1 training sets contain about 914
   distinct parents and P2 training sets about 773, so P1 also has more training
   diversity. Study 2 crosses leakage with diversity at matched crop and parent counts
   over one common test set (2C, H3).
4. **One rescored benchmark.** Study 2 rescores CRIC (H2), whose crop-level split places a
   training parent behind 100% of test cells.

Study 1 results are used only as pilot estimates for the power analysis. No Study 1 run
enters any Study 2 test, and the Study 2 seeds are disjoint from the Study 1 seeds.

While validating the analysis code, the H1 model was fitted to the Study 1 runs
(‘results/study2/power/power_pilot.json‘): P1 - P2 = 1.37 pp, Satterthwaite p = 0.0054.
That reanalysis was chosen after the Study 1 data were seen. It is reported as a post hoc,
exploratory observation and is not a confirmatory result.

## 1. Fixed training recipe

Every run uses the Study 1 classification recipe: ImageNet initialisation, a new linear
head, AdamW (learning rate 1e-4, weight decay 1e-4), batch size 64, 20 epochs, cross
entropy, final-epoch endpoint, no validation split, no early stopping, no checkpoint
selection. Training augmentation is RandomResizedCrop(224, scale 0.8-1.0, ratio 3/4-4/3),
horizontal flip (p = 0.5), vertical flip (p = 0.5), ImageNet normalisation; evaluation uses
Resize(224 x 224) and the same normalisation.

Execution changes relative to Study 1, none of which alters a model, split, seed, or
endpoint:

* The training host limits each user to four CPU cores (cgroup quota 400%). Crops are
  therefore decoded once into checksummed caches on node-local storage, uploaded to the
  GPU, and augmented there (‘cytofair.gpu_augment‘). Crop parameters come from an exact
  port of torchvision’s ‘RandomResizedCrop.get_params‘ (verified identical for 100
  seed/shape pairs, ‘tests/test_gpu_augment.py‘); resizing is antialiased bilinear.
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```
* cuDNN deterministic mode, `torch.use_deterministic_algorithms(True, warn_only=True)`,
  and `CUBLAS_WORKSPACE_CONFIG=:4096:8`.
* Every run writes, beside its JSON: per-crop test logits, a float16 copy of the final
  weights, penultimate features for the balanced 424-parent probe crops, and logits for
  any registered extra evaluation set (Section 7).

At most four GPUs (indices 0, 3, 4, 5 on the training host) run concurrently, one job per
GPU.

## 2. Architectures

Twelve ImageNet-initialised backbones (exact weights in `cytofair.backbones.WEIGHT_NAMES`):
the six Study 1 models (ResNet-18, ResNet-50, DenseNet-121, EfficientNet-B0,
MobileNet-V2, ViT-B/16) and six added for breadth (ConvNeXt-T, Swin-T, RegNetY-1.6GF,
EfficientNetV2-S, MaxViT-T, and self-supervised DINOv2 ViT-S/14 at 224 px).

## 3. Seeds and stopping rule

`STUDY2_SEEDS` = 137, 173, 211, 257, 307, 349, 401, 457, 503, 557, 601, 653, 701, 751,
809, 853, 907, 953, 1009, 1051. The first five were named as robustness seeds in the
original preregistration and were never run. Phase 1 uses all twenty; the mechanism arms
and CRIC use the first ten (`MECHANISM_SEEDS`).

No seed, architecture, or cell is added, removed, or replaced after any result is seen.
A failed run (infrastructure failure or non-finite loss) is retried once with the identical
seed and split; a cell that fails twice is reported as missing, and every test that needs it
fails closed rather than substituting a run.

## 4. Splits

All splits were generated before registration and are committed under `splits/study2/`
with SHA-256 hashes in `study2_split_manifest.csv`. Trainers refuse a split whose hash
differs. P1, P1-prime, P2, and P3 follow the original definitions. For all twenty SIPaKMeD
seeds, P2 has exactly 810 test crops, so P1-prime is byte-identical to P1 and is not trained
separately (P1-prime - P2 equals P1 - P2 by construction). For CRIC, P1-prime differs from
P1 by 2-11 training crops and is trained.

## 5. Phase 1: powered SIPaKMeD replication (H1, the primary endpoint)

**Design.** 12 architectures x 20 seeds x {P1, P2, P3} = 720 runs.

**Primary endpoint.** Test accuracy in percentage points.

**Confirmatory model (H1).** On P1 and P2 runs:

    accuracy ~ x + (1 + x | architecture) + (1 | seed) + (1 | split)

with `x = +0.5` (P1) or `-0.5` (P2), so the fixed effect is P1 - P2; `split` is the
seed-by-protocol partition shared by all architectures (omitting it would treat twelve
evaluations of one test set as independent). REML, `bobyqa`, lme4 with lmerTest
Satterthwaite degrees of freedom, two-sided alpha = 0.05.
**Decision rule:** H1 is supported if p < 0.05 and the estimate is positive.

**Convergence.** A singular (boundary) fit is accepted. Only on convergence failure, the
model is simplified in this order: uncorrelated architecture intercept and slope; then
dropping the seed intercept. The architecture slope and the split effect are never dropped.

**Assumptions and fallback.** Conditional residual skewness, excess kurtosis, Shapiro-Wilk,
architecture-slope BLUPs, and residual SD by protocol are reported. If no model converges,
or residual |skewness| > 1, or excess kurtosis > 3, the primary result is the two-way
cluster bootstrap (architectures and seeds resampled independently, 10,000 replicates,
percentile CI and two-sided bootstrap p).

**Sensitivity (reported, not decisive).** Kenward-Roger degrees of freedom; binomial GLMM
on correct counts with an observation-level random effect; seed-level exact sign-flip
test; the cluster bootstrap.

**Power.** `analysis/study2/power_simulation.R` simulated the confirmatory model 1,000
times per design from the Study 1 pilot (effect 1.37 pp; architecture intercept SD 0.44,
slope SD 0.36; split SD 0.31 for P1 and 0.59 for P2; residual SD 0.69; seed SD 0;
intercept-slope correlation set to 0 because six architectures put it on its -1
boundary). The registered rule was: choose the smallest seed count in {10, 15, 20, 25, 30},
with twelve architectures, whose power for a 0.8-pp effect is at least 0.90 under pilot
variances and at least 0.75 with every variance inflated by 1.5. Twenty seeds give 0.977
and 0.782 (fifteen give 0.949 and 0.698). 0.8 pp exceeds the 0.62-pp spread between
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published SIPaKMeD scores that motivates the question.

**Secondary analyses (non-confirmatory, no multiplicity claim).** P2 - P3 accuracy (same
model); P1 - P2 for macro-F1 and balanced accuracy; per-architecture paired differences
(descriptive t intervals); Kendall's tau between P1 and P2 architecture rankings with a
seed bootstrap; the P3 <= P2 <= P1 ordering per architecture; pooled confusion matrices,
NLL, and ECE; and two crop-level analyses from the saved predictions:
**M1** within-crop P1 - P2 accuracy for crops that appear in both P1 and P2 test sets
(crop-level bootstrap), and **M2** within-P1 accuracy as a function of the number of
training siblings and literal source-pixel overlap (logistic GLM with crop-clustered
standard errors, adjusting for class and the crop's P2 accuracy).

**Gate A.** If H1 is supported, the inflation effect is reported as certified. If not, the
paper is framed as a careful null replication. The mechanism arms are run regardless
(they are registered now), but their interpretation follows Gate A.

## 6. Arm 2C: leakage x diversity at matched size (H3)

**Design** (`cytofair.factorial`, `FACTORIAL_CONFIG`). For each of the ten mechanism seeds
and each class: 30% of parents with at least two crops are evaluation parents E; each E
parent contributes floor(k/2) test crops (the common test set, about 527 crops) and its
remaining crops are donor siblings. Six parents per class with at least three crops are
reserved as unseen background donors U (never trained on or tested). The rest form the
clean pool B. Four training sets each contain exactly 220 crops per class:

* `clean_high`: every B parent;
* `leak_high`: the E parents' donor siblings replace the last |E| parents of one fixed B
  permutation;
* `clean_low` and `leak_low`: the same construction on the first 70% of that permutation.

Within a diversity level, leak and clean have identical crop counts and identical numbers
of distinct parents (720 versus 504 in total; effective parent numbers 633 and 632 versus
432 and 427), and differ only in whether the test parents' siblings are present. All test
crops have a training sibling in the leak cells (mean 2.0 high, 3.0 low) and none in the
clean cells. Low-diversity parent sets are nested in high-diversity sets. Configurations
with 240 crops per class were infeasible for seed 307 and were rejected before any training.

**Models.** ResNet-50, DenseNet-121, ConvNeXt-T, ViT-B/16; 4 x 10 x 4 = 160 runs.

**H3.** `accuracy ~ leak * div + (1 + leak | architecture) + (1 | seed) + (1 | training_set)`
with leak and div coded +-0.5. H3 is the leak main effect (leakage at matched diversity),
predicted positive. Simplification order on convergence failure: uncorrelated slope, then
intercepts only.

**Secondary.** The diversity effect is estimated only within clean cells
(`clean_high - clean_low`), because the leak dose (siblings per test crop) is higher in the
low-diversity leak cell; the leak x diversity interaction is exploratory for the same
reason. Embedding diversity of every training set is reported in the space of a retrained
RGB parent-probe encoder (`cytofair.fingerprint`). A descriptive decomposition of the
Phase 1 P1 - P2 gap combines the H3 estimate (scaled by the 91.7% leak rate of natural P1)
with the clean diversity effect scaled log-linearly from the 720/504 contrast to the natural
914/773 contrast; it is labelled model-based and approximate.

## 7. Arm 2B: counterfactual background transplant (H4)

Every 2C model is also evaluated on composites of its seed's common test set
(`cytofair.transplant`, plans in `splits/study2/sipakmed_2b/`). The cell foreground is kept
(the Study 1 background-only mask: Otsu on saturation and luminance, 5 x 5 closing) and
feathered (Gaussian sigma 2) onto a donor background inpainted with Telea (radius 5):

* `original`; `self` (own inpainted background: seam and inpainting artifacts only);
* `sibling` (background of a donor sibling of the same parent: own acquisition context);
* `unseen_same_class` (background from a U parent of the same class);
* `unseen_other_class` (one composite per other class, from U parents).

**H4 (context-route difference-in-differences).** For each run,
`route = acc(sibling) - acc(unseen_same_class)` measures how much the recipient's own
acquisition context contributes beyond class-typical context. Leak models have seen that
context with the correct label; clean models have not. H4 is the leak coefficient in
`route ~ leak + div + (1 + leak | architecture) + (1 | seed) + (1 | training_set)`,
predicted positive (fallback: intercepts only).

**Secondary.** The ratio of H4 to the leak effect on original images (the share of the
leakage inflation attributable to the context route); the donor-class probability shift for
other-class backgrounds relative to same-class backgrounds (whether backgrounds carry class
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information that classifiers use, independent of leakage); ‘self‘ versus ‘original‘
(artifact size); and H4 restricted to recipients whose background covers at least 30% of
the crop.

## 8. Arm 2A: adversarial fingerprint suppression (H5)

A class-conditional parent head (512-unit MLP over the training parents, logits of
other-class parents masked) sits behind a gradient-reversal layer whose coefficient follows
the DANN ramp ‘lambda * (2 / (1 + exp(-10 p)) - 1)‘. The class head is unchanged. Because
the adversary only distinguishes parents within a class, suppressing it never requires
removing class information.

**Design.** ResNet-50 and ConvNeXt-T; lambda in {0, 0.01, 0.03, 0.1, 0.3, 1, 3}; natural P1
and P2 splits of the ten mechanism seeds; 280 runs. Lambda = 0 trains the same head with no
reversed gradient and is the matched baseline. The full sweep, including any degenerate
region, is reported.

**Recoverability R.** Within-class parent identification accuracy of a multinomial logistic
regression (C = 1, standardised features) on the frozen penultimate features of the
balanced 424-parent probe split (two fitting and two evaluation crops per parent; seed 17),
reported with its class-only floor (the accuracy of a representation that knows only the
class). Overall 424-way and nearest-centroid accuracies are secondary.

**Non-degenerate regime.** For each architecture, the lambdas whose mean P2 accuracy is
within 1.0 pp of its lambda = 0 mean. **Manipulation check:** at the largest such lambda,
mean R must fall by at least 25% of its above-floor value at lambda = 0; an architecture that
fails is excluded from H5, and if both fail H5 is reported as untestable, not refuted.

**H5.** Within the regime, per architecture-seed-lambda pair,
‘gap = acc(P1) - acc(P2)‘ and ‘R_pair‘ = mean R of the two runs;
‘gap ~ architecture + R_pair + (1 | seed)‘. H5 is the R_pair slope, predicted positive (the
inflation shrinks as the fingerprint is suppressed).

## 9. CRIC rescore (H2)

CRIC cells are 100 x 100 windows centred on the published nucleus coordinates (shifted
inward at image borders) from the 400 Figshare slide images (MD5-verified,
‘data_manifests/cric_images_verified.json‘). Six Bethesda classes. Models: ResNet-18,
ResNet-50, DenseNet-121, EfficientNet-B0, ConvNeXt-T, ViT-B/16; ten mechanism seeds;
P1, P1-prime, P2; 180 runs. **H2** is the H1 model applied to CRIC accuracy (P1 - P2),
predicted positive. Secondary: P1-prime - P2, balanced accuracy and macro-F1 (the classes
are imbalanced), and the CRIC versus SIPaKMeD effect sizes against their structural
leakage, which is exploratory.

## 10. Multiplicity

H1 is the single primary endpoint, tested alone at alpha = 0.05. H2, H3, H4, and H5 form
one Holm family (family-wise alpha = 0.05); each is supported only if its Holm-adjusted p
is below 0.05 and its estimate has the predicted sign. Everything else is secondary or
exploratory and carries no confirmatory claim.

## 11. Exclusions and deviations

Runs are excluded only for logged infrastructure failure, non-finite loss, or a failed
integrity check (wrong split hash, cache hash, or missing artifact); never for magnitude or
direction. Every attempt is logged. Any deviation from this document is recorded as a dated
amendment before the affected analysis is run; a deviation discovered afterwards is
reported as such.

## 12. Gate B and venue framing

Gate B is interpretive, not statistical: the causal evidence is "clean" if H3 and H4 are
supported and H5 is either supported or untestable for a stated reason, "murky" otherwise.
It determines framing only. All registered results are reported whatever the gates show.

## 13. Registered artifacts

The analysis code (‘analysis/study2/*.R‘, ‘analysis/study2/*.py‘), split manifest, task
builder, and caches are fixed by this commit; their SHA-256 hashes are listed in
‘reports/study2/registration_hashes.txt‘.

## Amendment 7 -- failure handling (2026-10-18, Asia/Kolkata)

Registered while Phase 1 was running and before any 2A run had started; no Study 2
result had been analysed or inspected. Two cases were not handled by Section 3:
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1. **Infrastructure failures.** A run that fails because of the host rather than the run
   itself (out-of-memory caused by another user's process on the same GPU, loss of the
   host, session, or controller, or a filesystem error, each identifiable from the
   attempt log) does not count toward the one-retry limit. It is rerun with the identical
   seed, split, and code, and every attempt stays in the log.
2. **Divergence in the 2A sweep.** Runs are deterministic, so a run whose training loss
   becomes non-finite diverges again on retry. In the gradient-reversal arm only, such a
   run is recorded as *diverged* rather than missing: a lambda with any diverged run of an
   architecture lies outside that architecture's non-degenerate regime, the diverged runs
   are excluded from H5, and the sweep table reports them with the epoch at which the loss
   became non-finite. A diverged run in any other arm still makes the tests that need it
   fail closed, as registered.

The collector (`analysis/study2/collect_runs.py --diverged-logs`) and `mechanism_lmm.R`
implement point 2; their new hashes are appended to `reports/study2/registration_hashes.txt`.

## Amendment 8 -- follow-up experiments (2026-10-19, Asia/Kolkata)

Registered after the Study 2 analysis was complete, before any follow-up model was trained
or any follow-up analysis was run on existing outputs. It adds eight new hypotheses (F1-F8)
on how leakage is used -- CRIC slide-context pull, test-time carrier ablations, carrier
removal during training, rank-limited LEACE erasure, a stain-normalisation dose, a
probe-matched linear adversary, and a pooled dose-response -- in their own Holm family. It
does not change H1-H5 or their verdicts. Full text: `PREREGISTRATION_STUDY2_AMENDMENT8.md`.

## Amendment 9 -- preparer bug fix (2026-10-20, Asia/Kolkata)

A labelling bug in `analysis/study2/followup/prepare_followup.py` (standard-recipe carrier runs
labelled `colour`) was fixed before any F3 or F4 estimate was seen. Details in
`PREREGISTRATION_STUDY2_AMENDMENT8.md`.
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Amendments 8 and 9, verbatim

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# Study 2, Amendment 8 -- Follow-up experiments on how leakage is used

Registered 2026-10-19 (Asia/Kolkata) as an amendment to ‘PREREGISTRATION_STUDY2.md‘.

## 0. Status, motivation, and what was known

This amendment was written **after** the registered Study 2 analysis was complete. Its
verdicts stand and are not revisited here: H1 (SIPaKMeD P1 - P2) and the Holm-family
hypotheses H2 (CRIC) and H3 (leakage at matched diversity) were supported; H4 (context route
through backgrounds) and H5 (gap tracks gradient-reversal suppression) were not.

The eight hypotheses below (F1-F8) are new questions, motivated by two observations from
Study 2: (i) leak-trained 2C models kept most of their advantage on backgrounds transplanted
from unseen fields, which points to a carrier inside the cell region rather than the
background; and (ii) for ConvNeXt-T the gradient-reversal adversary was evaded (independent
probe recoverability rose), so H5 had only one usable architecture.

**Known when writing:** every registered Study 2 output, including transplant condition
means by cell and the gradient-reversal sweep means. **Not known:** any per-cell CRIC result
broken down by slide composition; any evaluation of ablated images; any evaluation of erased
features; any Grad-CAM map; any model trained with masked backgrounds, colour augmentation,
stain normalisation, or the linear adversary. One-epoch smoke tests used only crop-level or
leak-cell runs, so no follow-up contrast could be observed before registration.

Results for F1-F8 are reported as registered follow-ups to Study 2, separately from H1-H5,
whatever they show.

## 1. Common elements

The Study 2 recipe (section 1 of the parent document) is unchanged: ImageNet initialisation,
AdamW 1e-4 / weight decay 1e-4, batch 64, 20 epochs, final-epoch endpoint, random resized crop
(0.8-1.0) and flips on the GPU, deterministic kernels. Only the factor each arm manipulates
differs. Every new training run saves per-crop logits, final weights, and probe-split
features. Failure handling follows Amendment 7: infrastructure failures are rerun; a
linear-adversary run whose attempts both end in a non-finite loss is recorded as *diverged*,
its lambda leaves that architecture's regime, and it is excluded from F7 and F8; any other
failure fails the arm closed.

**Recoverability R** (F5-F8) is within-class parent identification accuracy on the balanced
424-parent probe split (seed 17; two fitting and two evaluation crops per parent): multinomial
logistic regression on standardised features with sklearn's objective (C = 1), solved by
L-BFGS on the GPU (‘cytofair.followup_eval.fit_logistic‘), restricted at prediction to parents
of the crop's class. The same function scores every arm. Its agreement with the registered
H5 sklearn probe on the 280 Study 2 gradient-reversal runs is reported (secondary).

## 2. F1 -- CRIC slide-context pull (existing predictions; no training)

For every CRIC cell, the **context class** is the most frequent class among the *other*
cells of its slide image; cells whose slide has no other cell, or a tied majority, are
excluded. A cell is **discordant** if its own class differs from its context class. For each
of the 120 registered CRIC P1 and P2 runs (6 architectures x 10 seeds), the softmax
probability of the context class is computed for every test cell. Per cell and protocol,
probabilities are averaged over all runs in which the cell is a test cell.
```

```
**F1** is the mean, over discordant cells tested under both protocols, of
‘100 * (p_context(P1) - p_context(P2))‘, predicted positive: when a cell’s slide is in
training, predictions are pulled toward the slide’s majority label. Inference: percentile
cluster bootstrap over slides (10,000 replicates, seed 20261019), two-sided
‘p = 2 * min(P(boot <= 0), P(boot >= 0))‘.
Secondary: the within-cell P1 - P2 accuracy difference in concordant and discordant cells
and their contrast (predicted concordant > discordant); the rate of predicting the context
class; per-architecture F1.

## 3. F2 -- test-time carrier ablations (existing 2C checkpoints; no training)

The 160 registered 2C models are re-evaluated on their seed’s common test set under six
conditions, each rendered at native resolution (‘cytofair.carriers‘): ‘original‘; ‘grey‘
(luminance only); ‘stain_transfer‘ (Reinhard transfer of the crop’s LAB mean and standard
deviation to those of the 2B unseen same-class donor crop); ‘low_pass‘ (Gaussian blur, sigma =
2 px at the 224-px input); ‘illumination_flat‘ (subtract a sigma = 16 px Gaussian, add back the
crop mean); ‘background_grey‘ (pixels outside the Study 1 foreground mask set to the ImageNet
mean). Ablation plans are ‘splits/study2/followup/ablations/ablation_plan_seed*.csv‘.

For each ablation ‘a‘, with run-level accuracy in pp and ‘ablated = +-1/2‘:
‘acc ~ leak * ablated + div + (1 + leak | architecture) + (1 | seed) + (1 | training set) + (1 | run)‘
(fallback: random intercepts only). The reduction of the leak effect is ‘-coef(leak:ablated)‘.
**F2** is that reduction for ‘stain_transfer‘, predicted positive (the leak advantage depends
on the crop’s own colour statistics). Secondary: the reductions for the other four ablations;
the leak effect within each condition; agreement of re-evaluated originals with the recorded
predictions (fp16 weights).

**Grad-CAM (descriptive only).** For ResNet-50, DenseNet-121, and ConvNeXt-T 2C models, the
Grad-CAM map of the true-class logit at the last convolutional stage is upsampled to the crop
and the fraction of positive attribution outside the foreground mask is averaged per run. The
leak effect on that fraction is estimated with the H4 intercept model and reported with its
CI and no test. A sanity check compares each map with the map obtained after replacing the
classification head by a fixed random head (median Spearman correlation).

## 4. F3 and F4 -- carrier removal during training (new runs)

Ten seeds never used for any model, each checked feasible for the registered 2C
configuration: 1103, 1151, 1201, 1249, 1301, 1361, 1409, 1451, 1499, 1553. Their 2C designs
and cell splits are ‘splits/study2/followup/sipakmed_2c/‘ (manifest
‘splits/study2/followup/followup_split_manifest.csv‘). ResNet-50, DenseNet-121, ConvNeXt-T,
and ViT-B/16 are trained on all four cells under three conditions: ‘standard‘ (Study 2
recipe), ‘masked‘ (every training and test crop has its background set to the ImageNet mean;
cache ‘sipakmed_masked‘), and ‘colour‘ (training batches additionally receive HED-strong
optical-density jitter, alpha in [0.8, 1.2] and beta in [-0.2, 0.2] per stain channel,
followed by brightness and contrast factors in [0.8, 1.2]; test crops unmodified). 480 runs.

**F3**: within ‘masked‘, the H3 model
‘acc ~ leak * div + (1 + leak | architecture) + (1 | seed) + (1 | training set)‘ (H3 fallback
ladder); the leak coefficient, predicted positive (leakage survives without any background).
**F4**: within ‘standard‘ and ‘colour‘, with ‘treated = +-1/2‘,
‘acc ~ leak * treated + div + (1 + leak | architecture) + (1 | seed) + (1 | training set)‘
(fallback: random intercepts); ‘-coef(leak:treated)‘, predicted positive (colour augmentation
shrinks the leak effect). Secondary: the H3 model on ‘standard‘ (an independent replication
of H3 on fresh seeds); the same contrast for ‘masked‘ versus ‘standard‘ (background share of
the leak effect); cell means.

## 5. F5 -- LEACE erasure of within-class parent identity (existing Phase 1 checkpoints)

For every registered Phase 1 P1 and P2 model (12 architectures x 20 seeds; 480 models),
penultimate features of all SIPaKMeD crops are extracted (eval transform). On the model’s own
training crops, with ‘W = Sigma^{-1/2}‘ (ridge 1e-6 x mean eigenvalue) and the within-class
between-parent scatter ‘S = sum_j (n_j / n)(mu_j - mu_class(j))(mu_j - mu_class(j))^T‘, the
rank-r eraser removes the top-r eigendirections ‘U_r‘ of ‘W S W‘:
‘x’ = x - (x - mu) W U_r U_r^T W^{-1}‘ (rank-limited LEACE; Belrose et al., 2023). Ranks
r in {0, 4, 16, 64, 256}. At each rank the linear class head is refitted on the erased
training features (the logistic objective above) and test accuracy is measured; R is
measured on the erased probe features.

Per architecture, the **regime** is the set of ranks whose mean P2 accuracy is within 1 pp of
rank 0 (refitted heads); r* is its largest rank. **Manipulation check**: mean R_pair at r* is at
least 25% of the above-floor rank-0 value below it (as for H5); architectures that fail, or
whose regime is {0}, are excluded. For each included architecture-seed,
‘reduction = gap(0) - gap(r*)‘ with ‘gap = acc(P1) - acc(P2)‘. **F5**: intercept of
‘reduction ~ 1 + (1 | architecture) + (1 | seed)‘ (fallback ‘(1 | architecture)‘), predicted
```

```
positive (erasing linearly decodable within-class field identity closes part of the gap).
If no architecture is included, F5 is untestable, not refuted.

## 6. F6 -- stain-normalisation dose (new runs)

Reinhard normalisation of every crop toward one reference (the mean over all SIPaKMeD crops of
per-crop LAB means and of per-crop LAB standard deviations), blended in LAB space at strength
s in {0.5, 1.0} and applied to training, test, and probe crops alike (caches
‘sipakmed_reinhard50‘, ‘sipakmed_reinhard100‘). ResNet-50, ConvNeXt-T, DenseNet-121, and
EfficientNet-B0 on natural P1 and P2 for the ten Study 2 seeds no mechanism arm used (601,
653, 701, 751, 809, 853, 907, 953, 1009, 1051): 160 runs. Strength 0 is the corresponding
registered Phase 1 run (identical deterministic recipe, split, and seed). **F6**: in
‘gap ~ architecture + s + (1 | seed)‘ over the three strengths, ‘-coef(s)‘, predicted positive.
The regime and manipulation check of section 5 decide which strengths enter F8.

## 7. F7 -- probe-matched linear adversary (new runs)

The 2A design with a different adversary: per-feature standardisation (BatchNorm without
affine parameters) followed by one linear layer over the training parents, class-conditional
masking, gradient reversal with the DANN ramp; the adversary has its own AdamW optimizer
(learning rate 1e-3, no weight decay) and is re-initialised, with fresh optimizer state, every
5 epochs from a forked random stream. ResNet-50, ConvNeXt-T, and DenseNet-121; lambda in
{0, 0.1, 0.3, 1, 3}; natural P1 and P2; seeds as in F6; 300 runs. Regime, manipulation check,
and model exactly as H5: ‘gap ~ architecture + R_pair + (1 | seed)‘ over included
architectures and in-regime lambdas. **F7** is the R_pair slope, predicted positive.

## 8. F8 -- pooled dose-response

All included architecture x level cells of F5 (ranks), F6 (strengths), and F7 (lambdas),
after their regimes and manipulation checks:
‘gap ~ cell + R_pair + (1 | manipulation:seed)‘ with one fixed intercept per
manipulation x architecture cell. **F8** is the R_pair slope, predicted positive (across
independent ways of removing the fingerprint, the inflation tracks how much of it remains).

## 9. Multiplicity and interpretation

F1-F8 form one Holm family (family-wise alpha = 0.05), separate from H1-H5. Each is supported
only if its Holm-adjusted p < 0.05 and its estimate is positive; estimates are oriented so the
predicted direction is positive. Everything else here is secondary or descriptive.

Pre-specified reading: F2 and F3 supported together locate the leaked signal in the cell’s own
appearance rather than its background; F4 supported identifies colour augmentation as a
partial mitigation; F5, F7, or F8 supported means the inflation tracks recoverable field
identity (the shortcut account); F5 not supported while R falls means the inflation is not
carried by linearly decodable within-class field identity in pooled features.

## 10. Analysis order and artifacts

Each arm is analysed once all of its registered tasks are complete (‘--allow-incomplete‘
outputs are progress reports only); the Holm family is computed once all eight are available.
The code (‘src/cytofair/carriers.py‘, ‘src/cytofair/followup_eval.py‘, the trainer options,
‘scripts/study2/build_followup_tasks.py‘, ‘analysis/study2/followup/*‘), split and ablation
manifests, and task files are fixed by this commit; their SHA-256 hashes are listed under
"Amendment 8" in ‘reports/study2/registration_hashes.txt‘, together with the hashes of the
derived caches. A synthetic dry run of ‘followup_lmm.R‘
(‘analysis/study2/followup/simulate_followup_inputs.py‘) recovered every planted effect before
registration.

## Amendment 9 -- preparer bug fix (2026-10-20, Asia/Kolkata)

When the registered models were first run on the completed carrier arm, ‘followup_lmm.R‘
stopped at F4 because ‘prepare_followup.py‘ had labelled all 160 standard-recipe runs as
‘colour‘: the label was derived from a DataFrame row in which a missing colour augmentation
(‘None‘ in the run record) had become a truthy NaN. The masked runs were labelled correctly.
The script halted before writing any output, so no F3 or F4 estimate was seen. The fix
derives each run’s condition from its own record (‘carrier_condition‘: background-masked
cache -> ‘masked‘; ‘color_augment == "hed_strong_bc"‘ -> ‘colour‘; otherwise ‘standard‘), which
is the registered definition; no model, rule, or data changes. A unit test
(‘tests/test_followup_prepare.py‘) covers it. The synthetic dry run had not exercised the
preparer, which is why the error was not caught before registration.

Known when this amendment was written: the arm-level results of F1, F2 (with the Grad-CAM
description), and F5, analysed with the registered code once their arms were complete, as
section 10 allows. Not known: any F3, F4, F6, F7, or F8 estimate.
```

S

Phase 0 checklist, Cx22 literature check, and excluded attempts

S.1 Phase 0 checklist

```
# Phase 0: arXiv v1 readiness

Status of each pre-posting item from the scaling roadmap, as checked on 2026-10-18.
"Verified" means checked directly; "Authors" means only the authors can do it.

| Item | Status | Evidence or action |
|---|---|---|
| Public repository installs and tests pass | Verified | Fresh ‘git clone‘ of ‘github.com/optimisticexquisite
    /cytofair‘ at ‘2c5d5c3‘: ‘uv sync --extra dev‘ succeeds, 45/45 tests pass, ‘cytofair --help‘ runs. |
| Repository ships provenance maps and split manifests | Verified | ‘provenance/*.parquet‘ (SIPaKMeD Branch A
    /B, P3 map, Cx22, CRIC, Herlev) and ‘splits/generated/‘ (20 SIPaKMeD split CSVs + manifest) are tracked.
    |
| Repository contains the work the paper reports | **Gap** | The public ‘main‘ stops at the 22 August results.
     The 11 Cx22-extension commits and all Study 2 work exist only locally (branch ‘study2‘). Push before
    posting so the paper’s repository link resolves to the reported code. |
| Software licence | **Authors** | The repository has no ‘LICENSE‘ file, so the toolkit is legally all-rights-
    reserved and cannot be reused. Choose a code licence (MIT or Apache-2.0 are common) and state that
    derived artifacts follow each source’s terms (below). |
| Redistribution terms for derived artifacts | **Authors** (partly verified) | No source images are
    redistributed. Derived parent IDs, coordinates, hashes, and split assignments: CRIC and the Herlev
    repackaging are CC BY 4.0 (attribution suffices); SIPaKMeD is "public for experimental use, cite Plissiti
     et al." (derived IDs very likely fine; confirm); Cx22 is non-commercial research and education only;
    CCEDD forbids redistribution of the images (the released maps contain only source file names and
    coordinates; confirm this is acceptable to the CCEDD authors). |
| CRIC images for Study 2 | Verified | All 400 slide images are CC BY 4.0 on Figshare (collection 4960286)
    and were fetched with Figshare MD5 checks; ‘data_manifests/cric_images_verified.json‘ records MD5 and SHA
    -256 per file. Images are not committed. |
| All co-authors approve the exact posted version | **Authors** | Required before posting. The current draft
    still has e-mail and repository-URL placeholders. |
| PDF text extraction | Fixed in sources | The project record’s extracted text shows broken small caps ("
    CytoF air"), ligature glyphs, and split equations. The Study 2 manuscript preamble sets T1 font encoding
    and Unicode glyph mapping; the same two lines are suggested for the ISBI draft (see ‘paper/v2/README.md‘)
    . |
| "Replication in preparation" sentence | Suggested text | "A preregistered, adequately powered replication
    and a causal decomposition of the effect are in progress." (Study 2 is registered in this repository as ‘
    PREREGISTRATION_STUDY2.md‘.) |
| Anonymity policy of the target venue | **Authors** | Posting a preprint is usually allowed; promoting it
    during double-blind review may not be. Check the chosen venue’s policy before announcing the preprint. |

## Note on public timestamps

The Study 2 preregistration is committed locally before any Study 2 result exists, matching
how Study 1 was registered (its plan commit predated its first result commit, and both were
pushed later). A local commit is weak evidence of timing. For an independent timestamp,
push the registration commit, or deposit ‘PREREGISTRATION_STUDY2.md‘ on OSF, as early as
possible; the commit hash and file hashes in ‘reports/study2/registration_hashes.txt‘ let a
```

```
reader confirm that the deposited text is the one that governed the analysis.
```

S.2 Cx22 literature check

```
# Do published Cx22 studies pool the release subsets?

**Question (roadmap, Part 5).** Cx22's official Multi-Train / Multi-Test partition is
parent-disjoint, but the separately released Pair subset shares 233 CCEDD source images
with Multi-Train and 55 with Multi-Test. Pooling the 1,320 released ROIs and re-splitting
them at the ROI level therefore reintroduces source-image leakage. Is that risk realised in
published work, or only theoretical?

**Method (2026-10-18).** Europe PMC search `"Cx22" AND (cervical OR cytology)` (43 hits),
then the open-access full text (Europe PMC `fullTextXML`) of every hit that uses Cx22
images, searched for sentences describing training, testing, splitting, or folds. Hits
where "Cx22" denotes something else (a connexin, a compound) were discarded. Papers without
open full text were not assessed.

**Stated protocols (verbatim excerpts).** The table records what each paper states; it
does not infer anything the text does not say.

| Paper | Use of Cx22 | Stated protocol |
|---|---|---|
| BMC Medical Imaging 2023, deep ensemble segmentation (PMC10514950) | train and test | "The Cx22 dataset
    contained a training dataset and a testing dataset with 400 and 100 samples, respectively." Official
    Multi split. |
| Scientific Reports 2025, Mask2Former with denoising (PMC12095581) | train and test | "The Cx22 dataset
    contains 400 images in the training dataset and 100 ones in the testing dataset." Official Multi split. |
| Cell Reports Medicine 2026, UniCAS foundation model (PMC12866161) | segmentation benchmark | "Cx22-multi:
    This public dataset consists of 500 cytology images, divided into 400 for training and 100 for testing."
    Official Multi split. |
| Cancer Cell International 2026, joint U-Net and CNN ensemble (PMC13041069) | train and validate | Describes
     Cx22 as "1,320 images" and states "a stratified five-fold cross-validation protocol, in which the data
    were divided into five mutually exclusive folds", with splitting "at the image level, ensuring that all
    cell instances derived from a single image remained within the same subset". It does not state that Pair
    was excluded or that CCEDD source images were kept within one fold. The same paper treats SIPaKMeD crops
    as "independent, pre-cropped single-cell samples" and separates them "at the image level". |
| Scientific Data 2024 (PMC11227563); Cureus 2025 (PMC11885184) | cited only | No Cx22 training or evaluation.
     |
| Frontiers in Oncology 2025 (PMC12929100) | unclear | Describes "Cx22 clinical data" modelled with an MLP;
    the text does not match the image dataset and was not assessed further. |

**Answer.** Of four open-access studies that train or evaluate on Cx22 images, three use
the official 400/100 Multi-Train/Multi-Test split, which is parent-disjoint. One describes
image-level five-fold cross-validation over the 1,320-image collection without stating that
the Pair subset or the CCEDD source images were kept apart, which is the downstream reuse
pattern the Cx22 stress test models. The risk is therefore realised in at least one recent
publication, while most users follow the safe official split, consistent with Cx22 serving
as an existence proof that a hierarchy-aware release is used in practice. The same study's
description of SIPaKMeD crops as independent samples is the misconception this project
addresses.

**Limits.** Open-access full texts only; closed-access Cx22 users (for example a 2026
Scientific Reports background-gating network, a 2024 Computers in Biology and Medicine
unsupervised instance-segmentation method, and a 2025 Medical & Biological Engineering &
Computing method) were not assessed. Whether a study pooled subsets can be stated only from
what the paper reports.

**Reporting choice.** In keeping with the project's rule that no individual paper is named
as deficient, the manuscript reports these counts in aggregate. This file keeps the
verbatim evidence so the count can be checked.
```

S.3 Excluded attempts (Study 2 smoke tests)

```
# Study 2 excluded attempts

No attempt below is eligible for any Study 2 analysis. Each was an execution check run
```

```
before registration; none trained a P2, P3, clean-cell, or CRIC P2 model, so none could
reveal a protocol contrast. Their metrics were checked only for finiteness and completeness
of artifacts, never compared across protocols.

| Attempt | Architecture | Split | Epochs | Purpose |
|---|---|---|---|---|
| `p1_<architecture>` (12 runs) | all twelve registered backbones | SIPaKMeD P1, seed 137 | 1 | pipeline,
    artifacts, throughput |
| `p1_resnet18_repeat` | ResNet-18 | SIPaKMeD P1, seed 137 | 1 | bitwise determinism (logits and features
    identical to `p1_resnet18`) |
| `leak_high_resnet50` | ResNet-50 | 2C `leak_high`, seed 137 | 1 | transplant extra-evaluation artifact |
| `p1_grl_resnet50` | ResNet-50 | SIPaKMeD P1, seed 137, lambda = 1 | 1 | adversary bookkeeping and DANN ramp
    |
| `cric_p1_resnet18` | ResNet-18 | CRIC P1, seed 137 | 1 | CRIC window cache and six-class labels |
| `p1_vit_b_16_math`, `p1_vit_b_16_math_repeat`, `p1_dinov2_vits14_math` | ViT-B/16, DINOv2 ViT-S/14 |
    SIPaKMeD P1, seed 137 | 1 | deterministic (math) attention kernels: speed and repeatability |

The first ViT-B/16 smoke run used fused memory-efficient attention, whose backward pass
PyTorch reports as non-deterministic. Registration therefore fixes the deterministic math
attention kernel for every run (Section 1 of `PREREGISTRATION_STUDY2.md`).

Outputs are kept on the training host under `smoke/` and are not copied into `results/`.

Before registration, the 2C factorial configuration with 240 crops per class was rejected
because it was infeasible for registered seed 307 (and for 10 of 100 unrelated seeds); no
model was trained on it.
```

T

Reproduction cookbook

T.1 Environment

Python 3.13 with the locked dependencies (`uv sync --extra dev`); PyTorch 2.9.0 with CUDA 12.8 on the GPU host; R 4.4 with lme4 2.0 and lmerTest 3.2 for the mixed models. Software versions recorded by every run are in Table 27.2.

T.2 Study 2

```
uv run cytofair study2-design                          # regenerate every registered split (hash-identical)
scripts/study2/node_prepare.sh ROOT PYTHON CRIC_DIR    # crop caches and transplant caches on the GPU host
python scripts/study2/build_tasks.py --root ROOT --repository REPO --python PYTHON --seeds 20
python scripts/study2/launch_matrix.py splits/study2/tasks/*.jsonl \
    --gpus 0,3,4,5 --logs ROOT/runs/logs --status ROOT/runs/status.json
analysis/study2/run_all.sh ROOT PYTHON RSCRIPT         # registered analysis; fails closed
python scripts/study2/paper_numbers.py                 # manuscript numbers and tables
python scripts/study2/figures_results.py               # Study 2 figures
```

T.3 Amendment 8

```
uv run cytofair followup-design                        # fresh-seed 2C designs and ablation plans
uv run cytofair build-carrier-caches --cache-dir ROOT/cache --cache-name sipakmed --output ROOT/cache
python scripts/study2/build_followup_tasks.py --root ROOT --repository REPO \
    --study2-runs ROOT/runs --python PYTHON
python scripts/study2/launch_matrix.py splits/study2/followup/tasks/*.jsonl \
    --gpus 4,5,6,7 --logs ROOT/runs/followup/logs --status ROOT/runs/followup/status.json
analysis/study2/followup/run_followup.sh ROOT PYTHON RSCRIPT
```

```
python analysis/study2/followup/exploratory_overlap.py --study2-runs ROOT/runs \
    --followup-runs ROOT/runs/followup --output results/study2/followup/exploratory/overlap.json
python scripts/study2/figures_followup.py            # follow-up figures
```

T.4 This record

```
python record/scripts/split_study1.py                # Study 1 chapters from monograph/main.tex
python record/scripts/export_run_records.py RECORD_DATA results/study2/run_records
python record/scripts/make_record_tables.py          # tables and record macros
python record/scripts/make_record_figures.py         # record figures
record/build.sh                                      # full record and standalone part PDFs
```

U

Where everything is

Table U.1: Key files and directories.

Path	Content
`PREREGISTRATION.md`	Study 1 plan and Amendments 1–6
`PREREGISTRATION_STUDY2.md`	Study 2 plan, Amendments 7–9 pointers
`PREREGISTRATION_STUDY2_AMENDMENT8.md`	Amendments 8 and 9 in full
`provenance/`	crop-to-parent maps for the four datasets
`splits/study2/`	registered Study 2 splits, designs, plans, tasks, manifests
`splits/study2/followup/`	Amendment 8 designs, ablation plans, tasks, manifests
`src/cytofair/train_study2.py`	the Study 2 trainer
`src/cytofair/gpu_augment.py`	GPU-resident augmentation
`src/cytofair/image_cache.py`	packed, hashed crop caches
`src/cytofair/factorial.py`, `transplant.py`	2C designs and 2B composites
`src/cytofair/carriers.py`, `followup_eval.py`	Amendment 8 transforms, LEACE, Grad-CAM, device probe
`src/cytofair/risk.py`	structural closed forms (`risk-report`)
`scripts/study2/`	task builders, scheduler, figures, numbers, archiving
`analysis/study2/`	registered Study 2 analysis (R and Python)
`analysis/study2/followup/`	registered Amendment 8 analysis, exploratory overlap
`results/study2/analysis/`	Study 2 registered outputs
`results/study2/followup/`	Amendment 8 registered and exploratory outputs
`results/study2/run_records/`	exported records of every run
`reports/study2/`	registration hashes, Phase 0, literature check, excluded attempts
`paper/v2/`	the single-column manuscript
`monograph/`	the Study 1 record source
`record/`	this record
`~/cytofair_study2_archive` (GPU host)	121 GB archive of every run artifact

V

Glossary

Parent
The acquisition unit a crop was cut from: a SIPaKMeD cluster field, a Cx22 (CCEDD) source image, or a CRIC slide image.

P1, P1′, P2, P3
Crop-level, size-matched crop-level, parent-disjoint, and component-disjoint splits.

Straddle
A parent with crops on both sides of a split.

Leak and clean cells
2C training sets that do or do not contain the test fields' sibling crops, at matched size and diversity.

Recoverability R
Within-class field-identification accuracy of a linear probe on frozen features.

Regime
The manipulation levels whose mean P2 accuracy stays within 1 pp of baseline.

Manipulation check
At least a quarter of the above-floor recoverability removed at the strongest in-regime level.

Holm family
A set of tests whose p-values are adjusted together by Holm's step-down procedure.

LEACE
Least-squares concept erasure: a linear projection after which no linear function of the features covaries with the concept.

GRL
Gradient-reversal layer: identity in the forward pass, negated and scaled gradient in the backward pass.

HED jitter
Random scaling and shifting of stain channels in optical-density space.

Reinhard transfer
Matching an image's LAB mean and standard deviation to target values.

Bibliography